

Basic Python workshop





Facebook interface for the page **somkiat.cc**. The top navigation bar includes the Facebook logo, the page name **somkiat.cc**, a search icon, and user profile icons for **Somkiat** and **Home**. The main navigation bar features **Page** (selected), **Messages**, **Notifications** (3), **Insights**, **Publishing Tools**, **Settings**, and **Help**.

The page cover image shows a man in a white Superman t-shirt with "SOMKIAT.CC" printed on it, posing against a white wall. The profile picture is a smaller version of the same image. The page name **somkiat.cc** and handle **@somkiat.cc** are displayed below the profile picture.

The left sidebar contains a menu with **Home** (selected), **Posts**, **Videos**, and **Photos**.

The main content area displays the cover image with a blue call-to-action button that says "Help people take action on this Page." and a blue button labeled "+ Add a Button". Below the cover image, there are buttons for **Liked**, **Following**, **Share**, and a three-dot menu icon.



**[https://github.com/up1/
course-python-workshop](https://github.com/up1/course-python-workshop)**

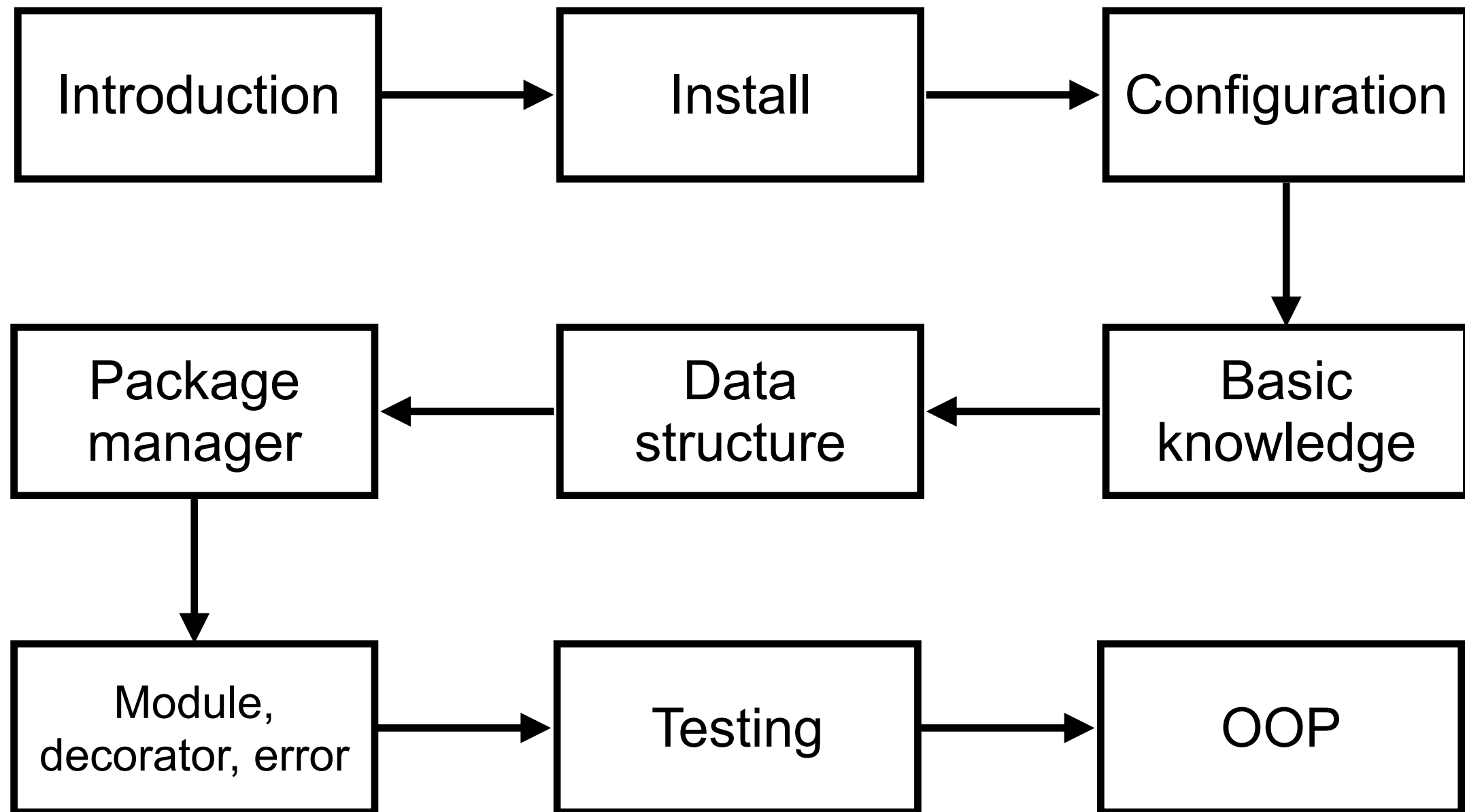


Topics

Introduction to Python
Install and configuration
Basic of Python
Setup environment
Data structure
Package manager
Testing
OOP (Object Oriented Programming)



Learning Path 1



Introduction to Programming



TIOBE Index May 2025

May 2025	May 2024	Change	Programming Language		Ratings	Change
1	1			Python	25.35%	+9.02%
2	3	▲		C++	9.94%	+0.41%
3	2	▼		C	9.71%	-0.27%
4	4			Java	9.31%	+0.62%
5	5			C#	4.22%	-2.27%
6	6			JavaScript	3.68%	+0.66%
7	8	▲		Go	2.70%	+1.10%
8	7	▼		Visual Basic	2.62%	+0.61%
9	11	▲		Delphi/Object Pascal	2.29%	+1.05%
10	9	▼		SQL	1.90%	+0.45%

<https://www.tiobe.com/tiobe-index/>



Basic of Python

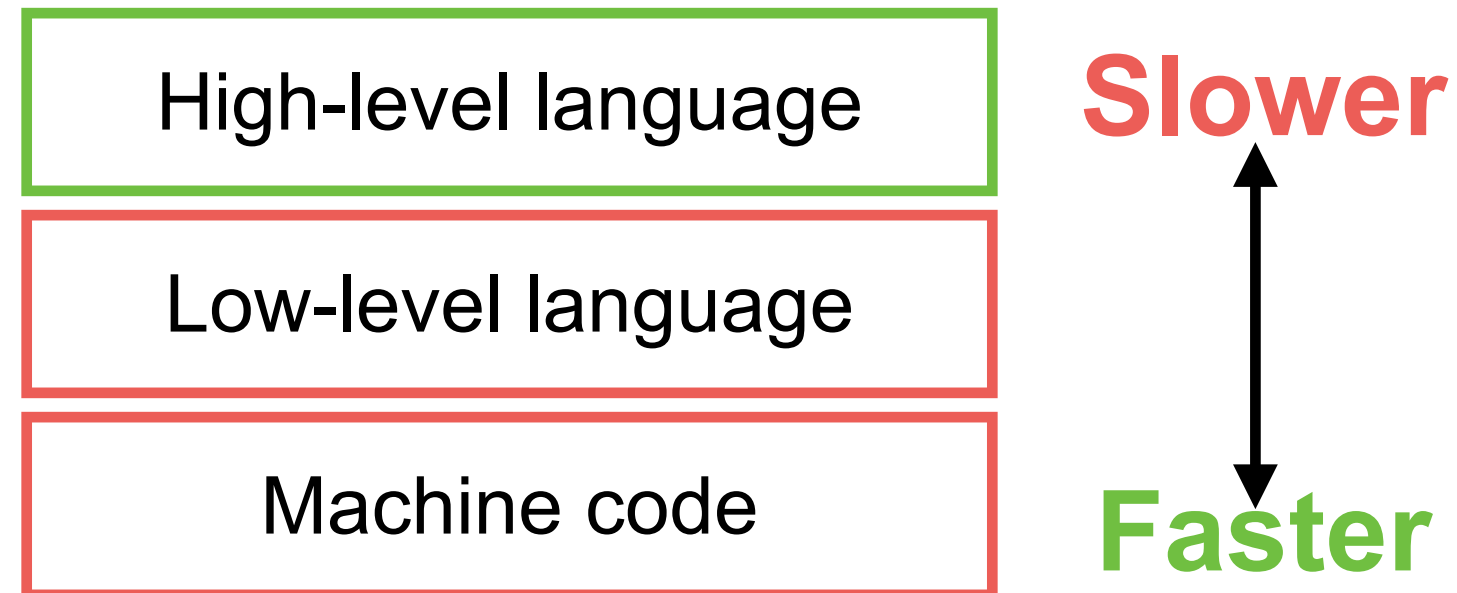
High-level programming language

More human readable

General purpose

Open-source

Interpreter



Python use cases

Data science

Machine learning

Web development

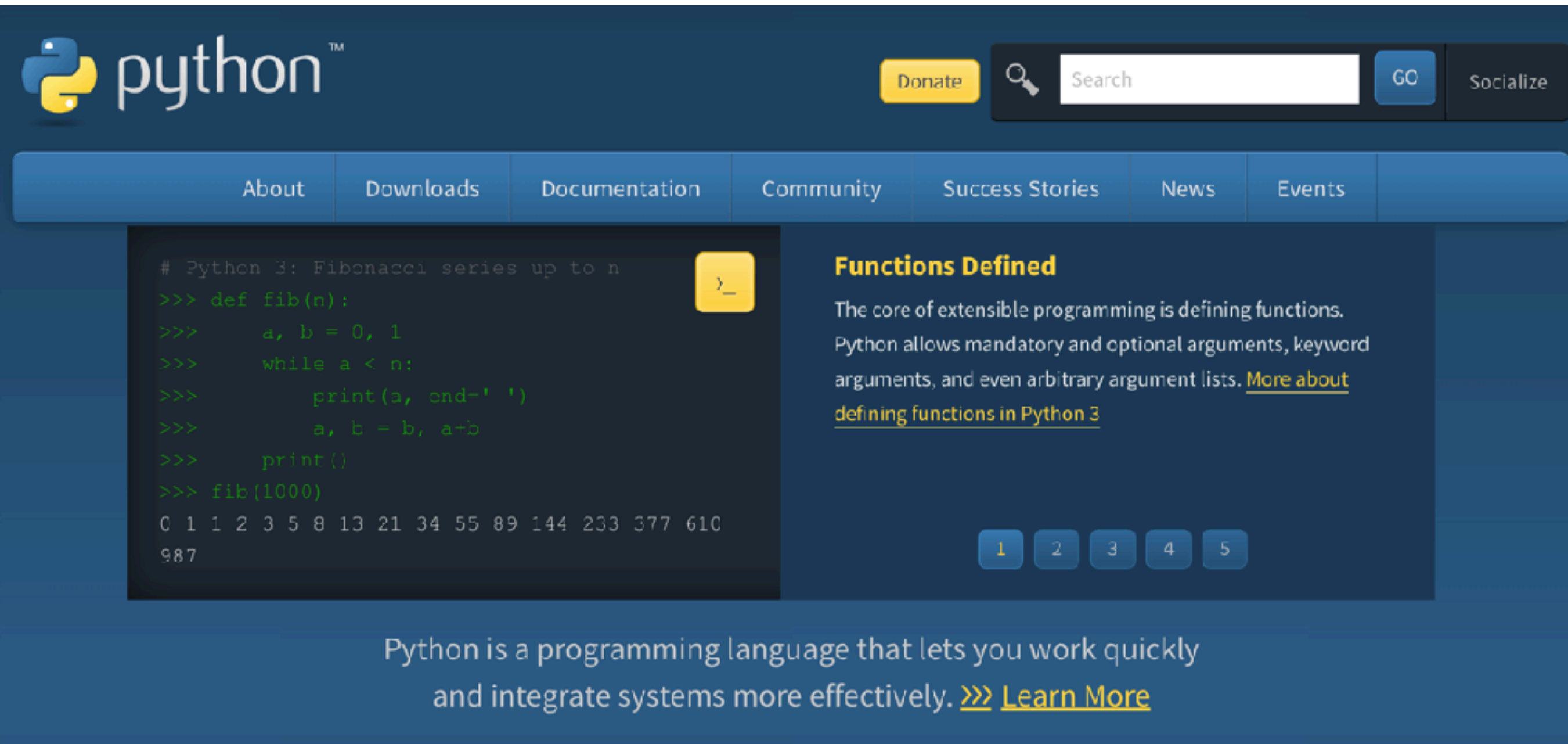
Desktop application

Task automation

Utility software



Install



The image is a screenshot of the Python.org homepage. At the top left is the Python logo and the word "python" with a trademark symbol. To the right are a yellow "Donate" button, a search bar with a magnifying glass icon and a "GO" button, and a "Socialize" button. Below these are navigation tabs: "About", "Downloads", "Documentation", "Community", "Success Stories", "News", and "Events". The main content area is split into two columns. The left column contains a code block with a yellow terminal icon, showing a Python script for a Fibonacci series and its output. The right column has a section titled "Functions Defined" with a description of functions and a link to "More about defining functions in Python 3". Below this are five numbered buttons (1-5). At the bottom, a large blue banner contains the text: "Python is a programming language that lets you work quickly and integrate systems more effectively. >>> [Learn More](#)".

python™

Donate

Search GO Socialize

About Downloads Documentation Community Success Stories News Events

```
# Python 3: Fibonacci series up to n
>>> def fib(n):
>>>     a, b = 0, 1
>>>     while a < n:
>>>         print(a, end=' ')
>>>         a, b = b, a+b
>>>     print()
>>> fib(1000)
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610
987
```

Functions Defined

The core of extensible programming is defining functions. Python allows mandatory and optional arguments, keyword arguments, and even arbitrary argument lists. [More about defining functions in Python 3](#)

1 2 3 4 5

Python is a programming language that lets you work quickly and integrate systems more effectively. >>> [Learn More](#)

<https://www.python.org/>



Check your Python

```
$python -V
```

```
$pip -V
```

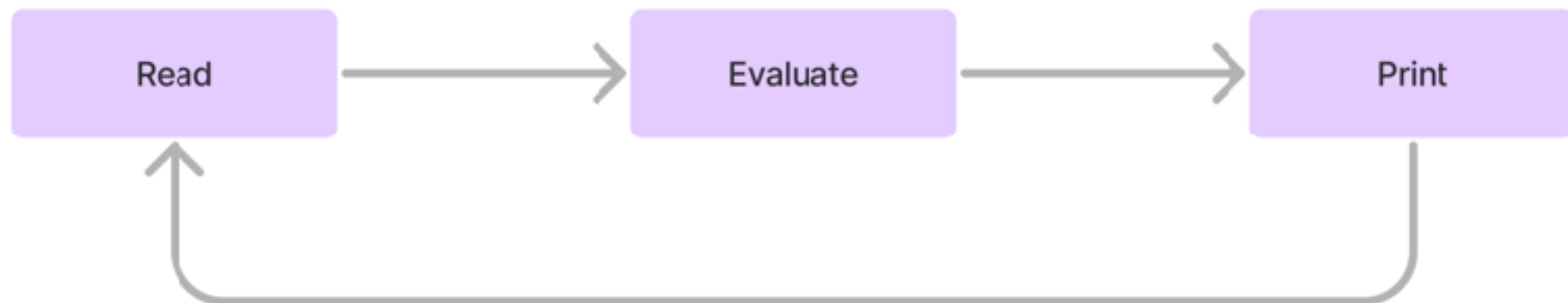


REPL mode

Read-Evaluate-Print-Loop

Interactive mode

Easy to code and fast feedback loop



IDE

VSCode

PyCharm

Jupyter Notebook (web-based)



Visual Studio Code



Linters for Python

Pylint
Mypy
Ruff
Flake8



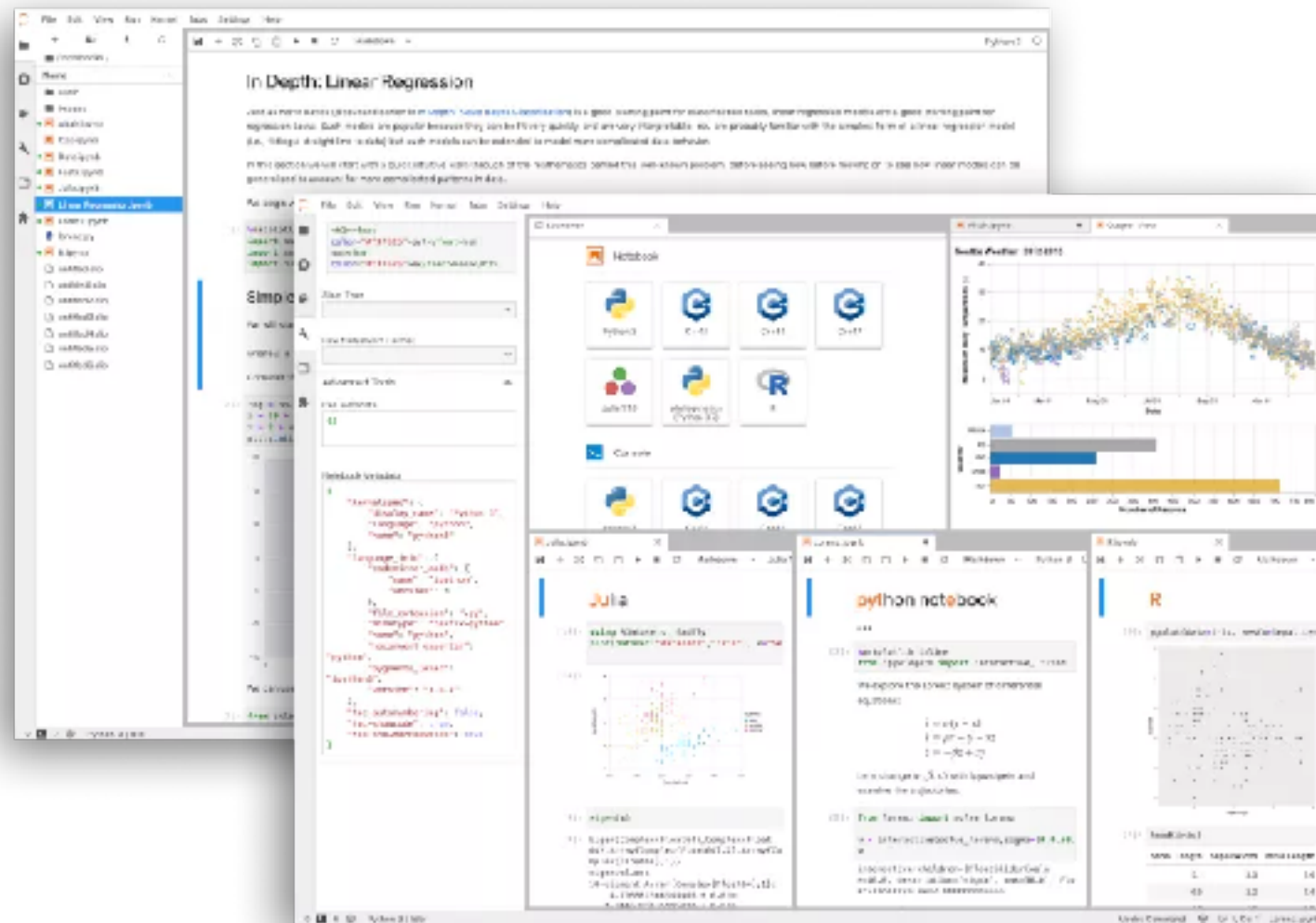
Visual Studio Code

<https://code.visualstudio.com/docs/python/linting>



Jupyter Notebook (web-based)

\$pip install jupyterlab
\$jupyter lab



<https://jupyter.org/>



First Program

Run as Interpreter process

\$python hello.py

Source code
hello.py

Virtual machine

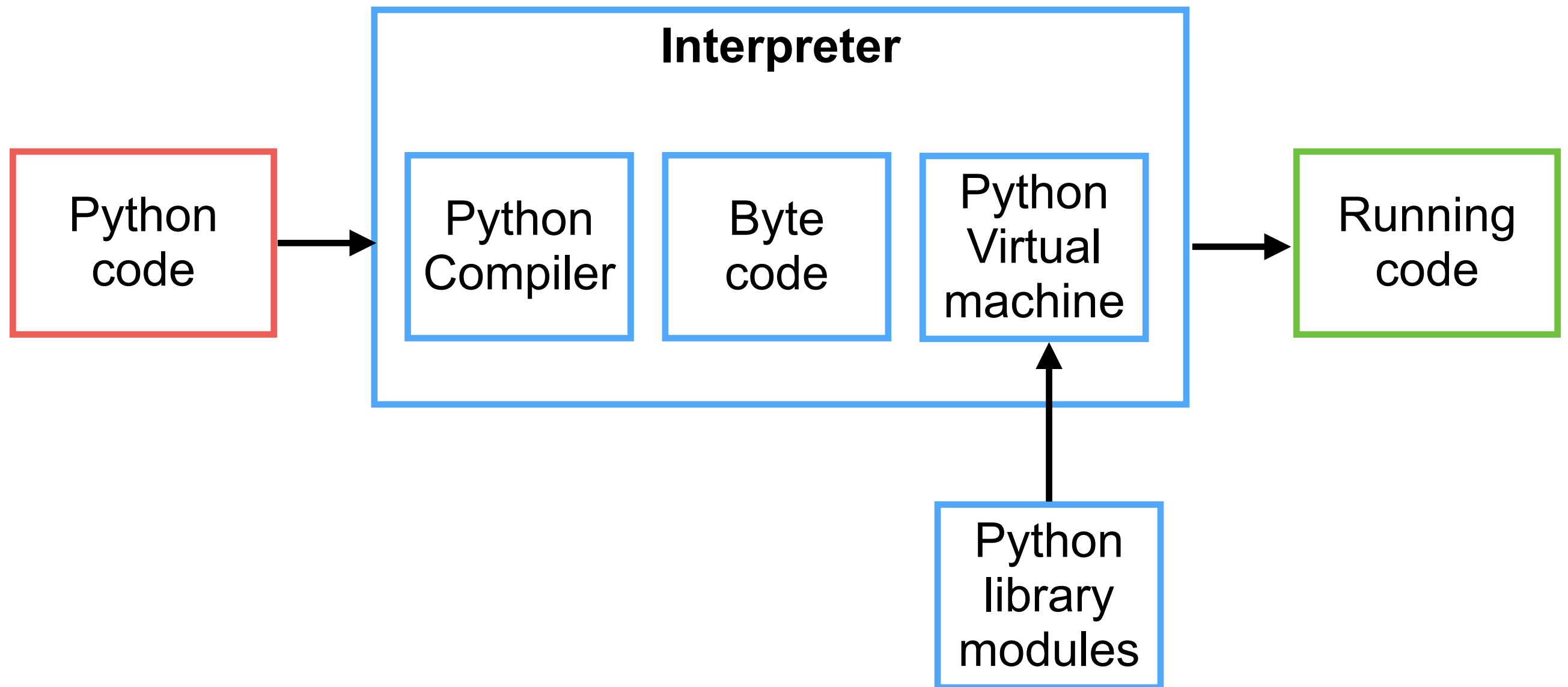
Machine code

```
print("Hello, World!")
```



Interpreter

Convert code into machine code
when the program is run



Run as Script

\$hello.py

```
#!/usr/bin/env python  
print("Hello, World!")
```



Run as Module

\$python -m hello

Compiled

Source code
hello.py

__pycached__
*.pyc

Machine code

```
print("Hello, World!")
```



Improve performance before calling/import from others



Better structure of program

```
def say_hello():  
    """Prints a greeting message."""  
    print("Hello, World!")  
  
if __name__ == "__main__":  
    say_hello()
```



Object Oriented Programming

```
class HelloWorld:
    def __init__(self, name):
        self.name = name

    def say(self):
        return f"Hello, {self.name}!"

if __name__ == "__main__":
    hello = HelloWorld("World")
    print(hello.say())
```



Whitespace !!

```
class HelloWorld:
    ↔ def __init__(self, name):
        ↔ self.name = name

    ↔ def say(self):
        ↔ return f"Hello, {self.name}!"

if __name__ == "__main__":
    ↔ hello = HelloWorld("World")
    ↔ print(hello.say())
```



Zen of Python

https://en.wikipedia.org/wiki/Zen_of_Python



Zen of Python

\$python
>>> import this

```
>>> import this
The Zen of Python, by Tim Peters

Beautiful is better than ugly.
Explicit is better than implicit.
Simple is better than complex.
Complex is better than complicated.
Flat is better than nested.
Sparse is better than dense.
Readability counts.
Special cases aren't special enough to break the rules.
Although practicality beats purity.
Errors should never pass silently.
Unless explicitly silenced.
In the face of ambiguity, refuse the temptation to guess.
There should be one-- and preferably only one --obvious way to do it.
Although that way may not be obvious at first unless you're Dutch.
Now is better than never.
Although never is often better than *right* now.
If the implementation is hard to explain, it's a bad idea.
If the implementation is easy to explain, it may be a good idea.
Namespaces are one honking great idea -- let's do more of those!
```

<https://peps.python.org/pep-0020/>



Variables and Data Types



Variable Name Rules

Start with letter or underscore character

Can't start with number

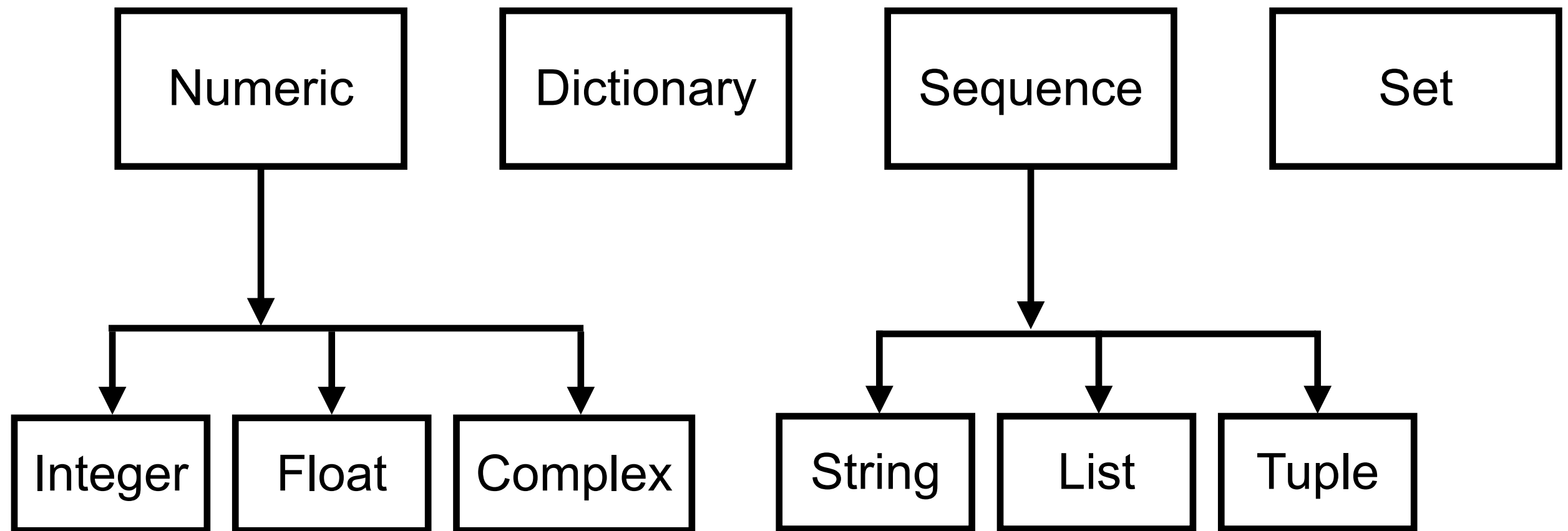
Only contain (A-z, 0-9 and _)

Case-sensitive

Python developer use **snake_case**



Data Types



Data Structure in Python

List

Dictionary

Tuple

Set

Queue

Linked list

Stack

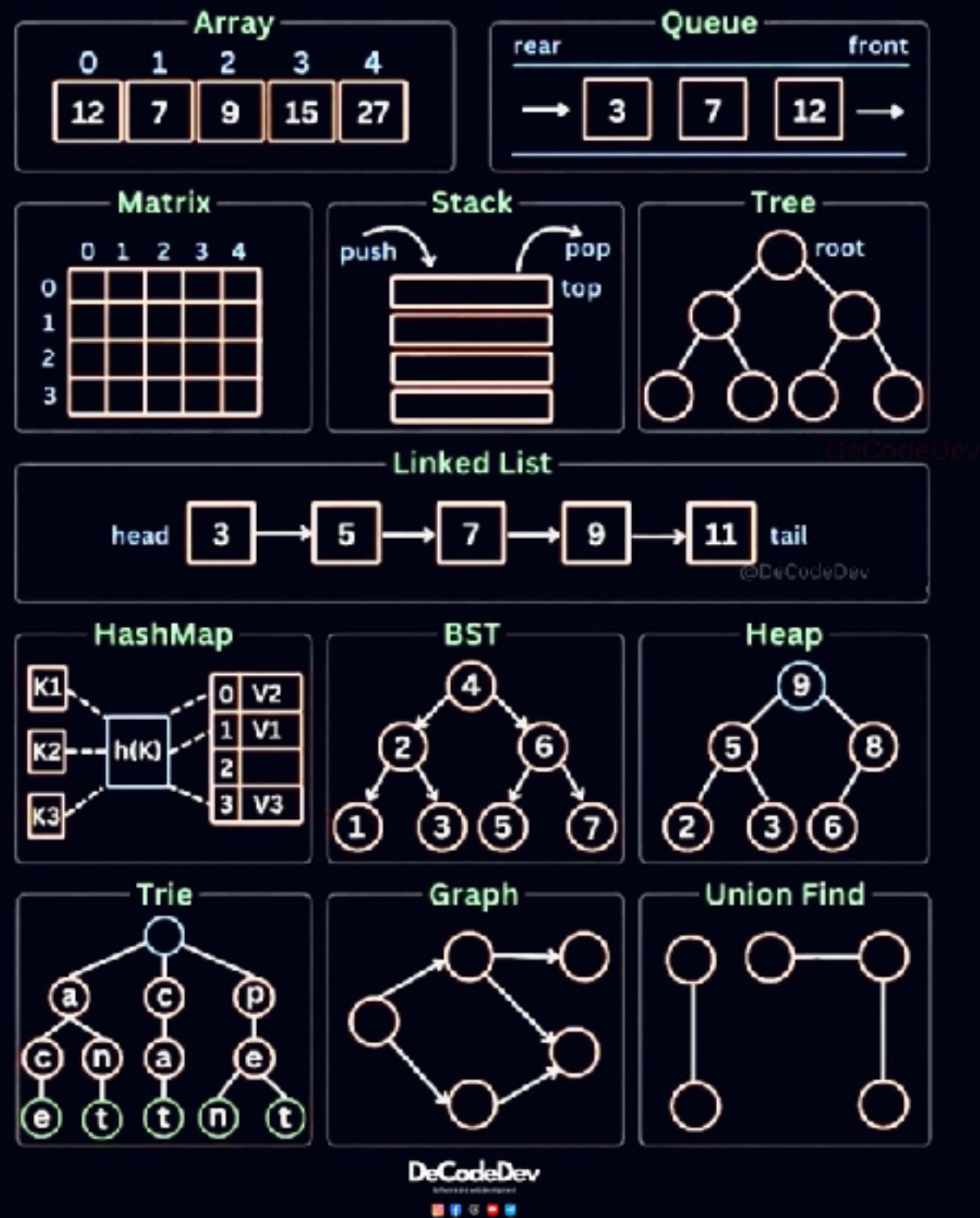
Tree

Graph

HashMap



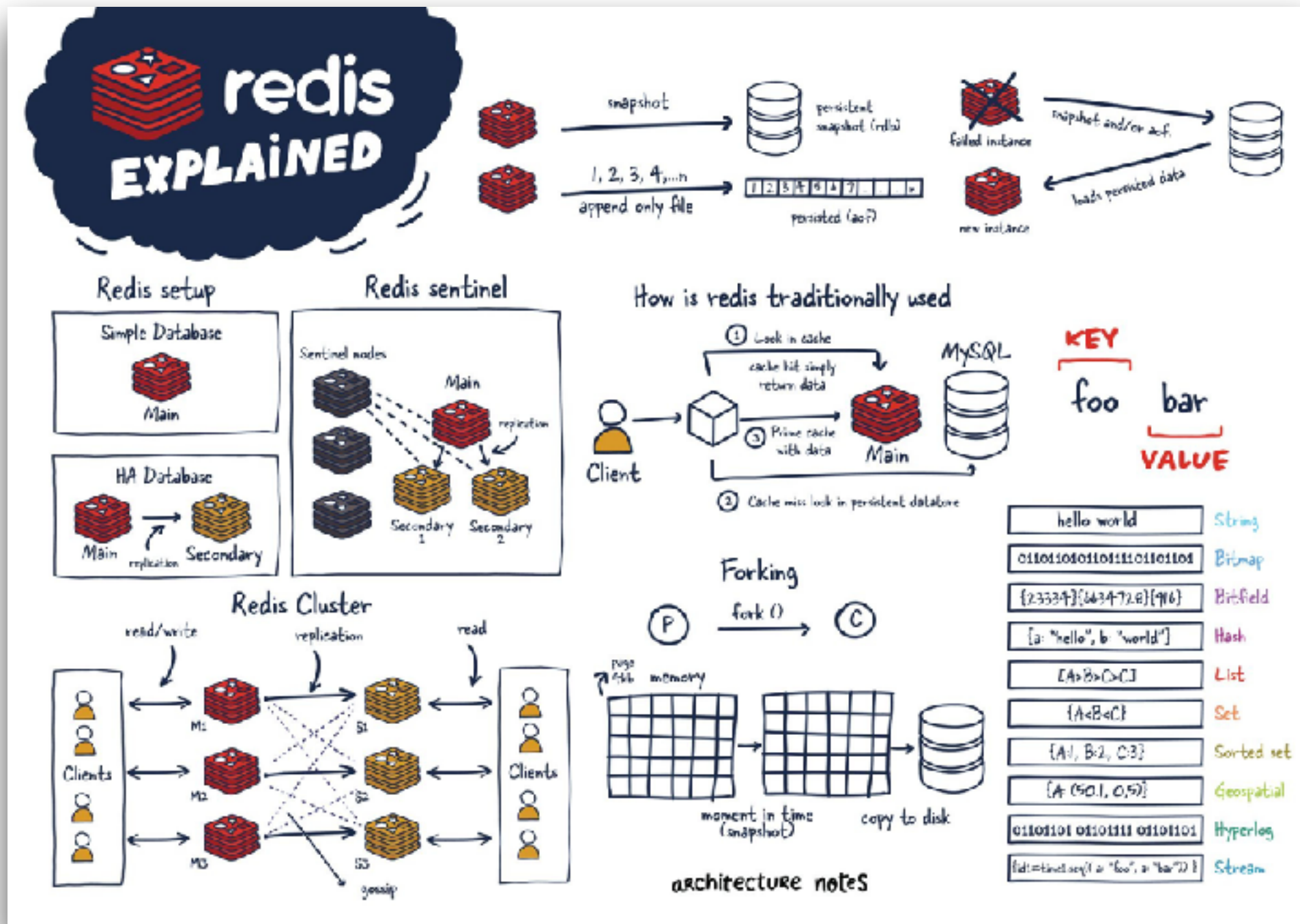
Type of Data Structure



<https://www.facebook.com/photo/?fbid=122234510876233789&set=a.122121113024233789>



Data Structure Database !!



<https://architecturenotes.co/p/redis>



Control flows

If

For

Range()

While

Break

Continue

Pass

For-else

Match

<https://docs.python.org/3/tutorial/controlflow.html>



Basic build-in functions

type()
dir()
vars()
help()
input()



Working with String

Strings are immutable (can't change)

Single quote vs Double quote !!

```
data = "Hello, World!"

print(data) # Print the string
print(len(data)) # Length of the string
print(type(data)) # Type of data is str (string)
print(data[0]) # First character
print(data[-1]) # Last character
print(data[0:5]) # First five characters
print(data[7:]) # From the 8th character to the end
print(data[7:12]) # Characters from index 7 to 11
print(data[7:12:2]) # Characters from index 7 to 11, stepping by 2
print(data[::-1]) # Reverse the string

data[0]= "H" # This will raise an error because strings are immutable
```



String formatting

```
a = "Hello"
b = "World"

c = f"{a}, {b}!" # Using f-string for formatting
print(c) # Output: Hello, World!

d = "{} {}".format(a, b) # Using format method for formatting
print(d) # Output: Hello World!

e = "%s %s!" % (a, b) # Using old-style formatting
print(e) # Output: Hello World!

f = "{0}, {1}!".format(a, b) # Using format method with positional arguments
print(f) # Output: Hello, World!

g = "{name}, {place}!".format(name=a, place=b) # Using format method with named arguments
print(g) # Output: Hello, World!
h = "Hello, {name}!".format(name=b) # Using format method with a single named argument
print(h) # Output: Hello, World!
i = "Hello, {0}!".format(b) # Using format method with a positional argument
print(i) # Output: Hello, World!
j = "Hello, {name}!".format(name="Alice") # Using format method with a named argument
print(j) # Output: Hello, Alice!

k = " ".join([a, b]) # Joining strings with a space
print(k) # Output: Hello World
```



List vs Tuple

Lists are mutable
Tuples are immutable

[1,2] (1,2)



List's operation

```
my_list = [1,2,3]

my_list.append(4) # Adding an element to the end of the list
my_list.insert(0, 0) # Inserting an element at the beginning of the list
my_list.remove(2) # Removing the first occurrence of an element
my_list.pop() # Removing the last element of the list
my_list.sort() # Sorting the list in ascending order
my_list.extend([5, 6]) # Extending the list with another list
my_list.reverse() # Reversing the order of the list
```



Multiply by 2 for all elements

```
my_list = [1,2,3]
```

```
for i in my_list:  
    i *= 2
```

```
my_list = [1,2,3]
```

```
for i in range(len(my_list)):  
    my_list[i] *= 2
```

```
my_list = [1,2,3]
```

```
# List comprehension to multiply each element by 2  
my_list = [x * 2 for x in my_list]  
print(my_list)
```

Readability code !!



Looping in Python !!

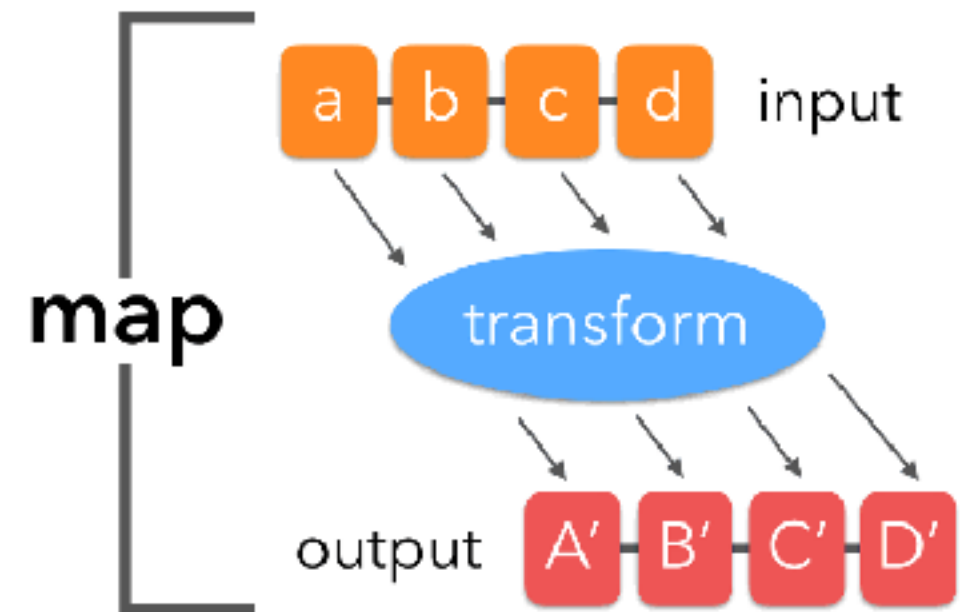
For loop

While loop

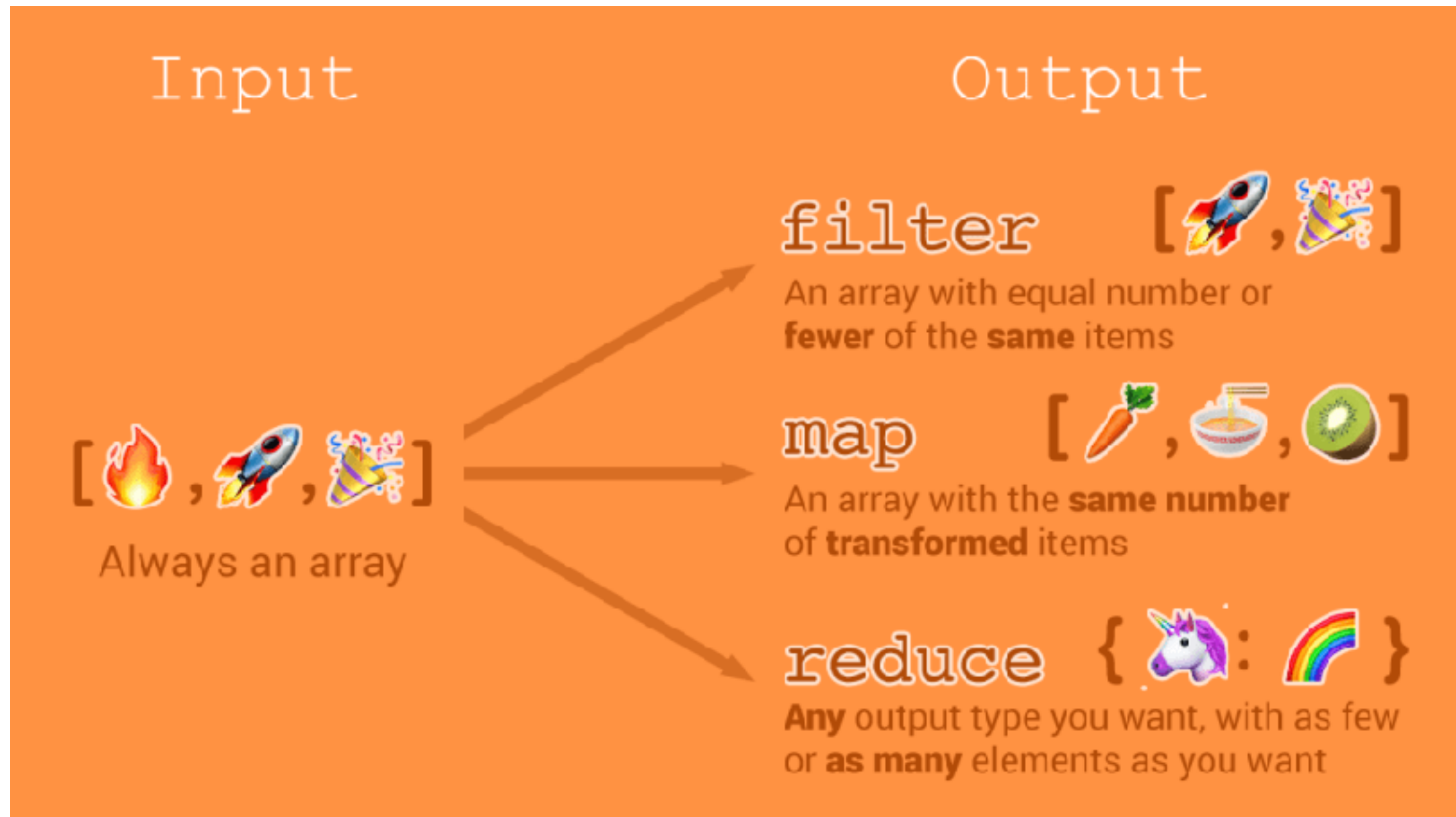


Functional programming !!

```
def multiply_by_two(x):  
    return x * 2  
  
def is_even(x):  
    return x % 2 == 0  
  
if __name__ == "__main__":  
    my_list = [1,2,3]  
  
    # Filter and Map with Functions  
    result_list = filter(is_even, my_list)  
    print(list(result_list))  
  
    # Map with Functions  
    result_list = map(multiply_by_two, my_list)  
    print(list(result_list))
```



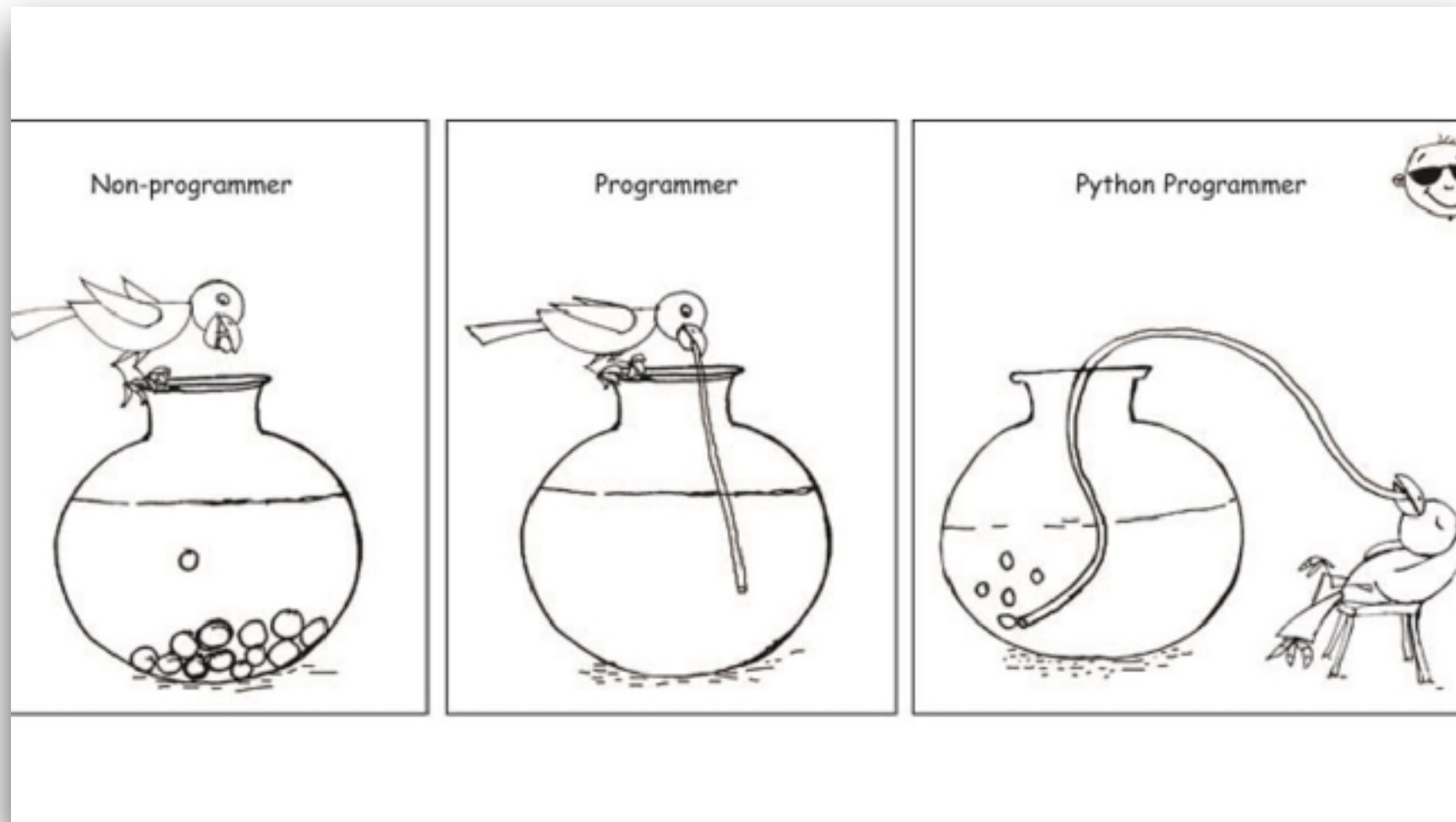
Filter, Map and Reduce



https://www.learnpython.org/en/Map%2C_Filter%2C_Reduce



Python programmer



Tuple

Tuples are immutable
Cannot change !!



Convert List to Tuple ?

Using tuple()

Using loop inside tuple()

Unpack list

```
data_list = ["Hello", "World", 1, 2, 3.14, True]

# Use tuple()
data_tuple = tuple(data_list)

# Use for loop in tuple
data_tuple = tuple(x for x in data_list)

# Use unpacking list to tuple
data_tuple = (*data_list,)

print(type(data_tuple))
print(data_tuple)
```



Set

Sets are mutable

No duplicate elements (unique)

Not maintain the order of the data insertion



Basic of Set

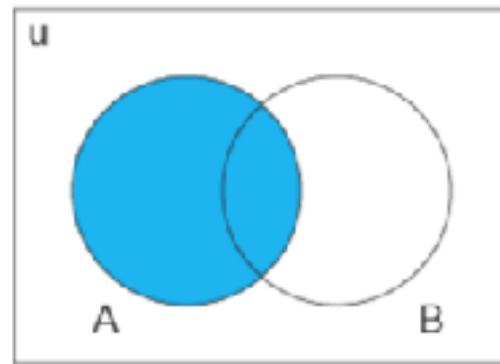
```
data_list = [1,1,1,2,2,3,4,5,6,7,8,9,10]
data_set = set(data_list)
print(type(data_set)) # <class 'set'>
print(data_set) # {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

# Add elements to the set
data_set.add(11) # Adding a single element
data_set.update([12, 13]) # Adding multiple elements
print(data_set) # {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13}

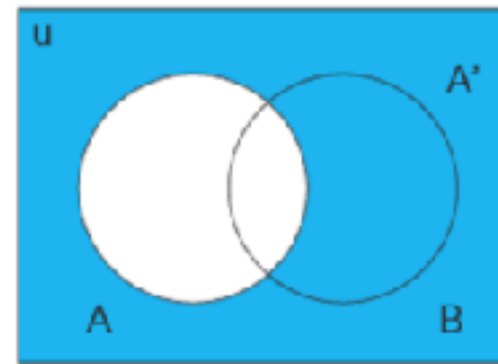
# Remove elements from the set
data_set.remove(1) # Removing a specific element
data_set.discard(2) # Discarding an element (no error if not found)
data_set.pop() # Removing an arbitrary element
print(data_set) # {4, 5, 6, 7, 8, 9, 10, 11, 12, 13}
```



Set's operations



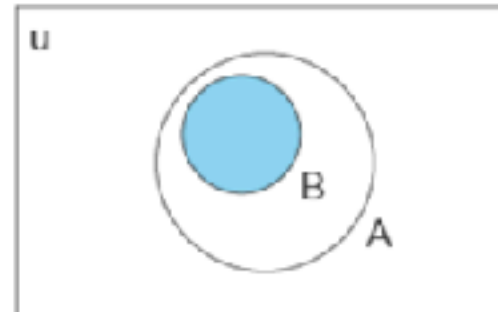
Set A



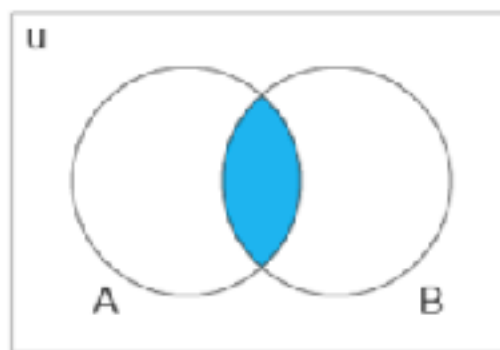
A' the complement of A



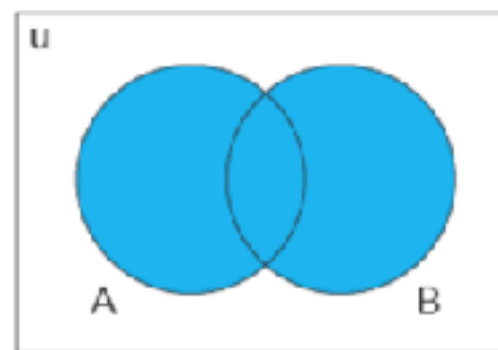
A and B are disjoint sets



B is proper subset of A
 $B \subset A$



Both A and B
A intersect B
 $A \cap B$



Either A or B
A union B
 $A \cup B$

<https://realpython.com/python-sets/>



Set's operations

```
# Check membership
print(3 in data_set)  # True
print(14 in data_set) # False

# Set operations
data_set2 = {5, 6, 7, 8, 9, 10, 11, 12}

# Union
union_set = data_set.union(data_set2)

# Intersection
intersection_set = data_set.intersection(data_set2)

# Difference
difference_set = data_set.difference(data_set2)

# Symmetric Difference
symmetric_difference_set = data_set.symmetric_difference(data_set2)

# Set comprehension
data_set_comp = {x * 2 for x in data_set if x % 2 == 0}
```



List vs Tuple vs Set

	Mutable	Ordered	Indexing / Slicing	Duplicate Elements
List	✓	✓	✓	✓
Tuple	✗	✓	✓	✓
Set	✓	✗	✗	✗

<https://towardsdatascience.com/15-examples-to-master-python-lists-vs-sets-vs-tuples-d4ffb291cf07/>



Workshop

Count unique words from text file

Read data
from file

Cleaning
data

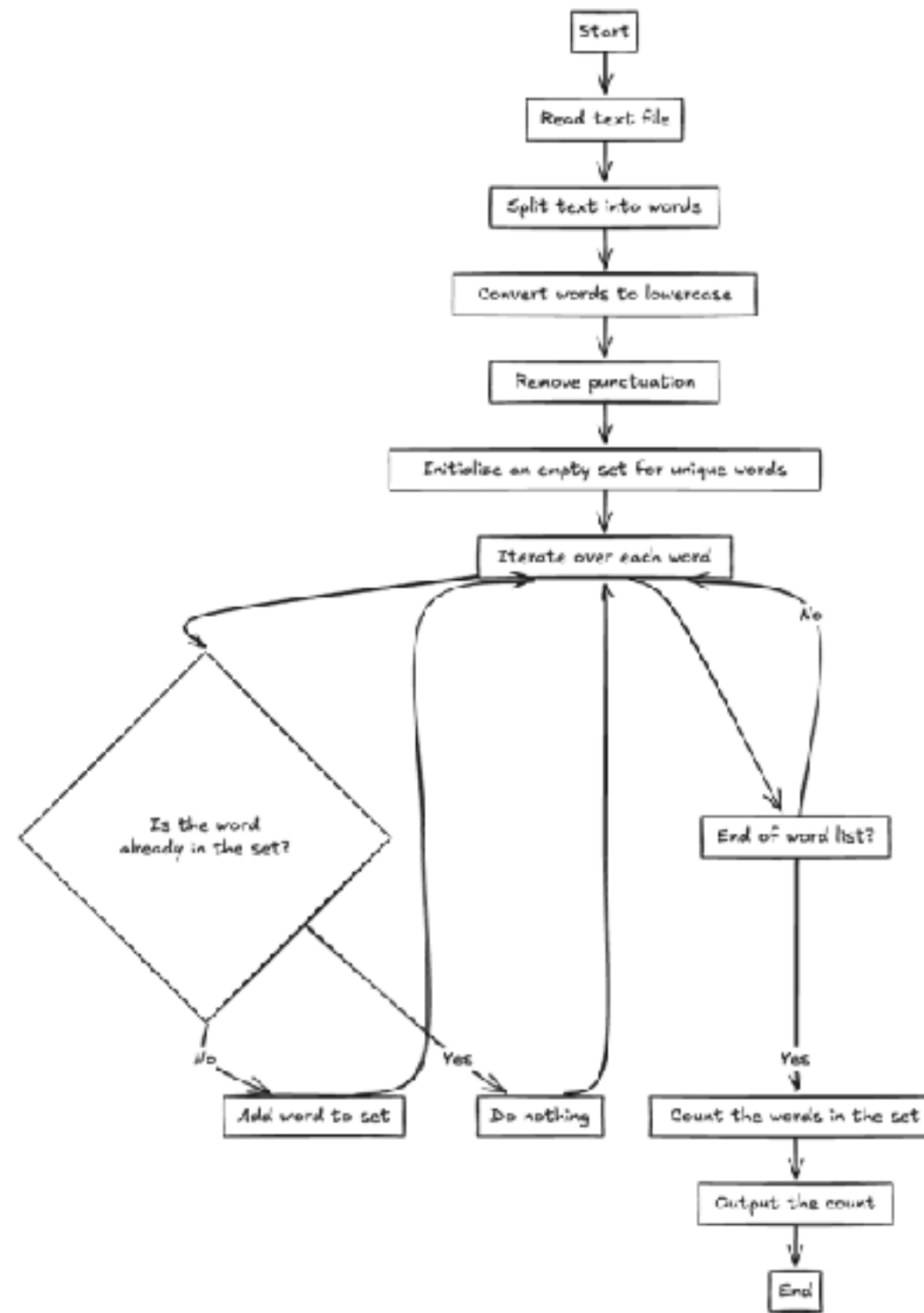
Splitting
data

Remove
duplicate
Words

Count all words



Workshop (Flow chart)



Error handling !!

Reliable of program to read data from file

```
def main():  
    file_path = 'data/message.txt'  
    try:  
        input_text = read_file(file_path)  
        unique_word_count = count_unique_words(input_text)  
        print(f"Number of unique words: {unique_word_count}")  
    except FileNotFoundError:  
        print(f"The file {file_path} does not exist.")  
    except Exception as e:  
        print(f"An error occurred: {e}")
```

<https://docs.python.org/3/tutorial/errors.html>



Working with logging

Keep stages of application

Unstructured

Structured

<https://docs.python.org/3/howto/logging-cookbook.html>



Logging !!

INFO - User John Doe with ID 123 made a purchase of \$200 on a Visa card with last four digits of 4312 on 2025-05-25.

```
{  
  "severity": "INFO",  
  "timestamp": "2025-05-25T12:34:56Z",  
  "userId": 123,  
  "userName": "John Doe",  
  "action": "purchase",  
  "card_type": "Visa",  
  "last_four_digits": 1414,  
  "amount": 200  
}
```

<https://newrelic.com/blog/how-to-relic/python-structured-logging>



Structured log

```
import structlog
import logging

# Configure structlog to output structured logs in JSON format
structlog.configure(
    processors=[
        structlog.stdlib.filter_by_level,
        structlog.processors.TimeStamper(fmt="iso"),
        structlog.processors.JSONRenderer()
    ],
    context_class=dict, logger_factory=structlog.stdlib.LoggerFactory()
)

# Configure the standard logging module to use structlog
logging.basicConfig(
    format="%(message)s",
    level=logging.INFO,
    handlers=[logging.StreamHandler()]
)

# Get a logger
logger = structlog.get_logger()

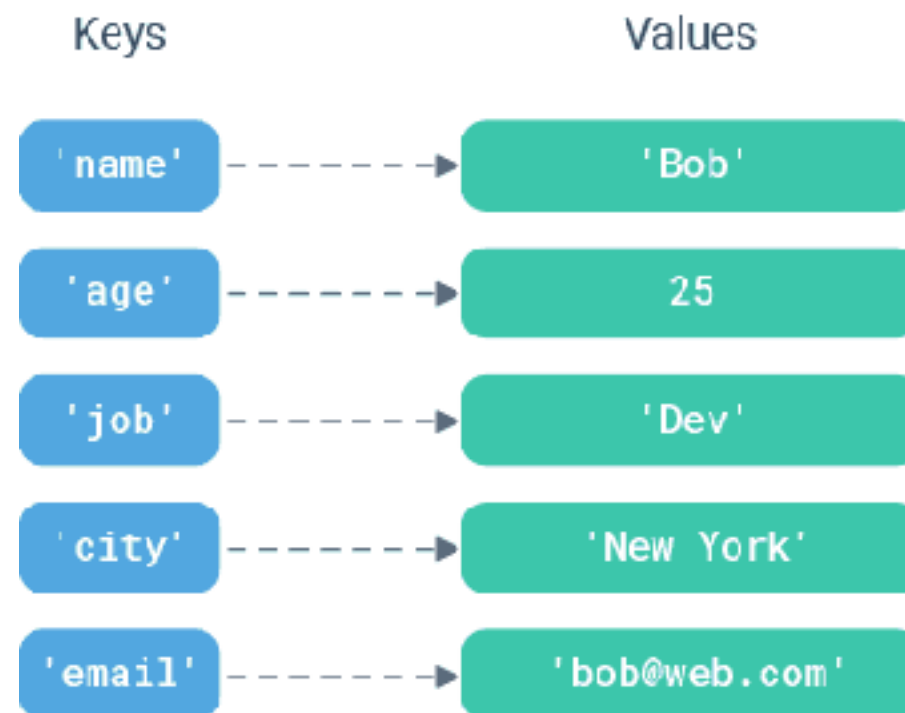
# Now we can log structured messages!
logger.info("User logged in", user_id="1234", ip="192.0.2.0")
```

<https://www.structlog.org/>



Dictionary

Ordered collections of unique values (key-value)
Dictionaries are mutable



Basic of Dictionary

```
data_dict = {  
    "name": "Alice",  
    "age": 30,  
    "city": "New York"  
}  
# Accessing values  
print(data_dict["name"]) # Alice  
print(data_dict.get("age")) # 30  
  
# Adding a new key-value pair  
data_dict["country"] = "USA"  
  
# Updating an existing key-value pair  
data_dict["age"] = 31  
  
# Removing a key-value pair  
data_dict.pop("city")
```



Basic of Dictionary

```
data_dict = {  
    "name": "Alice",  
    "age": 30,  
    "city": "New York"  
}  
  
# Iterating through keys and values  
for key, value in data_dict.items():  
    print(f"{key}: {value}")  
  
# Checking if a key exists  
print("name" in data_dict)    # True  
print("city" in data_dict)    # False  
  
# Merging dictionaries  
data_dict2 = {"state": "NY", "zip": "10001"}  
data_dict.update(data_dict2)
```



Workshop

Merge lists into a list

Names = [A, B, C]

Ages = [10, 20, 30]

Cities = [X, Y, Z]



```
[  
    {'age': 10, 'city': 'X', 'name': 'A'},  
    {'age': 20, 'city': 'Y', 'name': 'B'},  
    {'age': 30, 'city': 'Z', 'name': 'C'},  
]
```

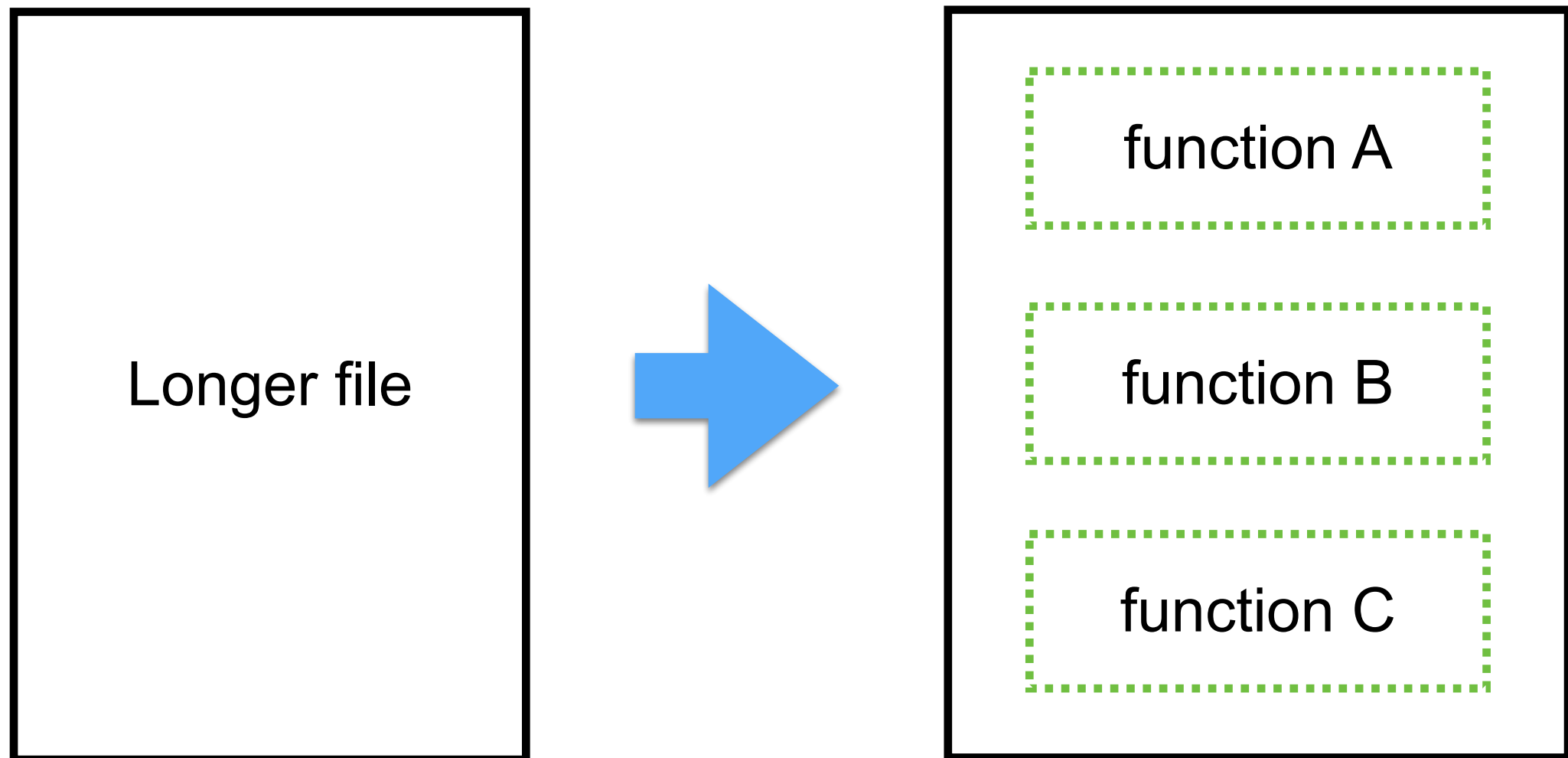


Functions in Python



Functions

Split long script to small groups
Manageable code



Build-in functions

Built-in Functions

The Python interpreter has a number of functions and types built into it that are always available. They are listed here in alphabetical order.

Built-in Functions			
A <u>abs()</u> <u>aiter()</u> <u>all()</u> <u>anext()</u> <u>any()</u> <u>ascii()</u>	E <u>enumerate()</u> <u>eval()</u> <u>exec()</u>	L <u>len()</u> <u>list()</u> <u>locals()</u>	R <u>range()</u> <u>repr()</u> <u>reversed()</u> <u>round()</u>
B <u>bin()</u> <u>bool()</u> <u>breakpoint()</u> <u>bytearray()</u> <u>bytes()</u>	F <u>filter()</u> <u>float()</u> <u>format()</u> <u>frozenset()</u>	M <u>map()</u> <u>max()</u> <u>memoryview()</u> <u>min()</u>	S <u>set()</u> <u>setattr()</u> <u>slice()</u> <u>sorted()</u> <u>staticmethod()</u> <u>str()</u> <u>sum()</u> <u>super()</u>
	G <u>getattr()</u> <u>globals()</u>	N <u>next()</u>	
		O	

<https://docs.python.org/3/library/functions.html>



Working with function (1)

```
# Using the Force formula F=ma

while True:
    mass = int(input("Enter the mass value: "))
    if mass > 0:
        break

while True:
    acceleration = int(input("Enter the acceleration: "))
    if acceleration > 0:
        break

print("The Force is", mass * acceleration)
```

<https://www.freecodecamp.org/news/best-practices-for-refactoring-code/>



Working with function (2)

```
# Using the Force formula F=ma

def get_positive_integer(prompt):
    while True:
        try:
            value = int(input(prompt))
            if value > 0:
                return value
        except ValueError:
            print("Please enter a valid positive integer.")

def calculate_force(mass, acceleration):
    return mass * acceleration

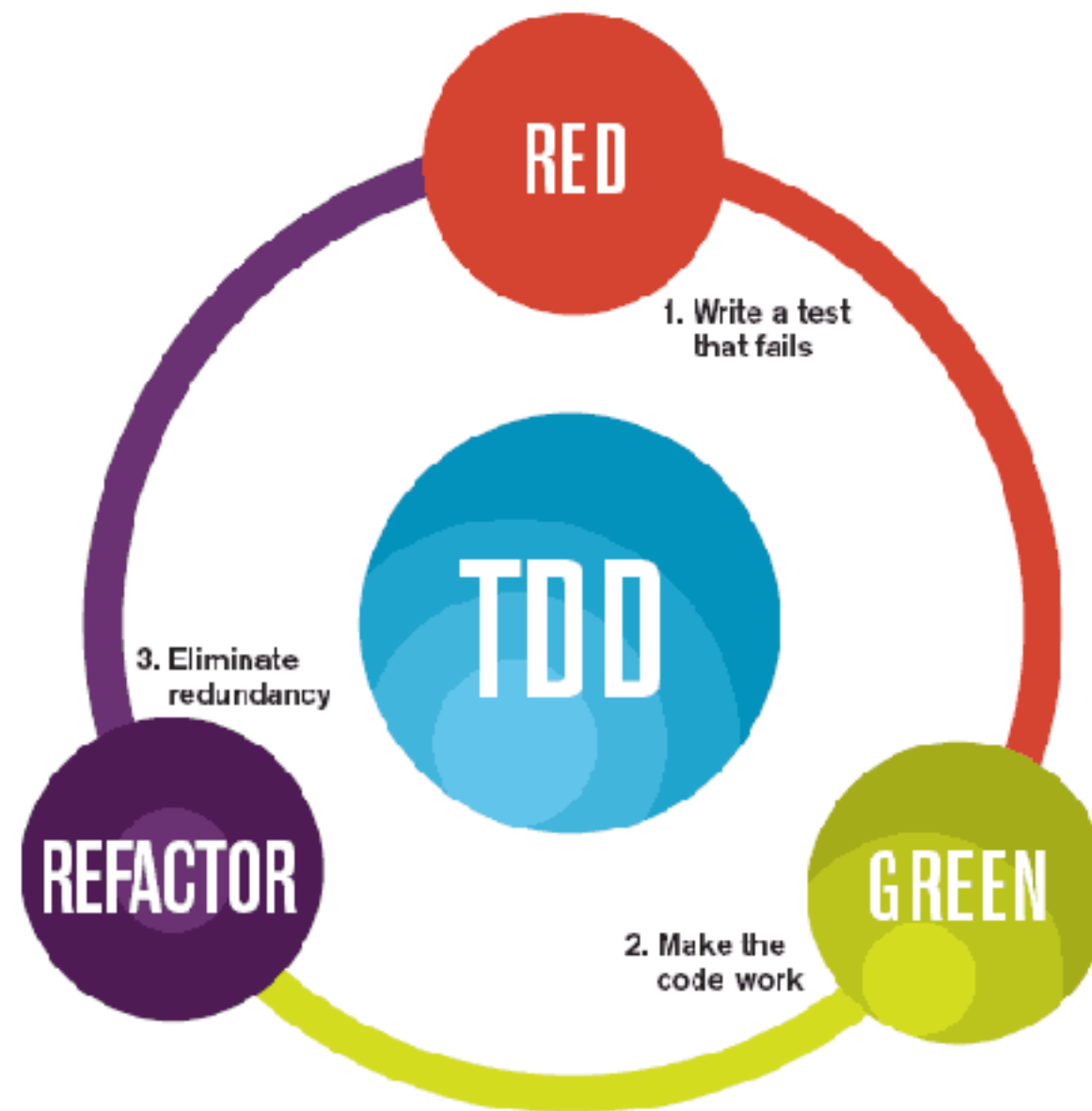
if __name__ == "__main__":
    mass = get_positive_integer("Enter the mass value: ")
    acceleration = get_positive_integer("Enter the acceleration: ")

    print("The Force is", calculate_force(mass, acceleration))
```

<https://www.freecodecamp.org/news/best-practices-for-refactoring-code/>



Trust in your code ?



The mantra of Test-Driven Development (TDD) is "red, green, refactor."



Write your first test case

unittest



pytest

<https://docs.pytest.org/en/stable/>



Run your test

\$pytest

```
from after import calculate_force

def test_with_valid_inputs_calculate_force():
    # Test with valid inputs
    result = calculate_force(2, 3)
    assert result == 6, f"Expected 6, but got {result}"

    # Test with zero
    result = calculate_force(0, 3)
    assert result == 0, f"Expected 0, but got {result}"

    # Test with negative numbers
    result = calculate_force(-2, -3)
    assert result == 6, f"Expected 6, but got {result}"

    # Test with mixed positive and negative numbers
    result = calculate_force(2, -3)
    assert result == -6, f"Expected -6, but got {result}"
```



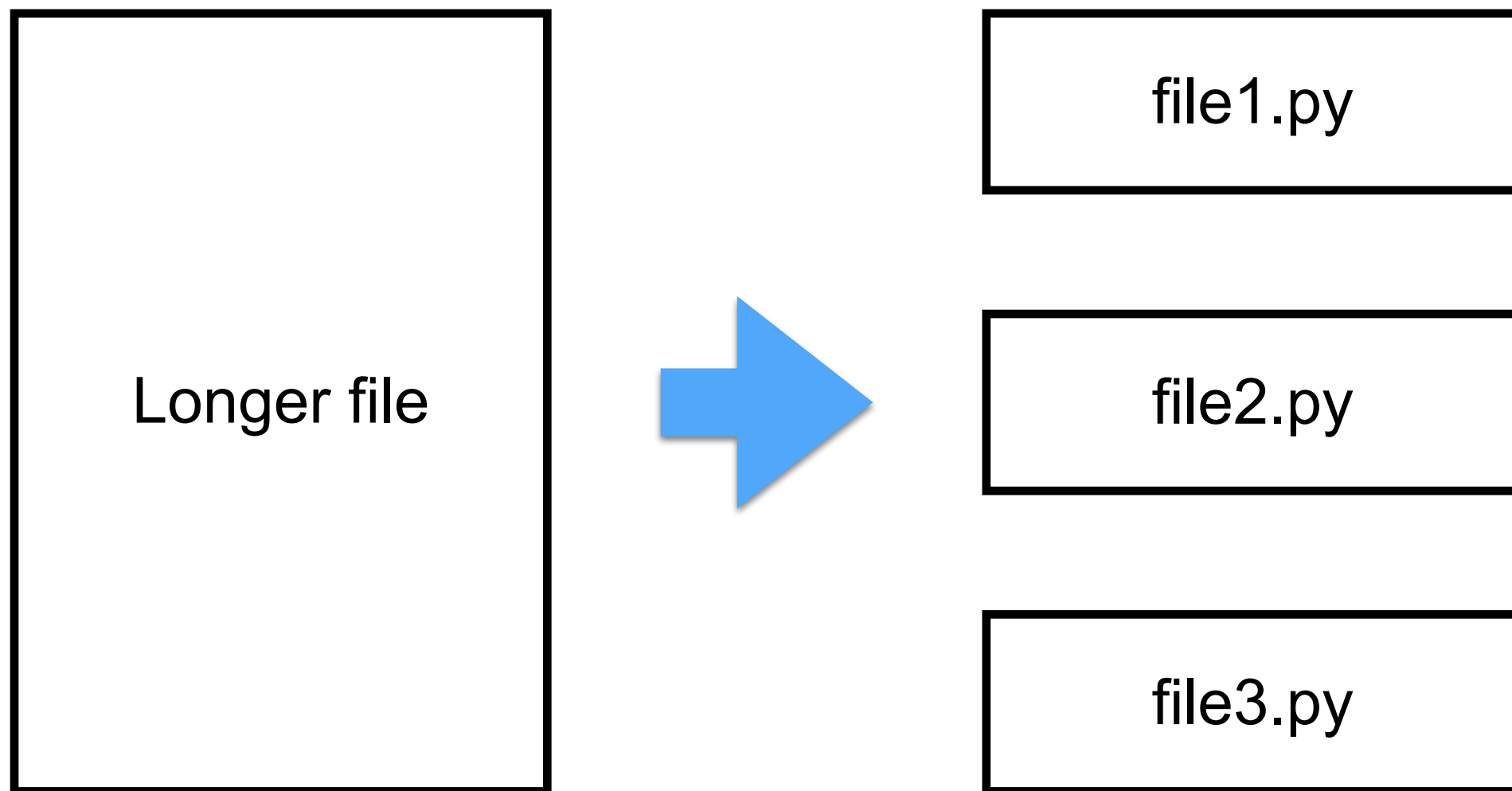
Modules in Python

<https://docs.python.org/3/tutorial/modules.html>



Modules

Split long script to small pieces of files



Example of Modules

Split long script to small pieces of files

hello.py

```
def say_hi(message: str) -> str:  
    return f"Hello, {message}!"
```

use

main.py

```
import hello  
  
if __name__ == "__main__":  
    result = hello.say_hi("Somkiat")  
    print(f'Result: {result}')
```



Modules vs Packages

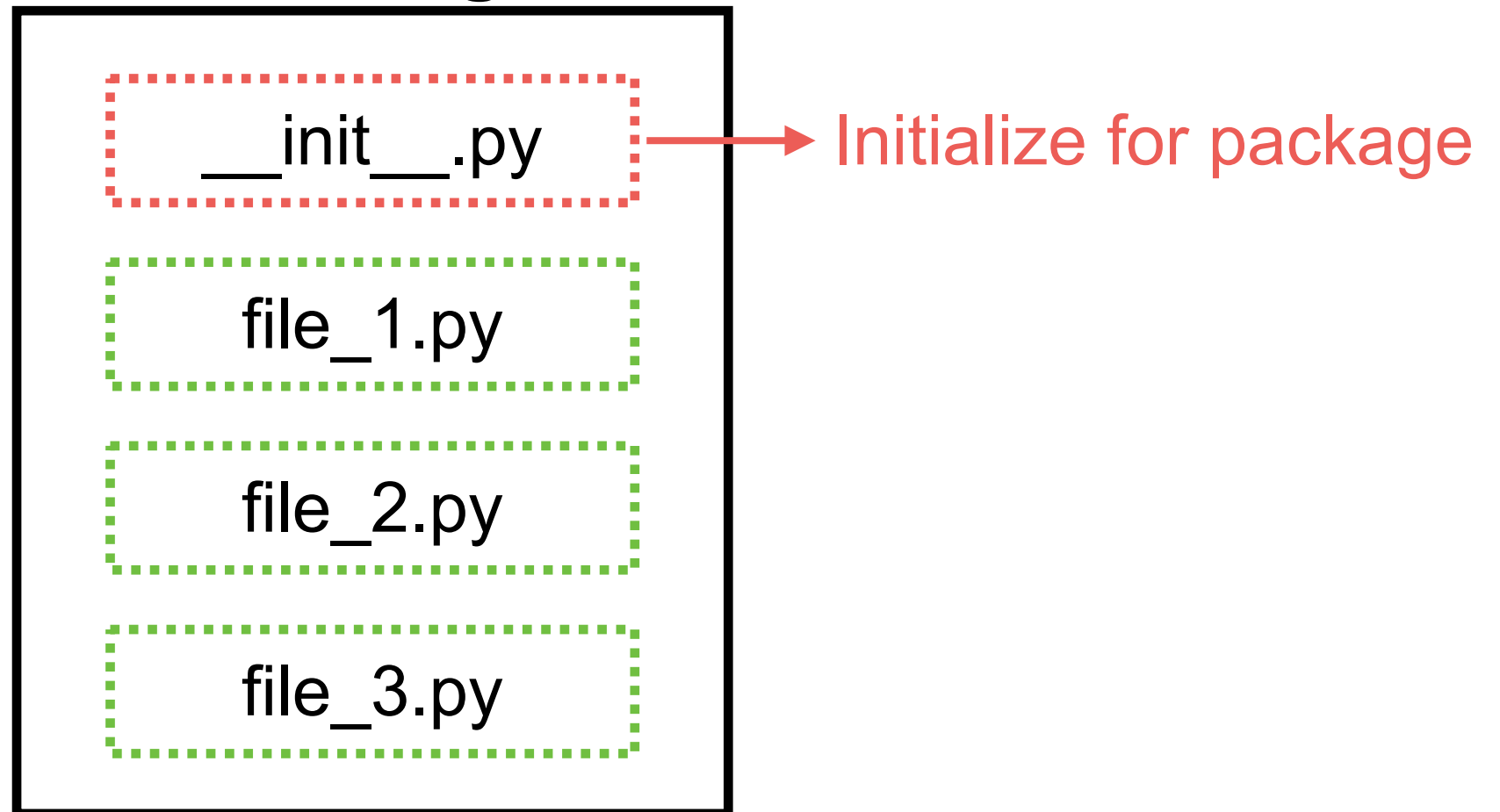
<https://docs.python.org/3/tutorial/modules.html>



Packages

Collection of modules that provide a set of related functionalities

Package



<https://docs.python.org/3/tutorial/modules.html#packages>



Demo package

demo_package

demo.py

```
def say_hi():  
    return "Hello from demo_package!"
```

__init__.py

```
__all__ = ["demo"]
```

main.py

```
import demo_package  
  
if __name__ == "__main__":  
    print(dir(demo_package))  
    print(demo_package.__file__)
```

```
from demo_package import *  
  
if __name__ == "__main__":  
    res = demo.say_hi()  
    print(res)
```

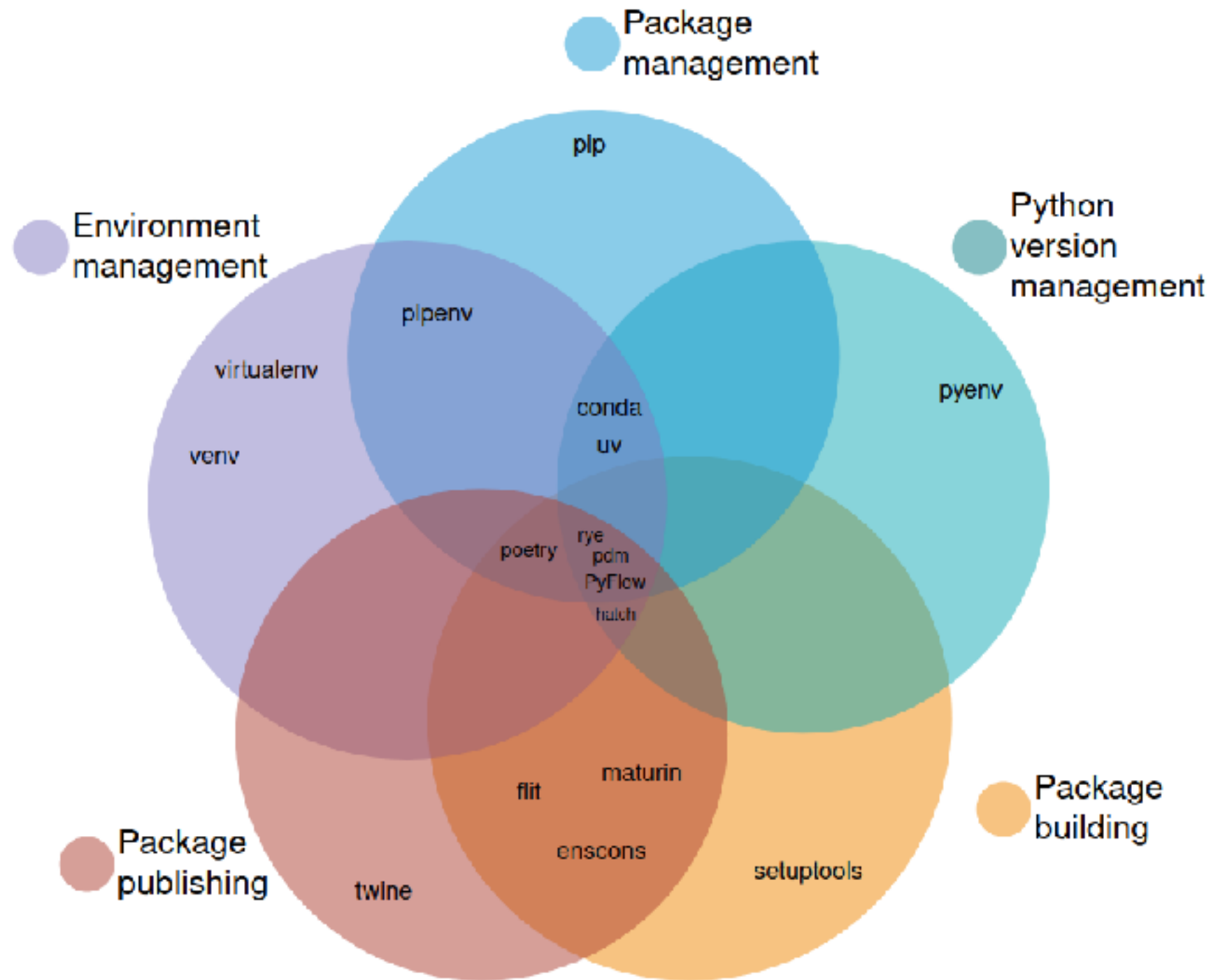
<https://sentry.io/answers/what-is-init-py-for-in-python/>



Environment Management

Package Management

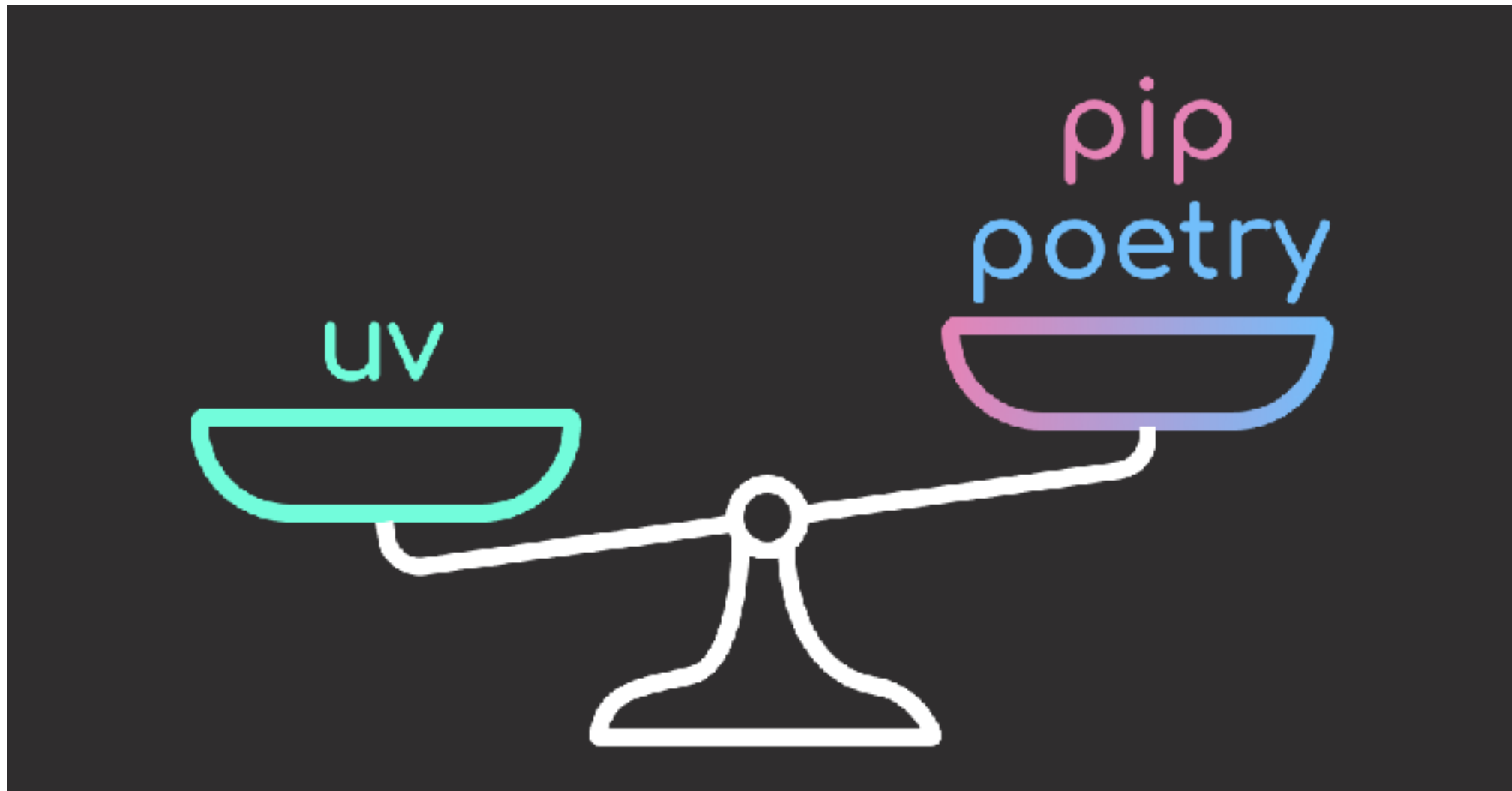




https://alpopkes.com/posts/python/packaging_tools/



Tools ?



UV

astral-sh/uv

An extremely fast Python package and project manager, written in Rust.



 384
Contributors

 2k
Issues

 56k
Stars

 2k
Forks



<https://github.com/astral-sh/uv>



Development tools

Code formatting
Linting

Build tools

Documentation

Testing



Object Oriented Programming

Classes and Object



OOP concepts



Class and Object

Class name

List of properties/attributes (data members)

List of behaviors/methods

Constructor



Single Responsibility Principle



Single Responsibility Principle

Just because you *can* doesn't mean you *should*.



Workshop

OOP with Harry Potter

<https://codingdojo.org/kata/Potter/>



Design class and relationship

Book

Order

OrderItem

Book store

PriceCalculator

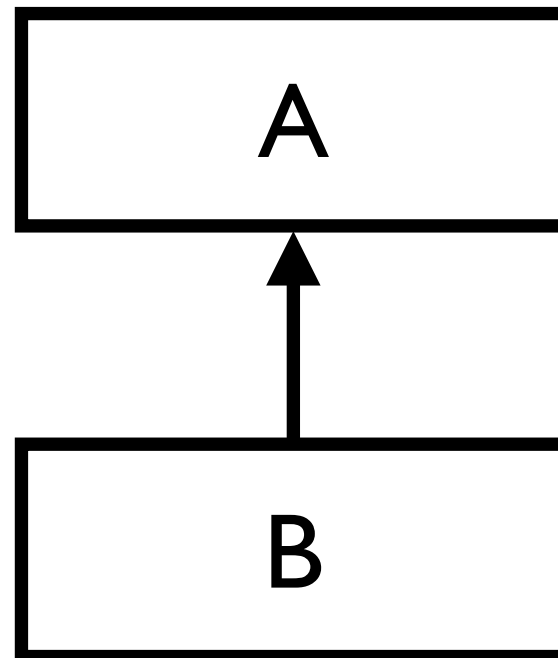
DiscountCalculator



Inheritance

Ability to derive a new class from existing class

superclass / base class

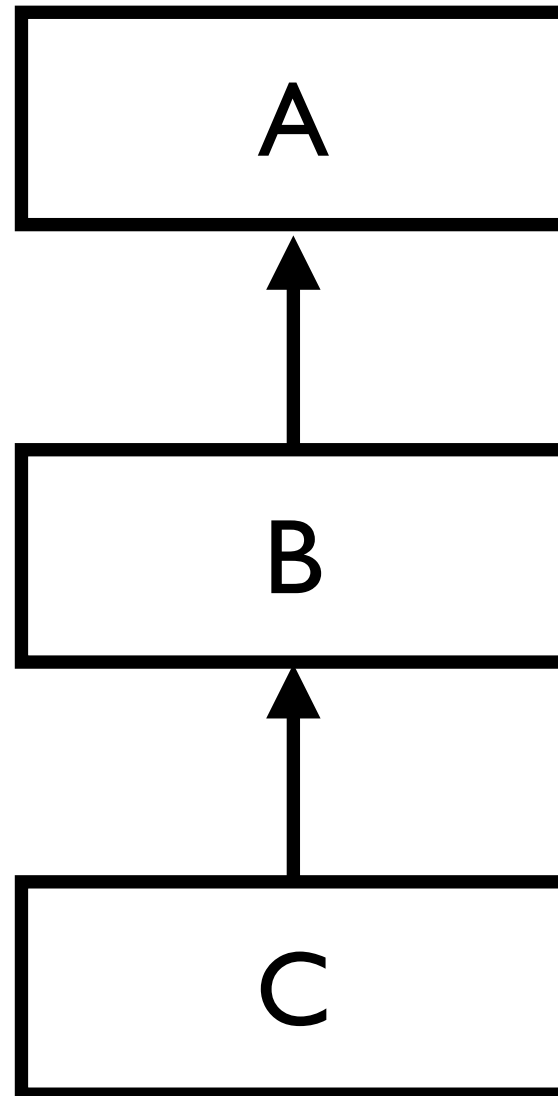


subclass / derived class

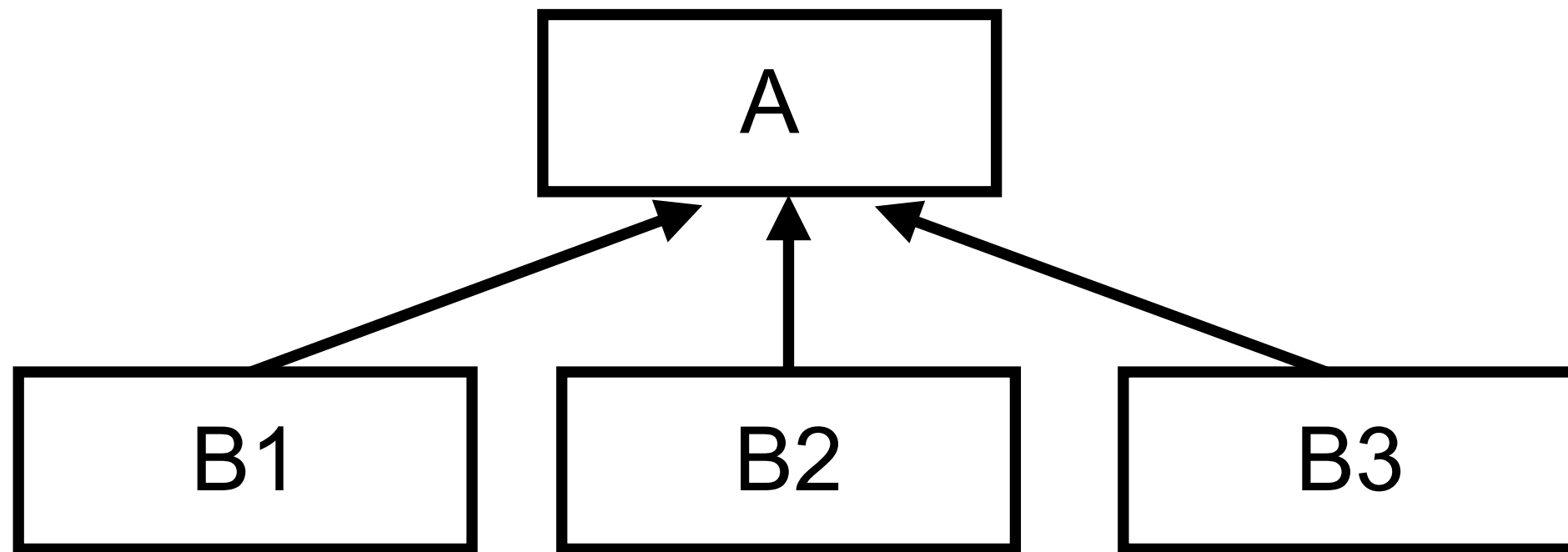
Additional properties and behaviors



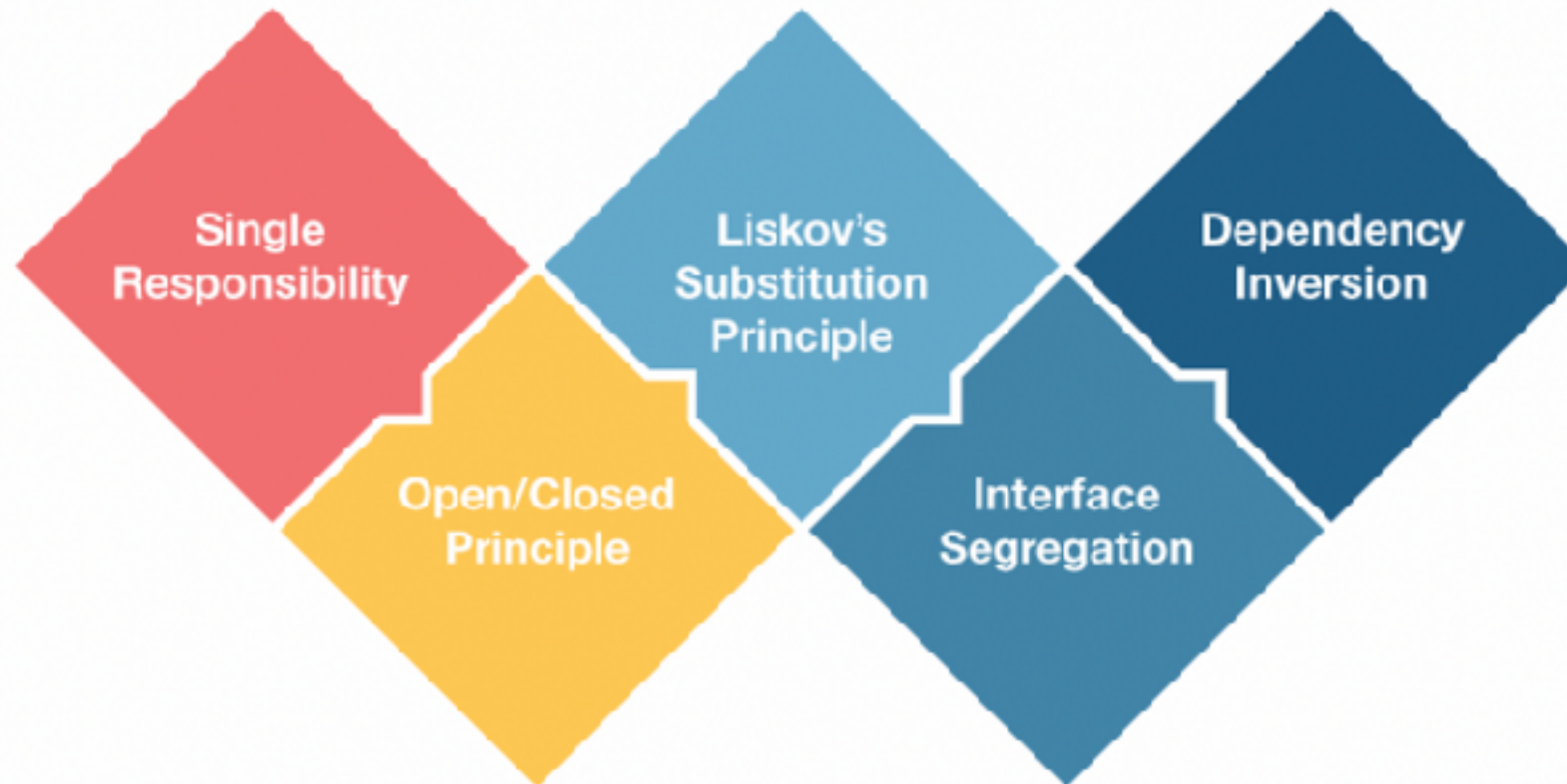
Multilevel Inheritance



Hierarchical Inheritance



S.O.L.I.D.



Single Responsibility Principle



Single Responsibility Principle

Just because you *can* doesn't mean you *should*.



Open-Closed Principle

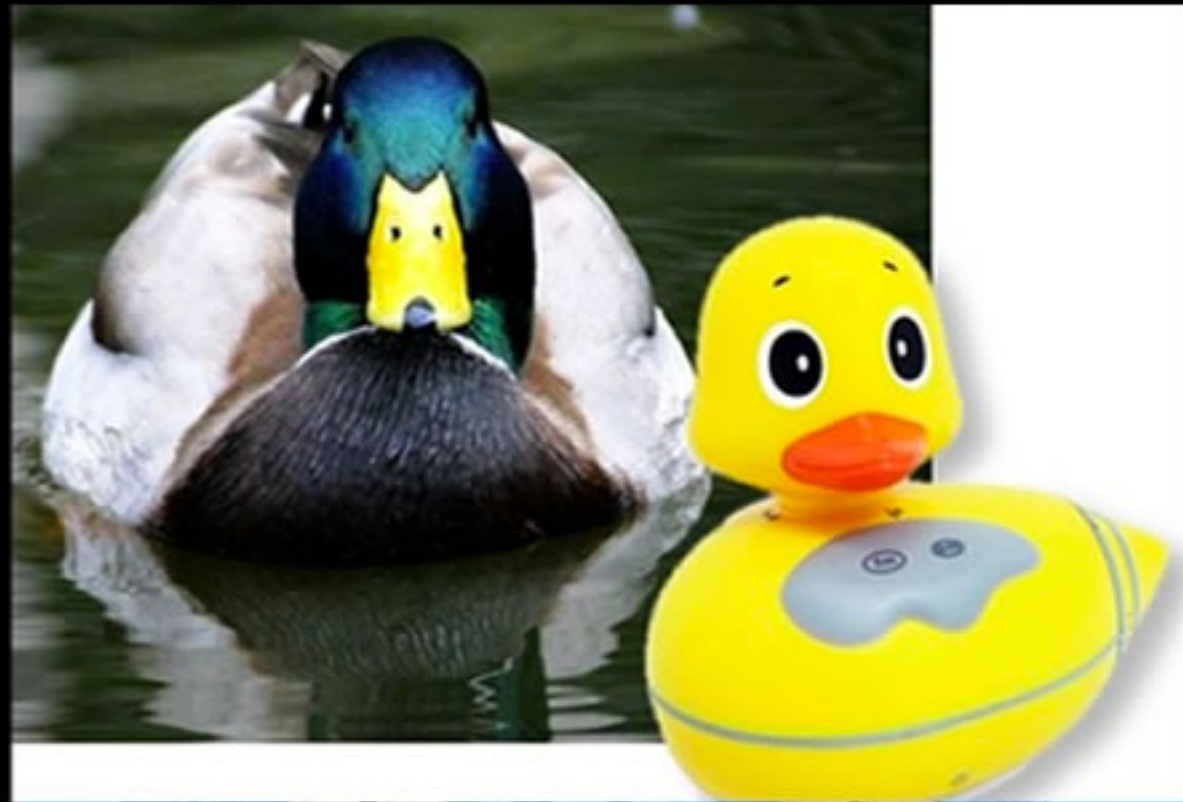


Open-Closed Principle

Open-chest surgery isn't needed when putting on a coat.



Liskov Substitution Principle



Liskov Substitution Principle

If it looks like a duck and quacks like a duck but needs batteries, you probably have the wrong abstraction.



Interface Segregation Principle



Interface Segregation Principle

You want me to plug this in *where*?



Dependency Inversion Principle



Dependency Inversion Principle

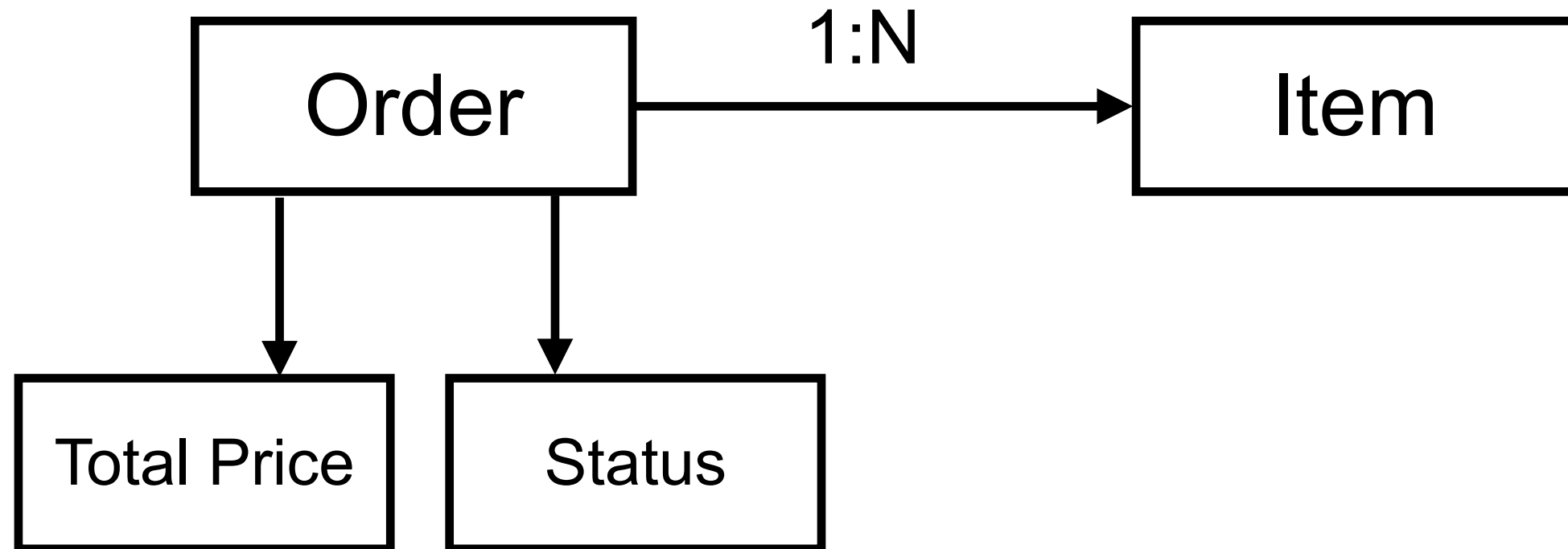
Would you solder a lamp directly to the electrical wiring in a wall?



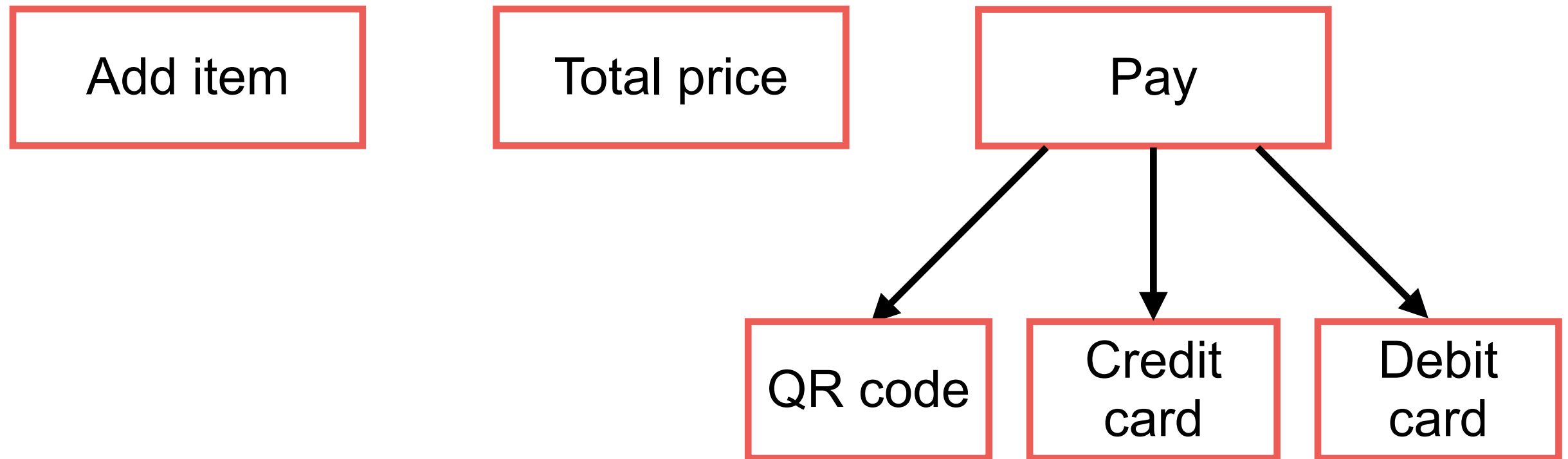
Workshop with Order



Design :: Data modeling



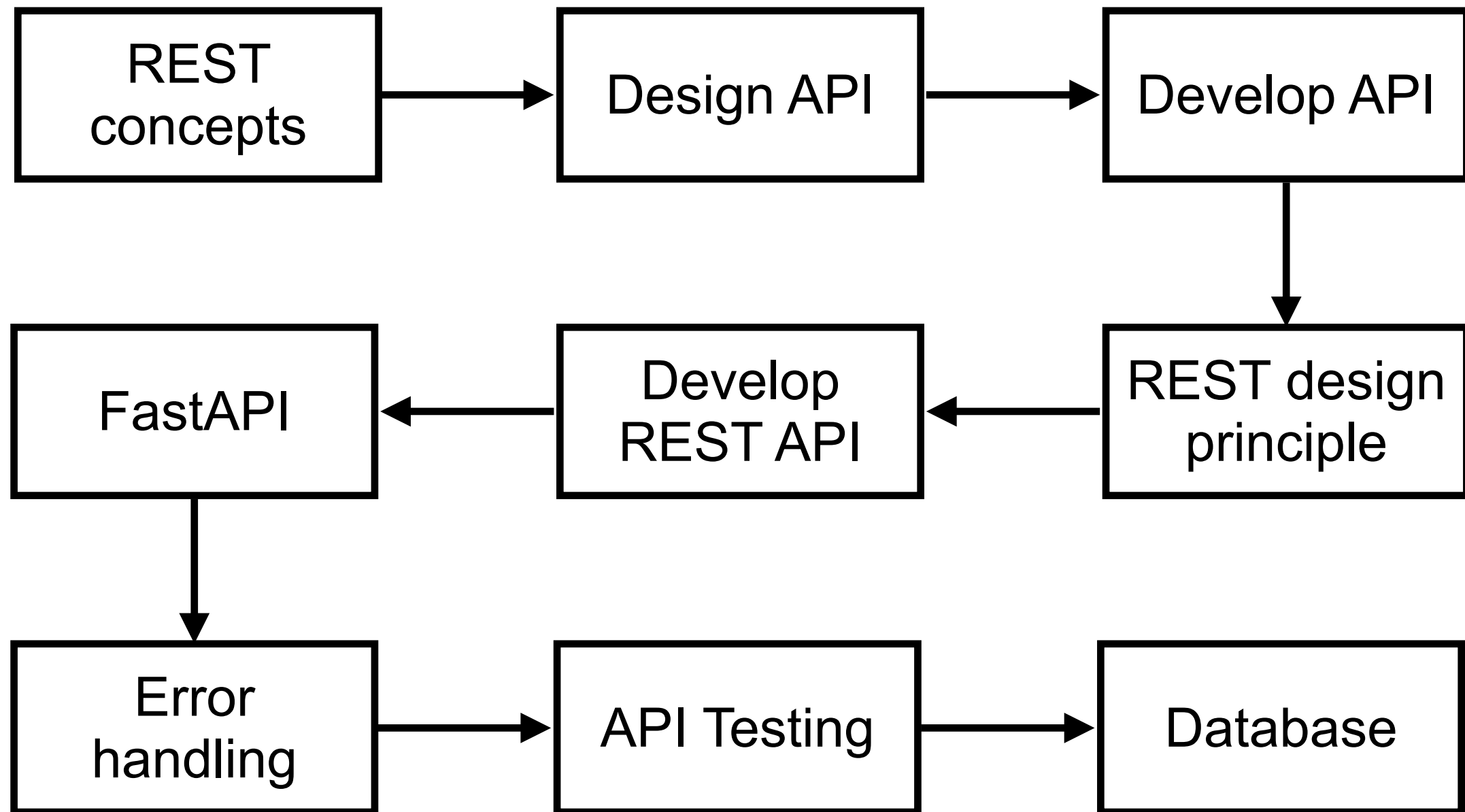
Design :: Operation



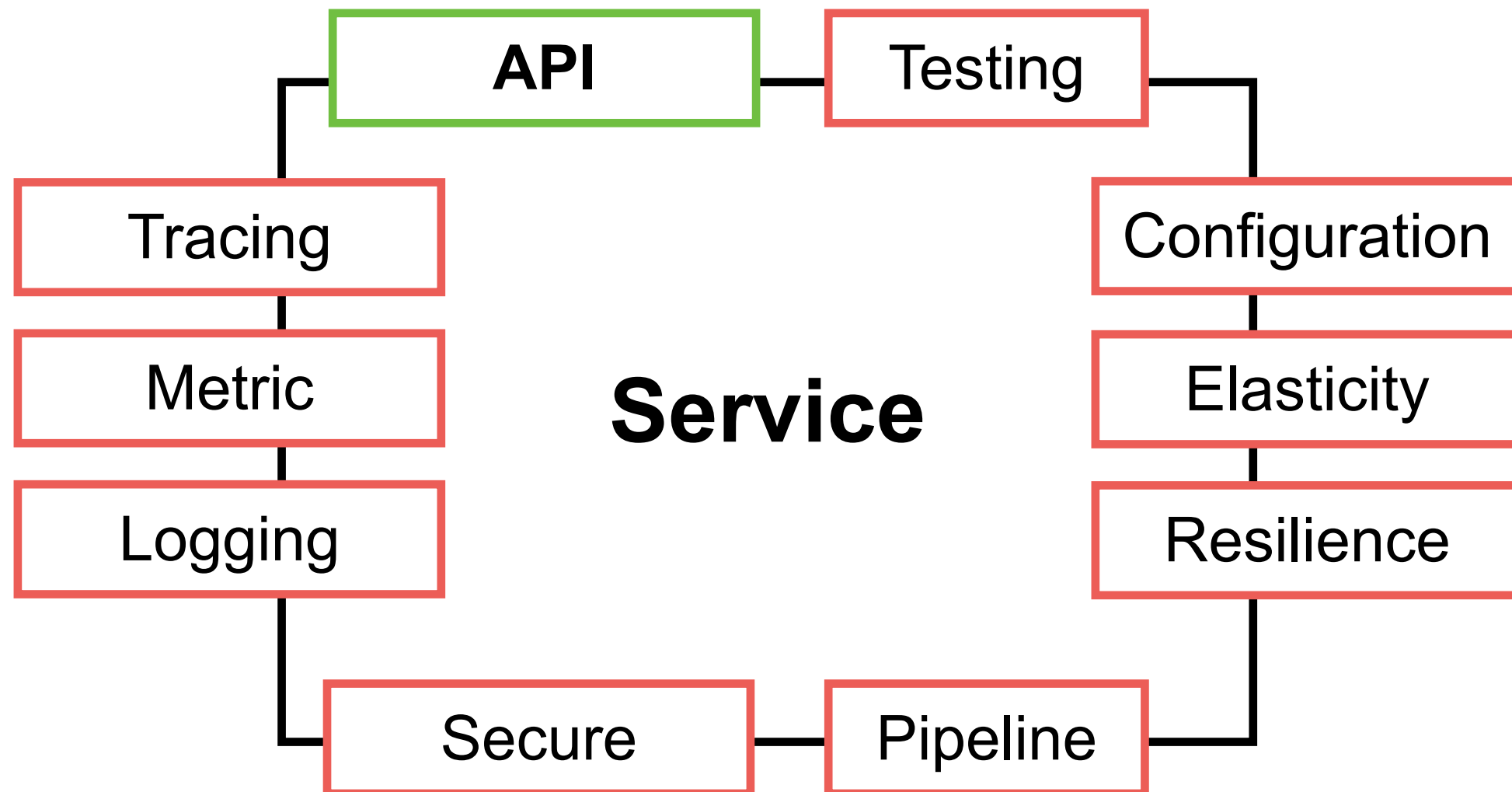
Part 2



Learning Path 2



Properties of Service



APIs

Application Programming Interface

REST API

gRPC

Messaging



REST

REpresentation **S**tate **T**ransfer

The style of **software architecture** behind RESTful services

Defined in 2000 by Roy Fielding

https://www.ics.uci.edu/~fielding/pubs/dissertation/rest_arch_style.htm



Goals

Scalability

Generality of interfaces

Independent deployment of components



REST properties

Performance

Scalability

Simplicity

Modification

Visibility

Portability

Reliability



REST constraints

Client-server

Stateless

Cacheable

Layer system

Uniform
interface

Code on
demand

eg. JavaScript

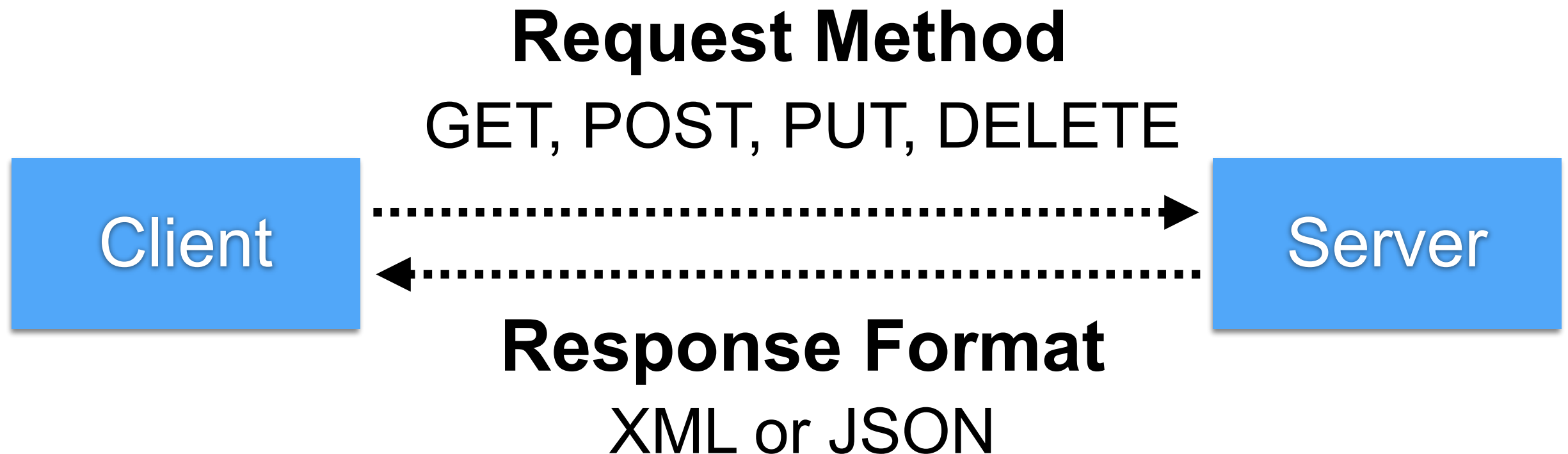


RESTful service

Implementation from REST



REST Request & Response



HTTP Request methods

Method	Meaning
GET	Read data
POST	Create/Insert new data
PUT/PATCH	Update data or insert if a new id
DELETE	Delete data

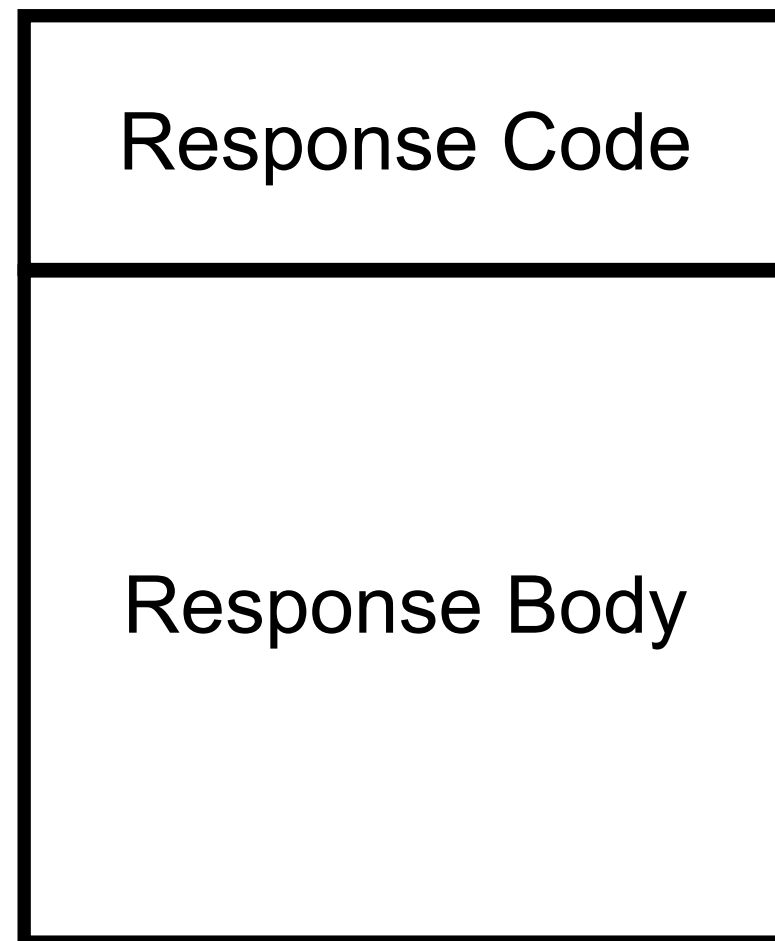


HTTP status codes

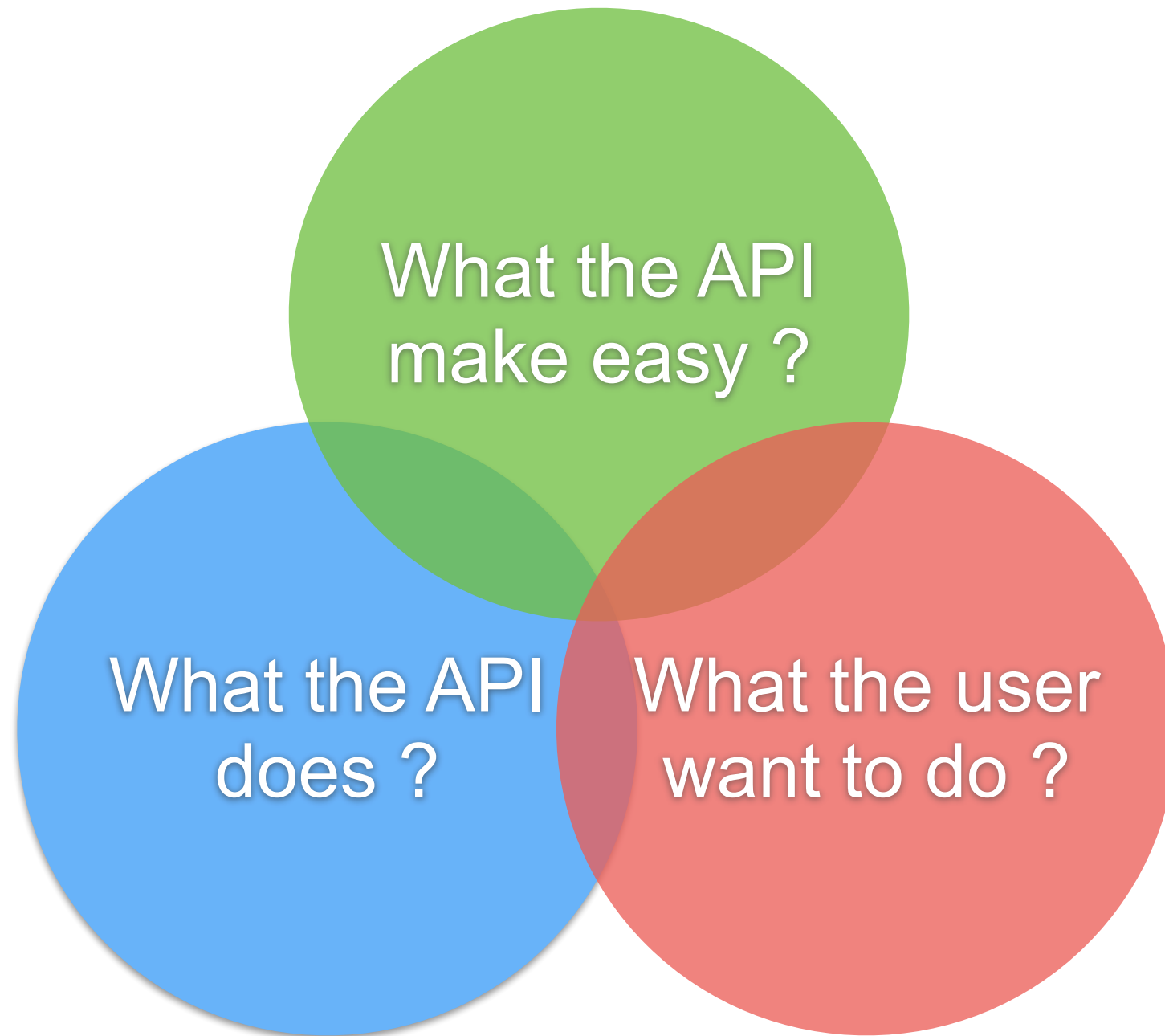
Code	Description	Example
1xx	Informational responses	100 = Continue 101 Switching protocols
2xx	Success	200 = OK 201 = Created
3xx	Redirection	300 = Multiple choices 301 = Moved permanently
4xx	Client errors	400 = Bad request 403 = Forbidden 404 = Not found
5xx	Server errors	500 = Internal server error 502 = Bad gateway 503 = Service unavailable



Response format ?



Good APIs ?



Principles of API Design



Principles of API Design

Consistency

Statelessness

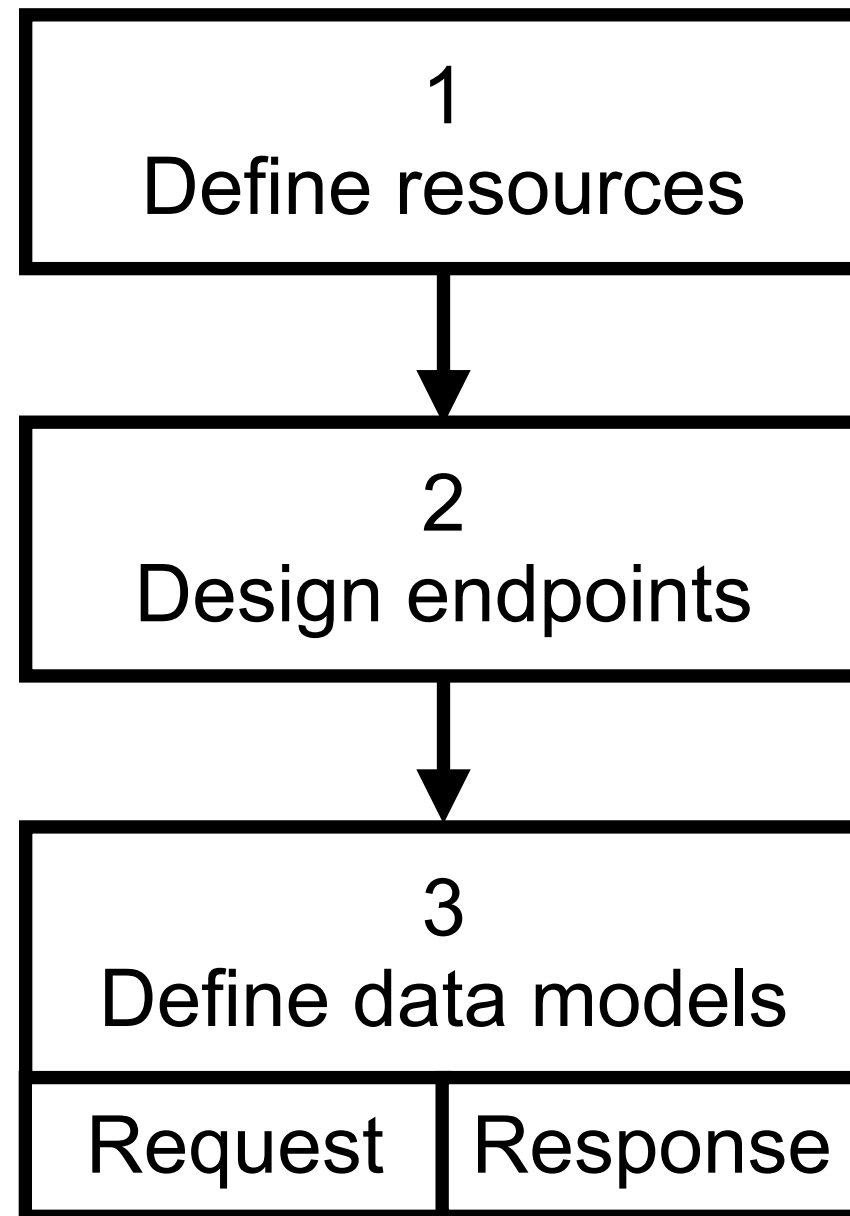
Resource-oriented design

Use standard
HTTP methods

Versioning



Steps to Design APIs



More !!

Authentication
Authorization

Pagination and
filtering

Rate Limiting

Pagination and
filtering

Error handling

Documentation

Testing

Monitoring and
Observability



Implementation

Coding

Testing

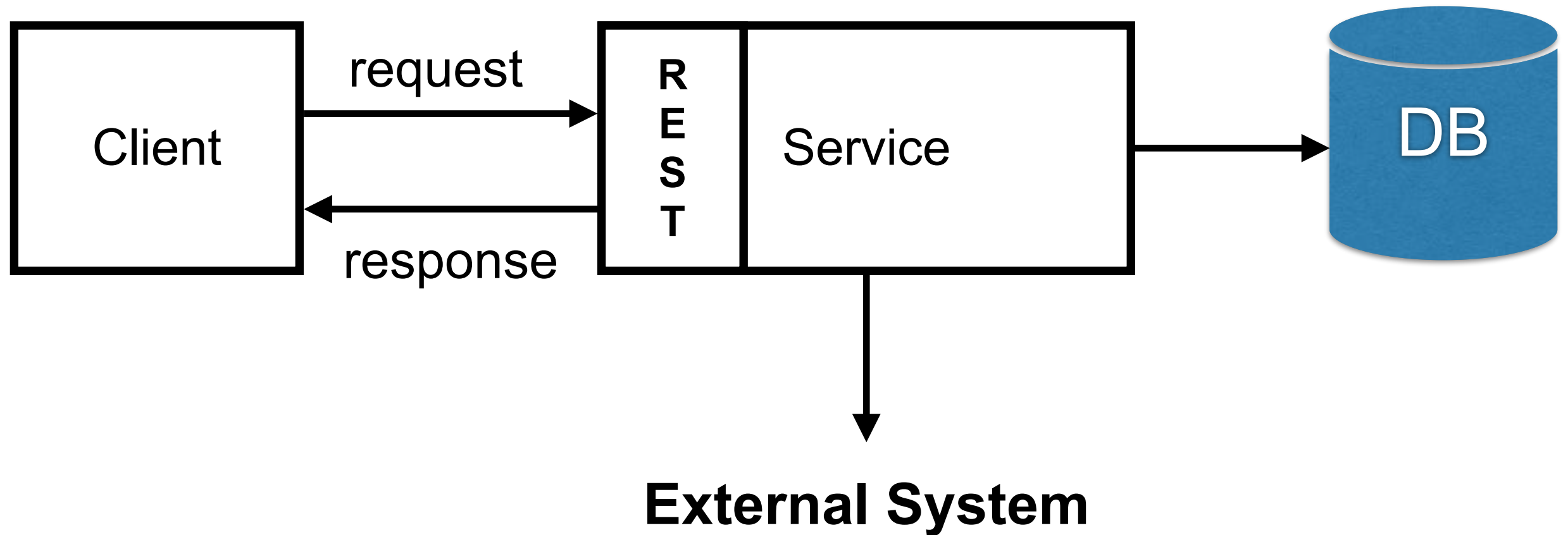
Observability

Performance

Security



Structure of Service



Design RESTFul APIs



Users

Method	Path	Description
GET	/users	Get all users
GET	/users/{id}	Get user by id
POST	/users	Create new user
PUT	/users/{id}	Update user
DELETE	/users/{id}	Delete user by id



API Documentation

pet Everything about your Pets		
POST	/pet/{petId}/uploadImage	uploads an image
POST	/pet	Add a new pet to the store
PUT	/pet	Update an existing pet
GET	/pet/findByStatus	Finds Pets by status
GET	/pet/{petId}	Find pet by ID
POST	/pet/{petId}	Updates a pet in the store with form data
DELETE	/pet/{petId}	Deletes a pet

<https://swagger.io/>



Develop RESTFul APIs



RESTFul API with Python



django



FastAPI



FastAPI framework, high performance, easy to learn, fast to code, ready for production

Test passing coverage 100% pypi package v0.115.12 python 3.8 | 3.9 | 3.10 | 3.11 | 3.12 | 3.13

<https://fastapi.tiangolo.com/>



FastAPI

Fast

Fewer bugs

Easy

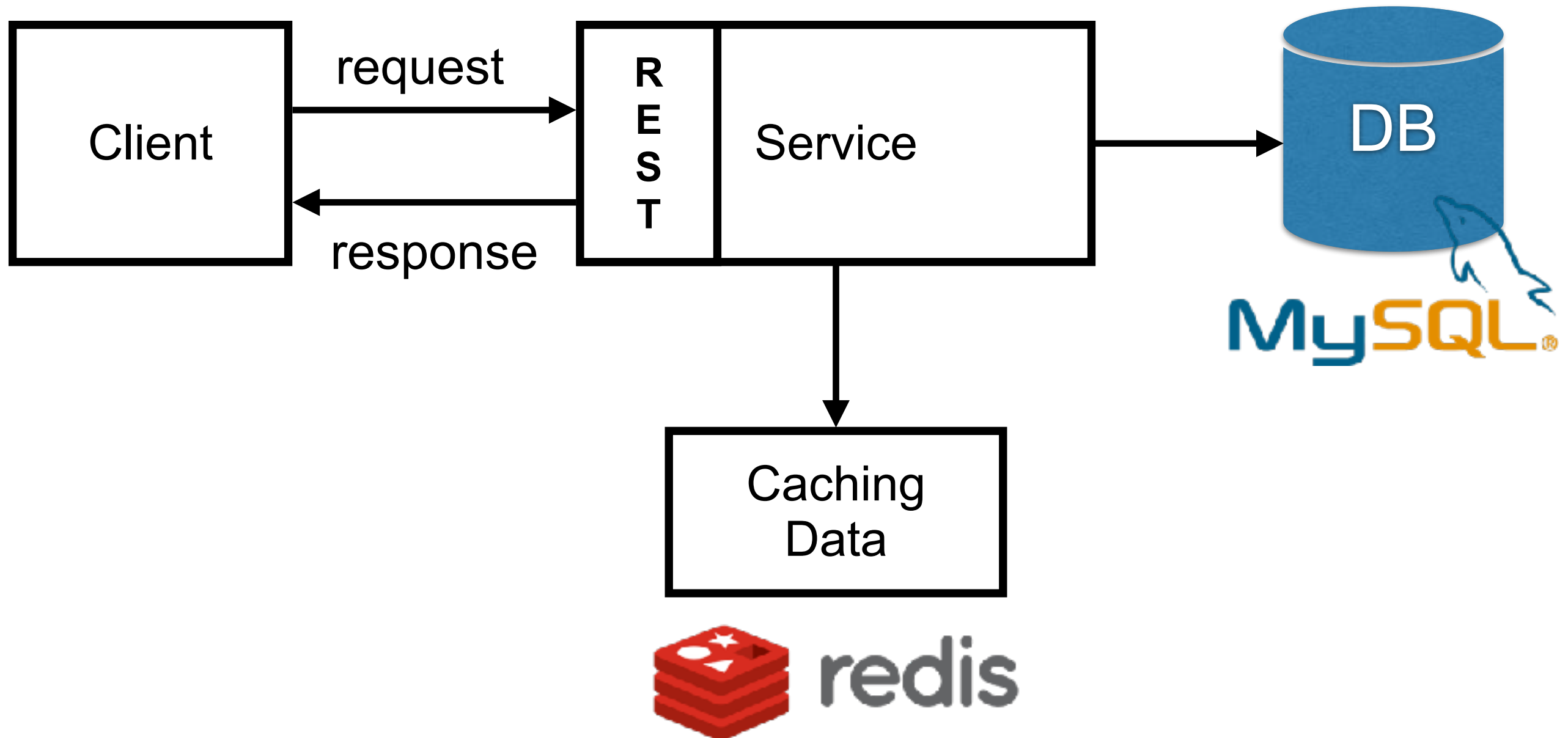
Production
ready

OpenAPI

JSON
Schema



Workshop



Working with Database



Database Ranking

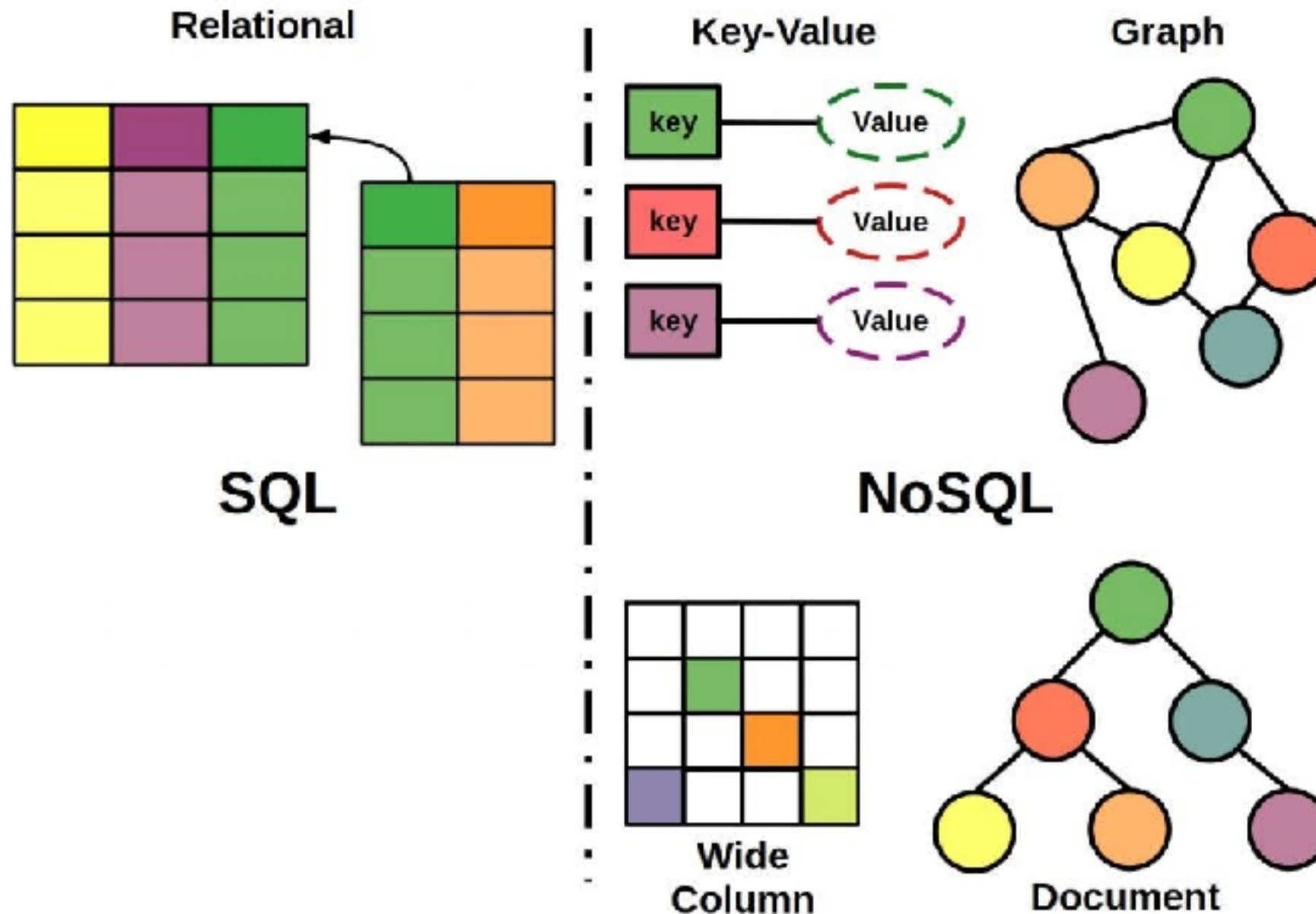
424 systems in ranking, June 2025

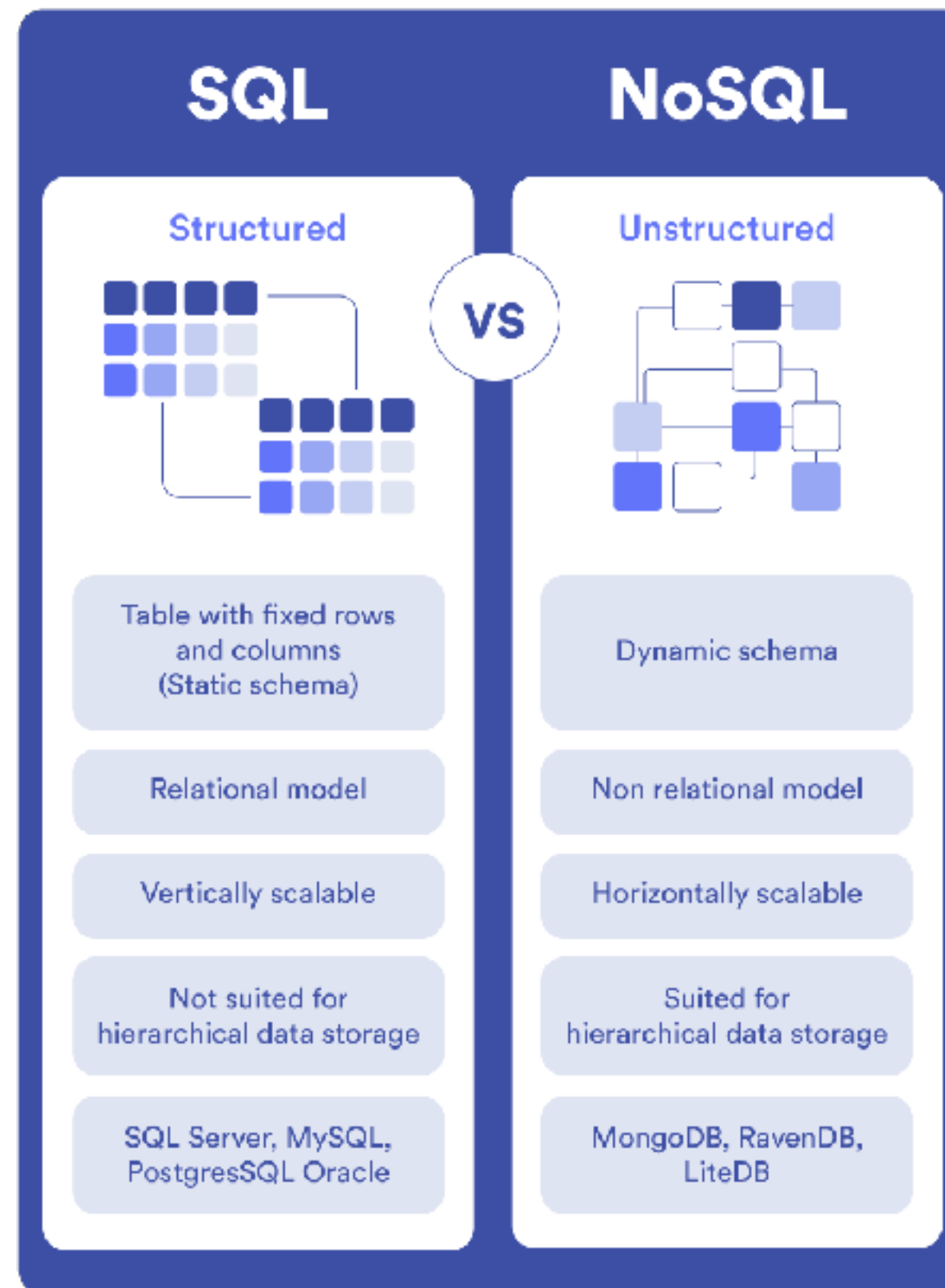
Rank			DBMS	Database Model	Score		
Jun 2025	May 2025	Jun 2024			Jun 2025	May 2025	Jun 2024
1.	1.	1.	Oracle	Relational, Multi-model ⓘ	1230.38	+3.82	-13.70
2.	2.	2.	MySQL	Relational, Multi-model ⓘ	953.57	-11.41	-107.77
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model ⓘ	776.75	+1.86	-44.81
4.	4.	4.	PostgreSQL +	Relational, Multi-model ⓘ	680.65	+6.34	+44.41
5.	5.	5.	MongoDB +	Document, Multi-model ⓘ	402.85	+0.33	-18.23
6.	6.	↑ 8.	Snowflake	Relational	174.49	+2.48	+44.13
7.	7.	↓ 6.	Redis	Key-value, Multi-model ⓘ	151.72	-0.47	-4.22
8.	8.	↑ 9.	IBM Db2	Relational, Multi-model ⓘ	125.13	-1.27	-0.77
9.	9.	↓ 7.	Elasticsearch	Multi-model ⓘ	121.28	-2.53	-11.55
10.	10.	10.	SQLite	Relational	117.03	-0.74	+5.63
11.	11.	↑ 12.	Apache Cassandra	Wide column, Multi-model ⓘ	108.27	+0.22	+9.44
12.	12.	↑ 15.	Databricks	Multi-model ⓘ	104.67	+2.02	+23.59
13.	↑ 14.	13.	MariaDB	Relational, Multi-model ⓘ	94.54	+0.93	+3.50
14.	↓ 13.	↓ 11.	Microsoft Access	Relational	88.28	-7.63	-12.88
15.	15.	↑ 17.	Amazon DynamoDB	Multi-model ⓘ	83.34	+4.04	+8.90

<https://db-engines.com/en/ranking>



Working with Database





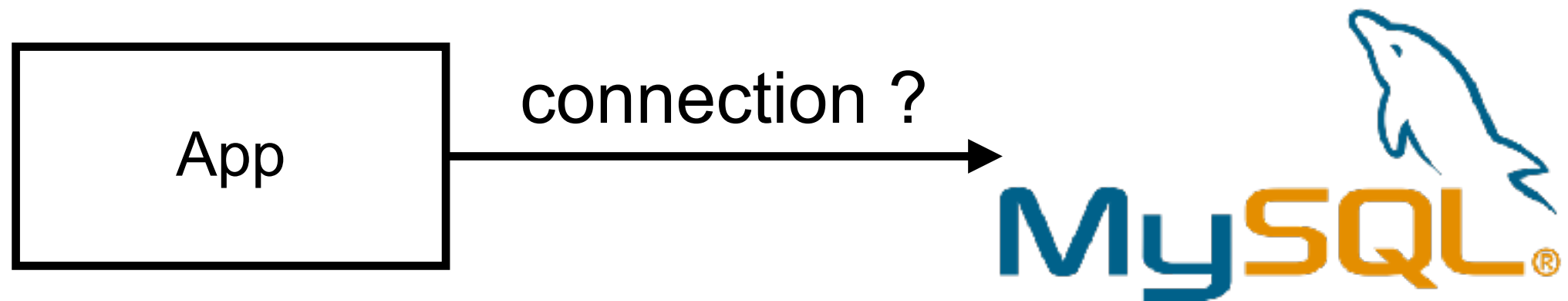
<https://pandorafms.com/blog/nosql-vs-sql-key-differences/>



Working with Database

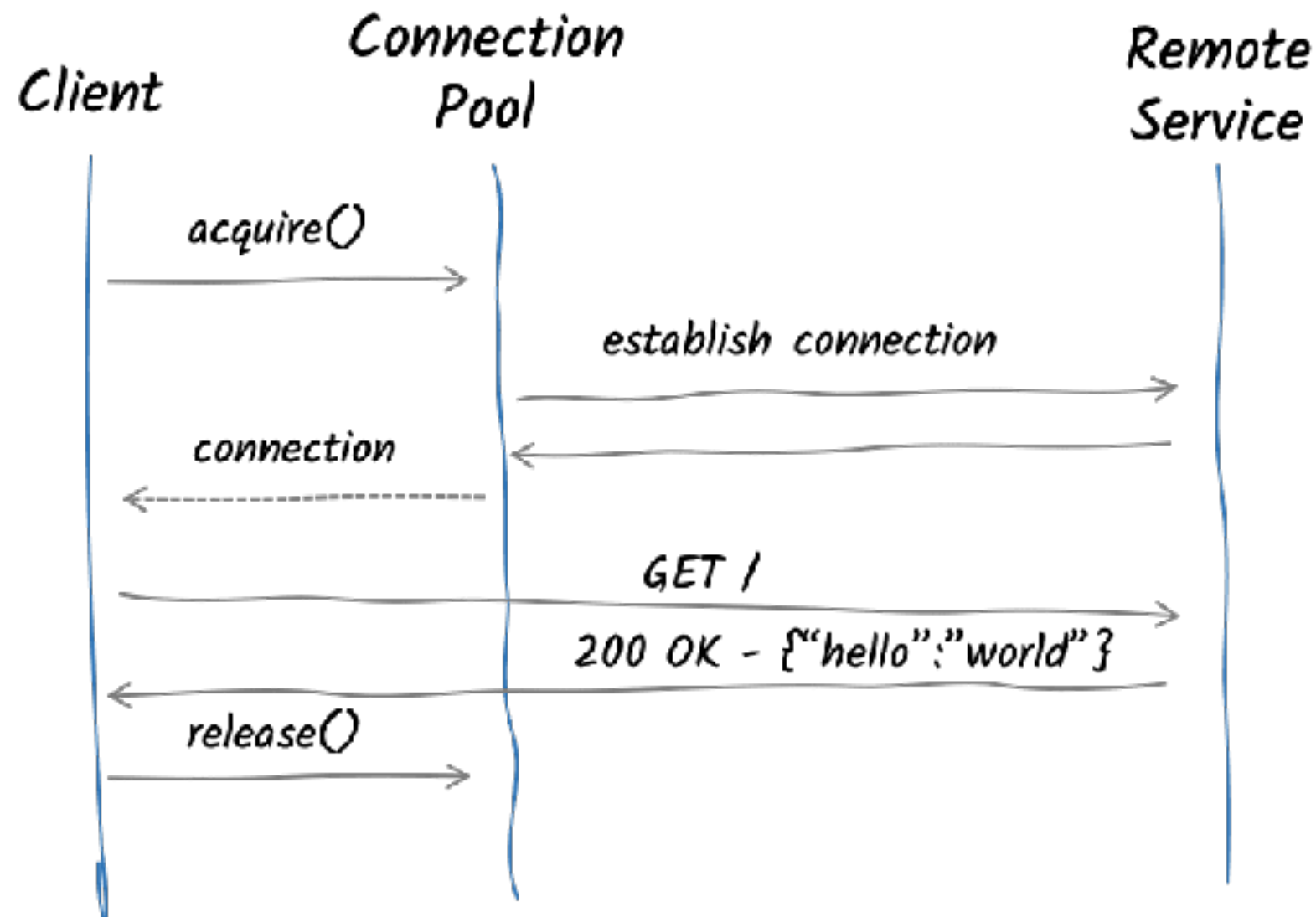


Manage connection ?



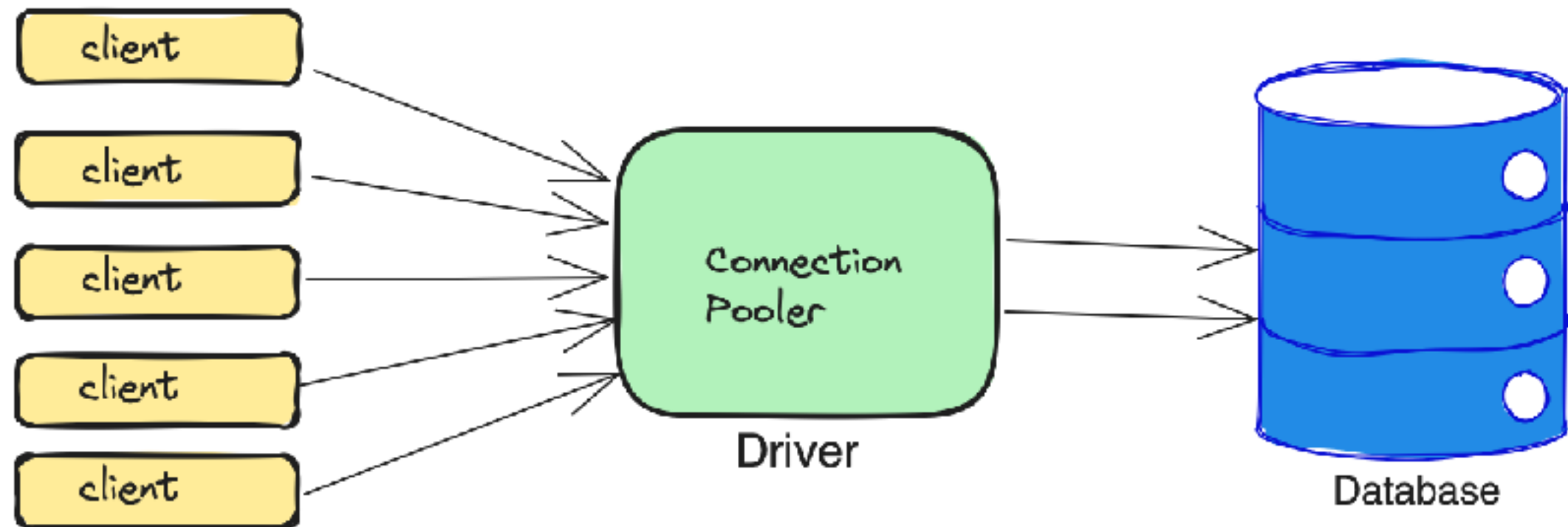
Connection pool

Cache and reuse database connections

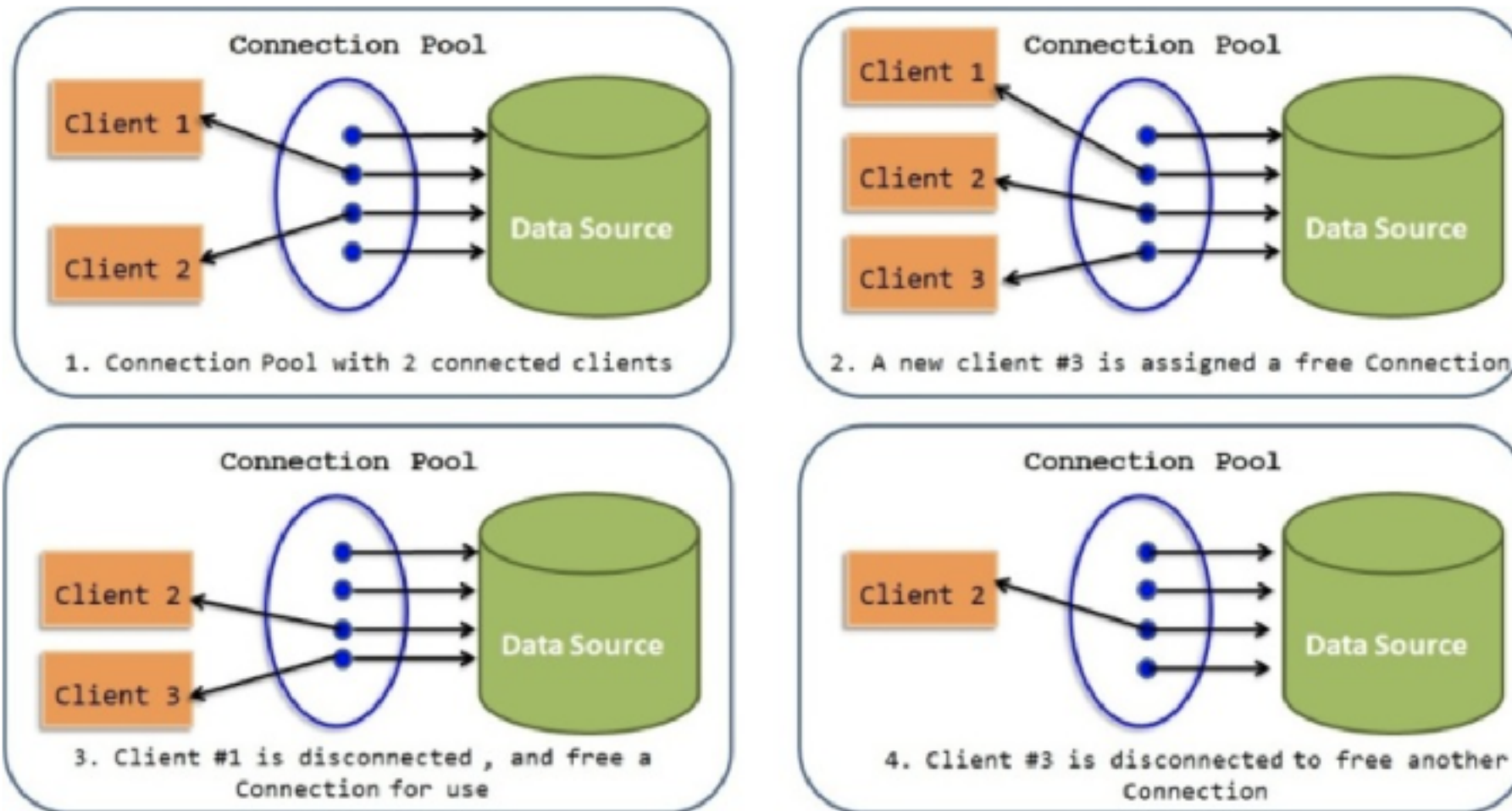


Connection pool

Cache and reuse database connections



Connection pool

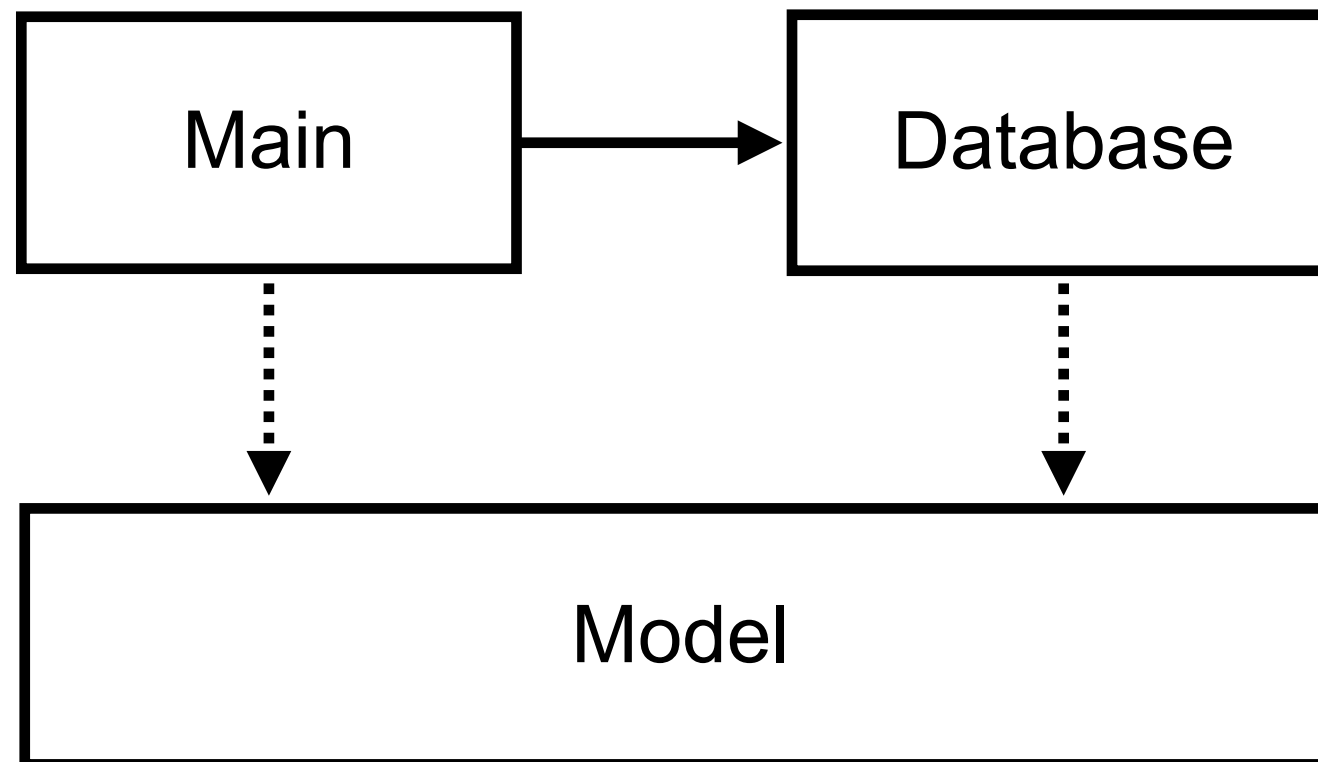


Start workshop



Basic project

CRUD (Create, Read Update, Delete) application



<https://github.com/up1/course-python-workshop/wiki/REST-API-with-FastAPI>

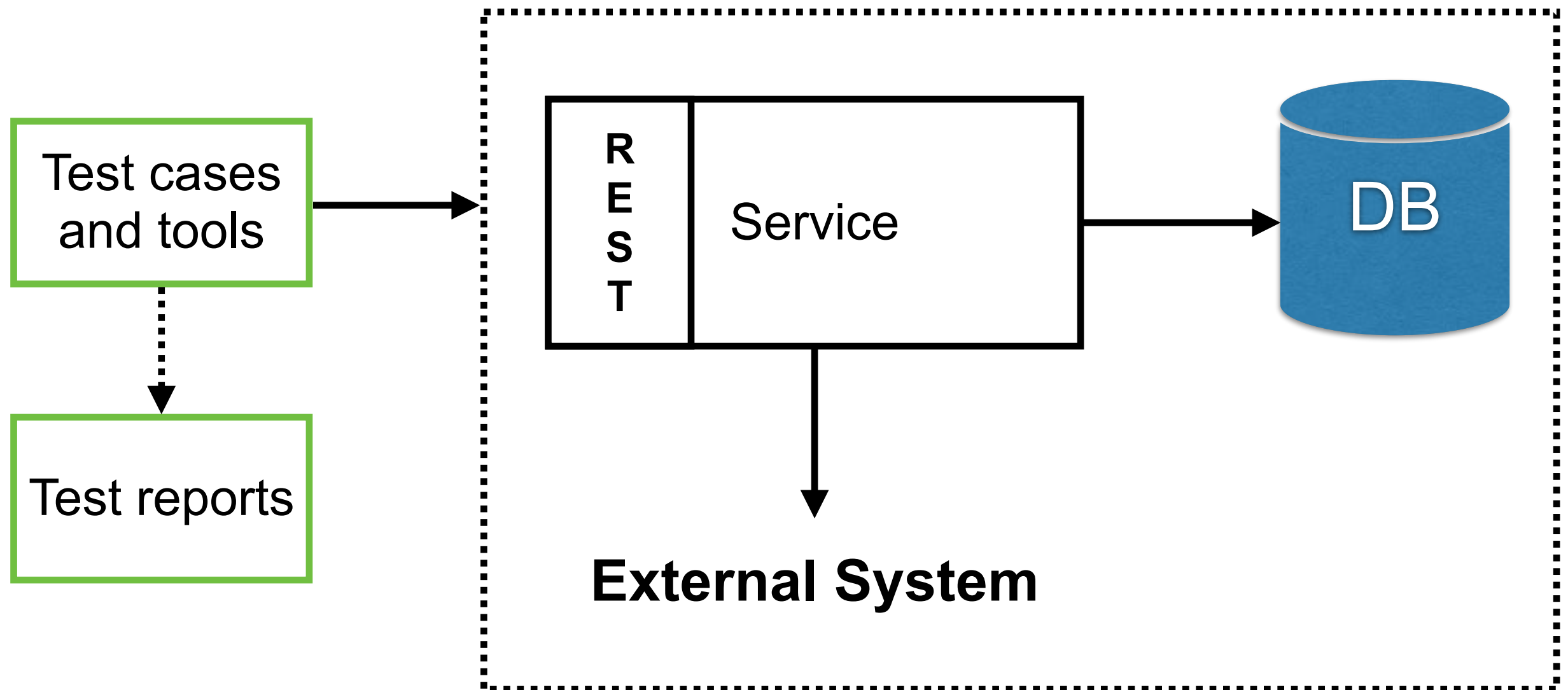


Testing with APIs



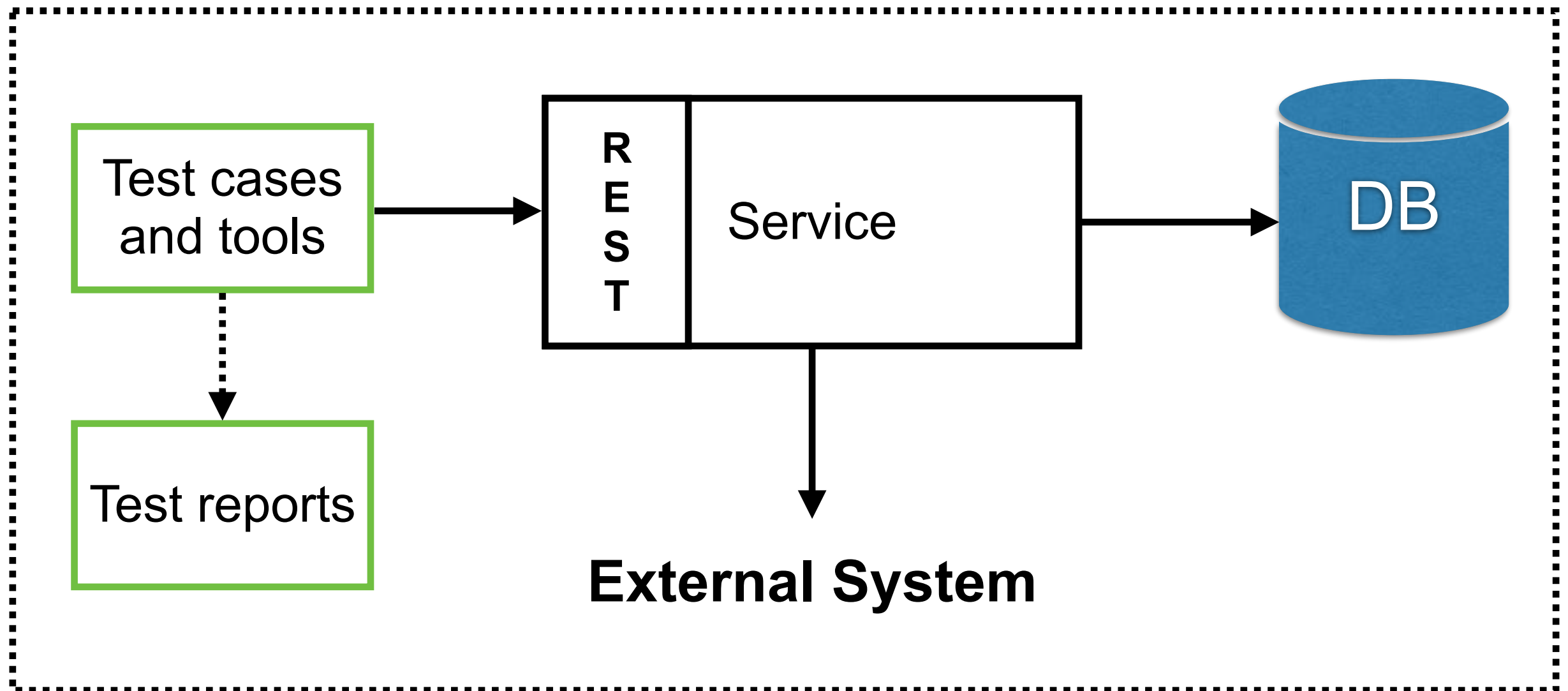
Test strategies

External testing



Test strategies

Internal testing



Testing tools



<https://www.postman.com/downloads/>



Postman CLI with newman

```
$npm install -g newman
```

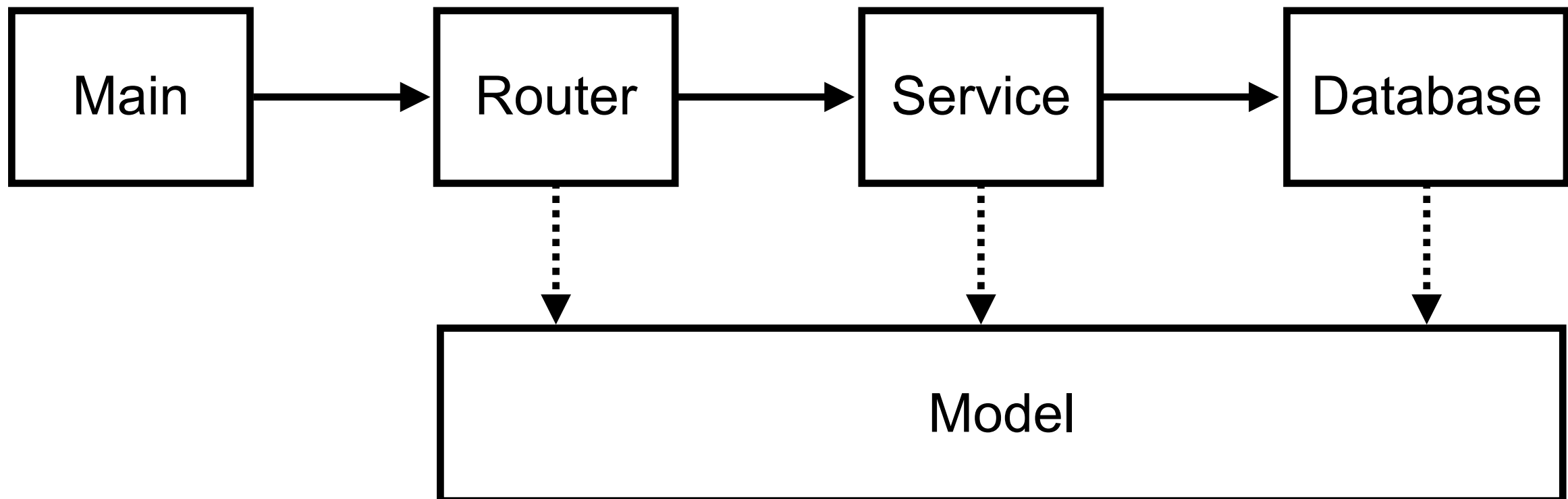


<https://www.npmjs.com/package/newman>



Improve structure

Single Responsibility Principle (SRP) Testability



<https://github.com/up1/course-python-workshop/wiki/REST-API-with-FastAPI>



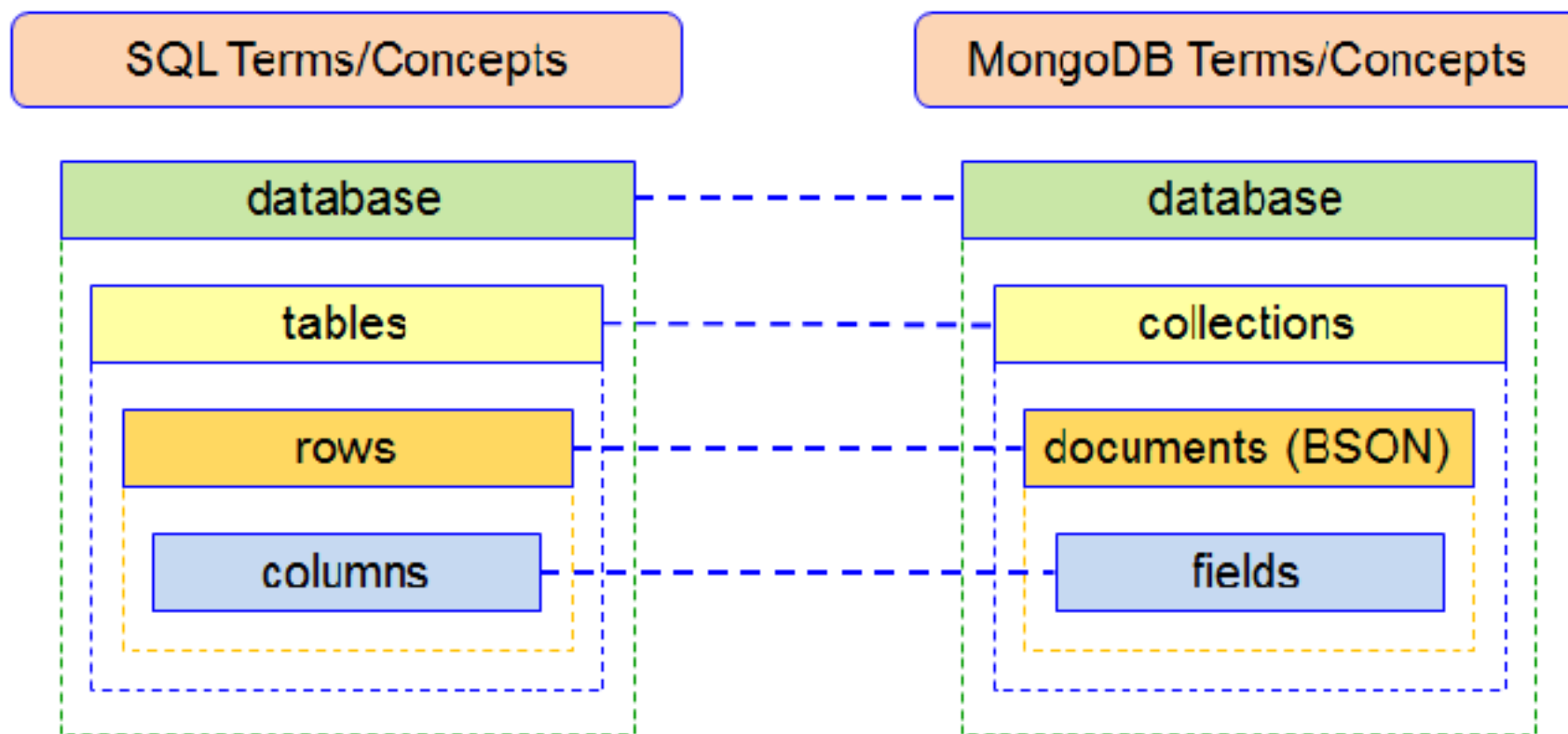
Working with Document database



MongoDB

NoSQL database

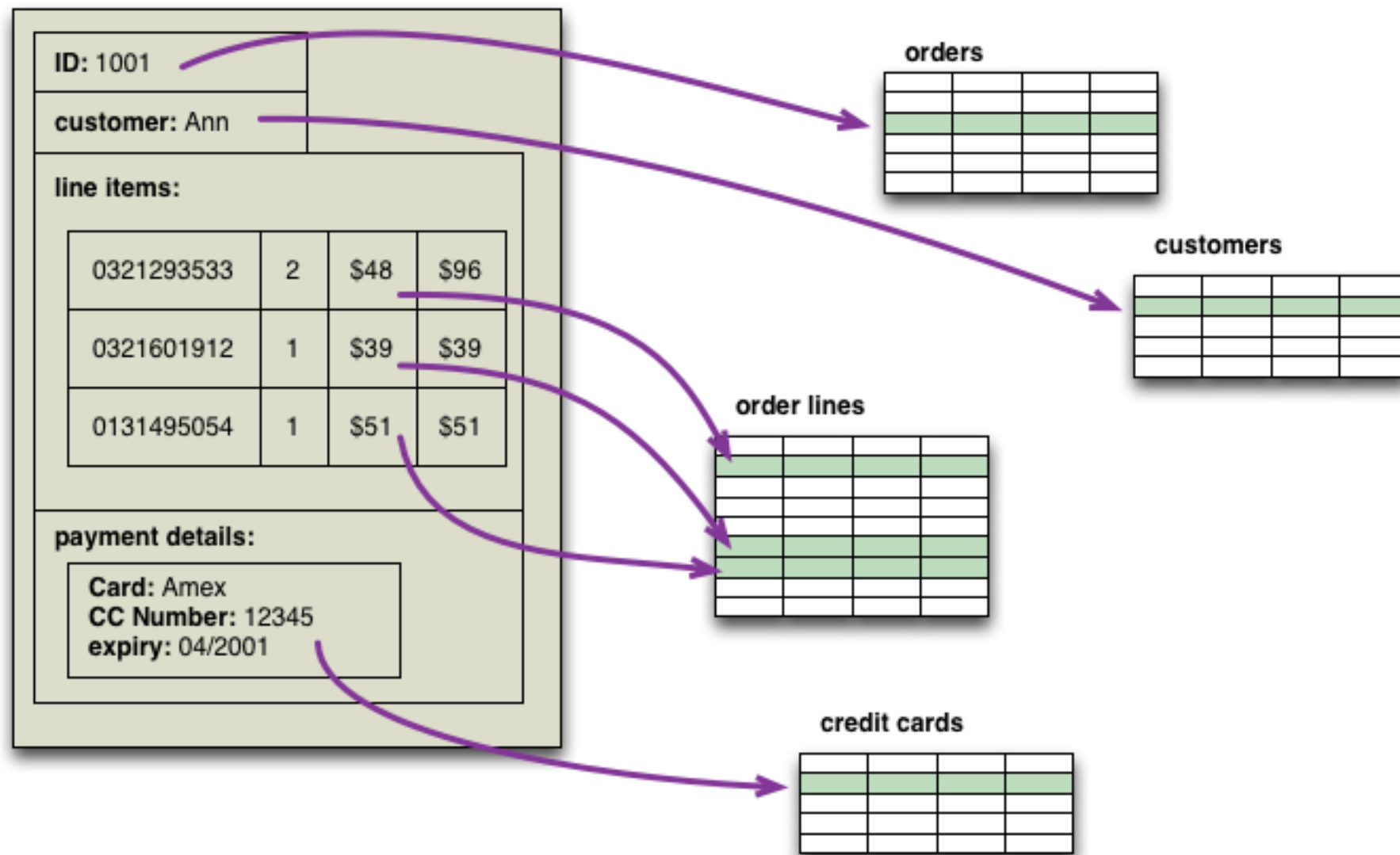
Document-oriented approach



<https://www.mongodb.com/>



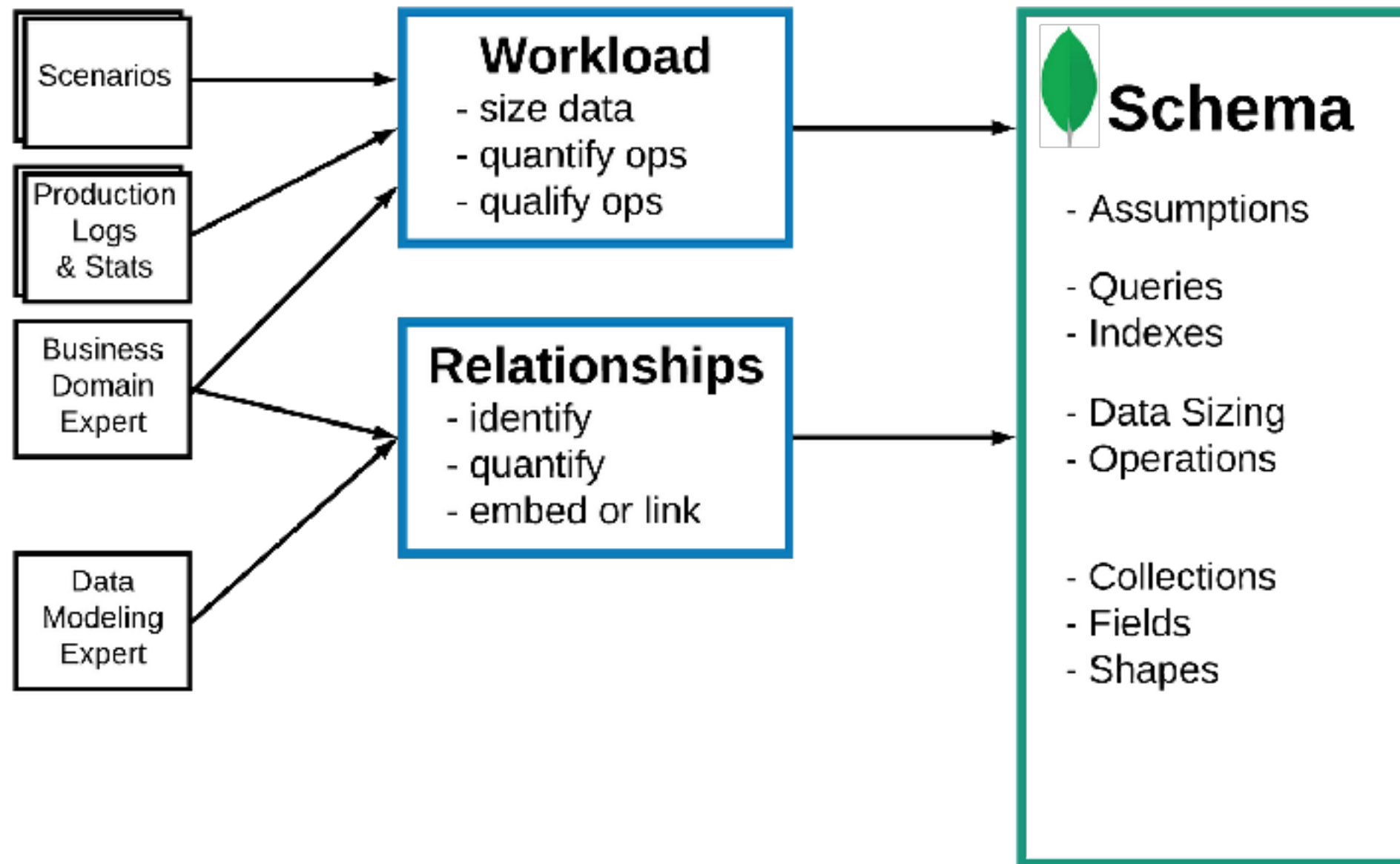
Document-based



<https://martinfowler.com/bliki/AggregateOrientedDatabase.html>



Data modeling for MongoDB



<https://www.mongodb.com/docs/manual/data-modeling/>



Working with Key-value database



Redis

REmote **D**ictionary **S**erver

Key-value database

Keep data in memory (save to disk)

Expiration data

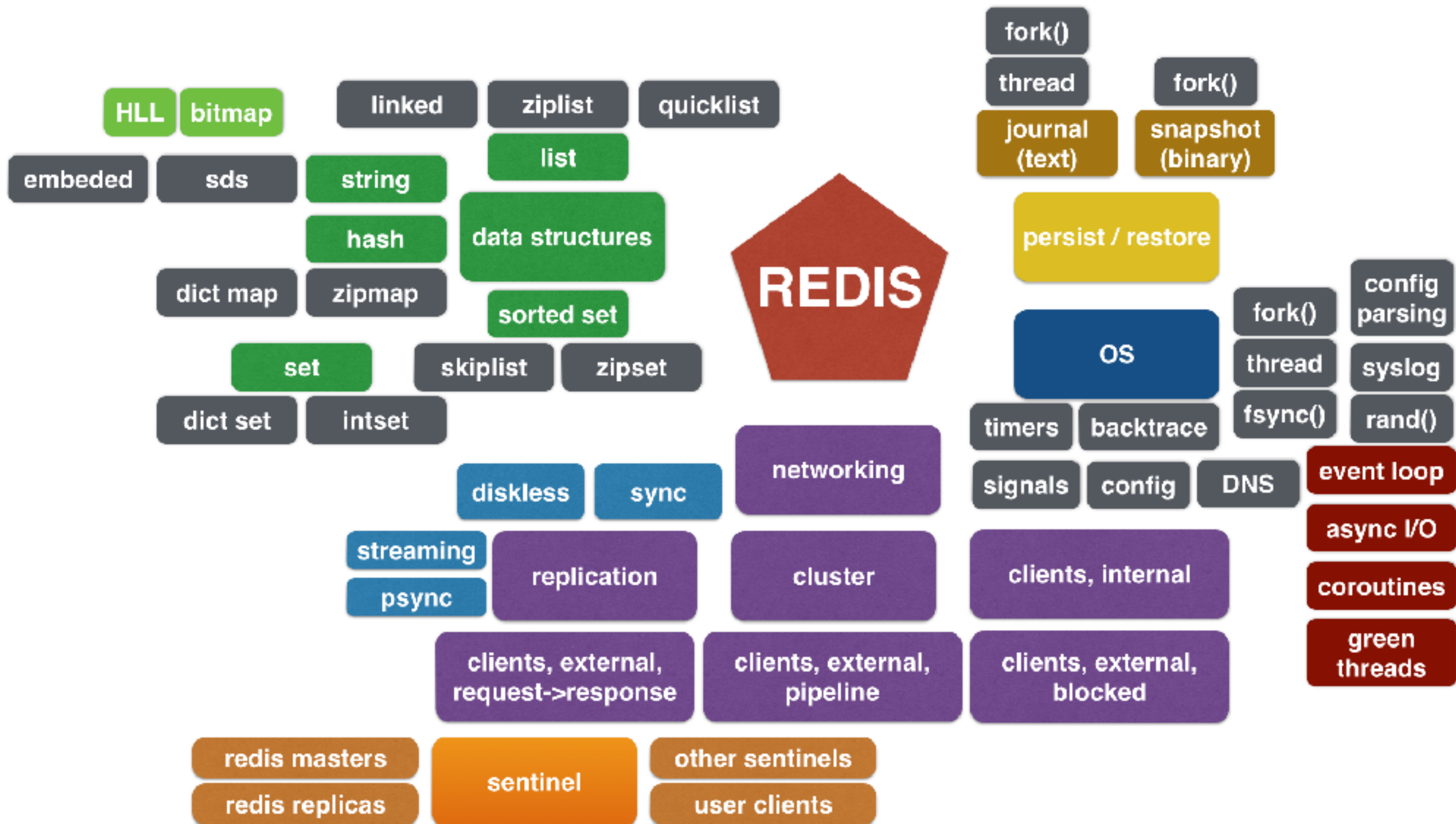
Data structure server



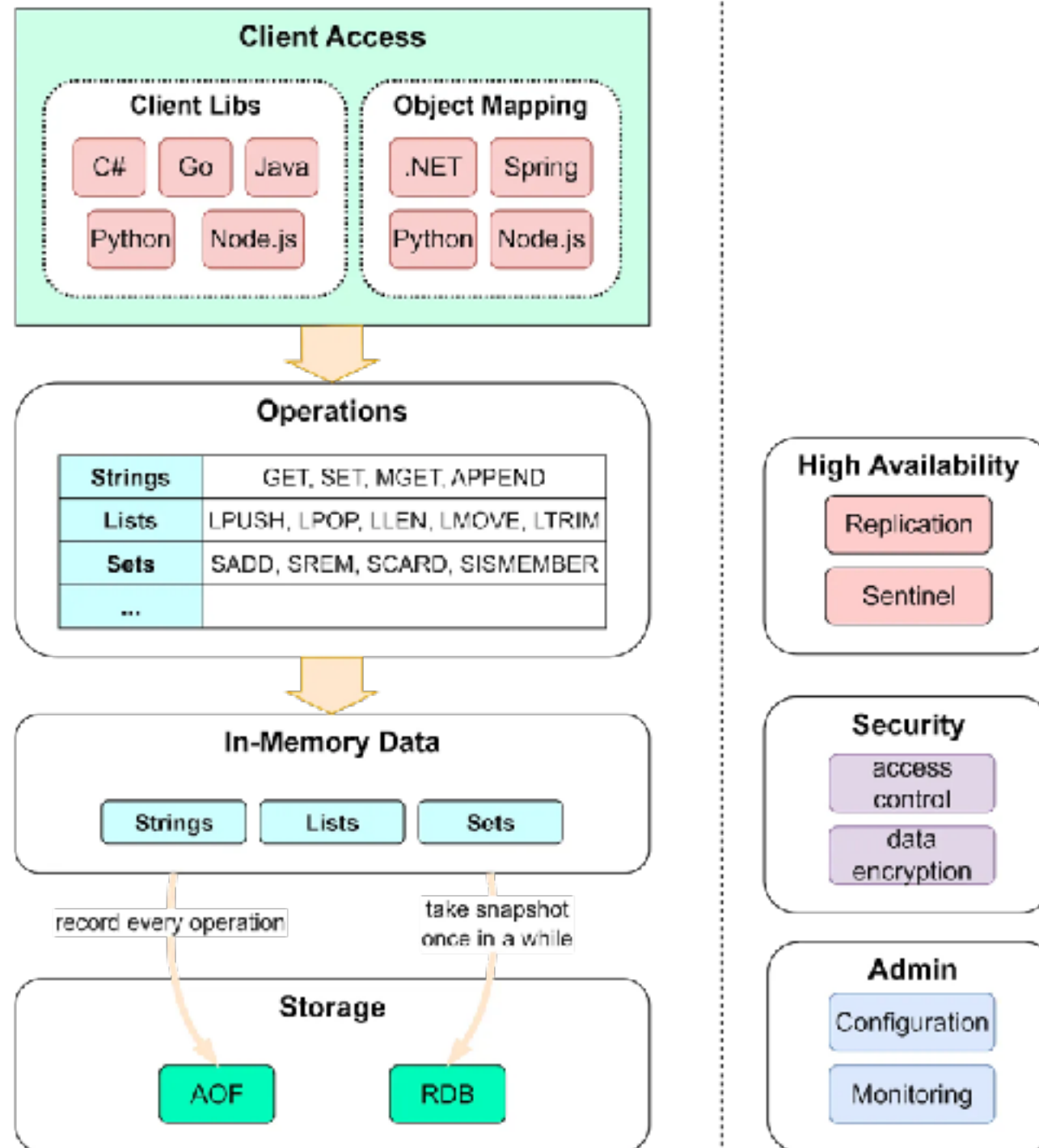
Efficient low-level data structure

<https://redis.io/technology/data-structures/>





High level view of Redis



<https://blog.bytebytego.com/p/a-crash-course-in-redis>



Why Redis Fast ?

RAM-based database

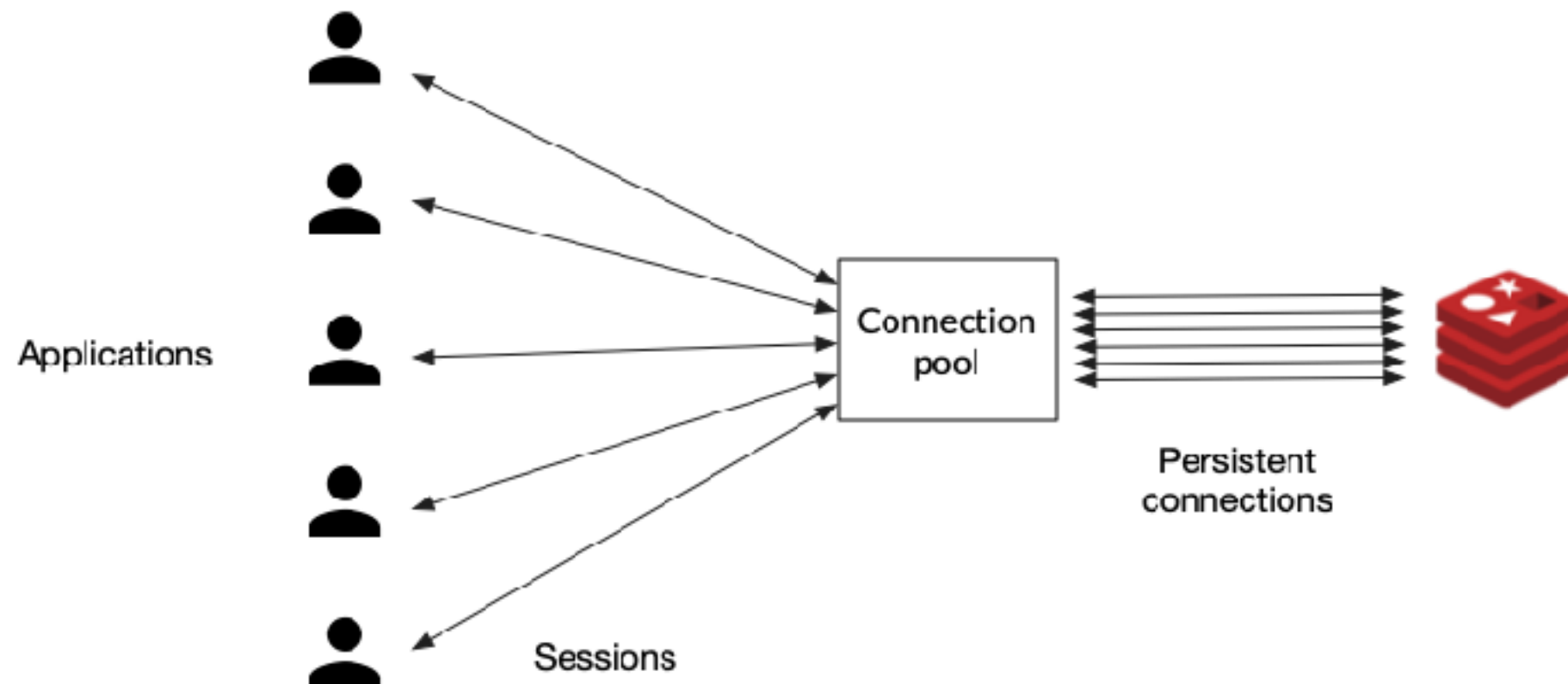
I/O multiplexing and single-thread read/write

Efficient low-level data structure



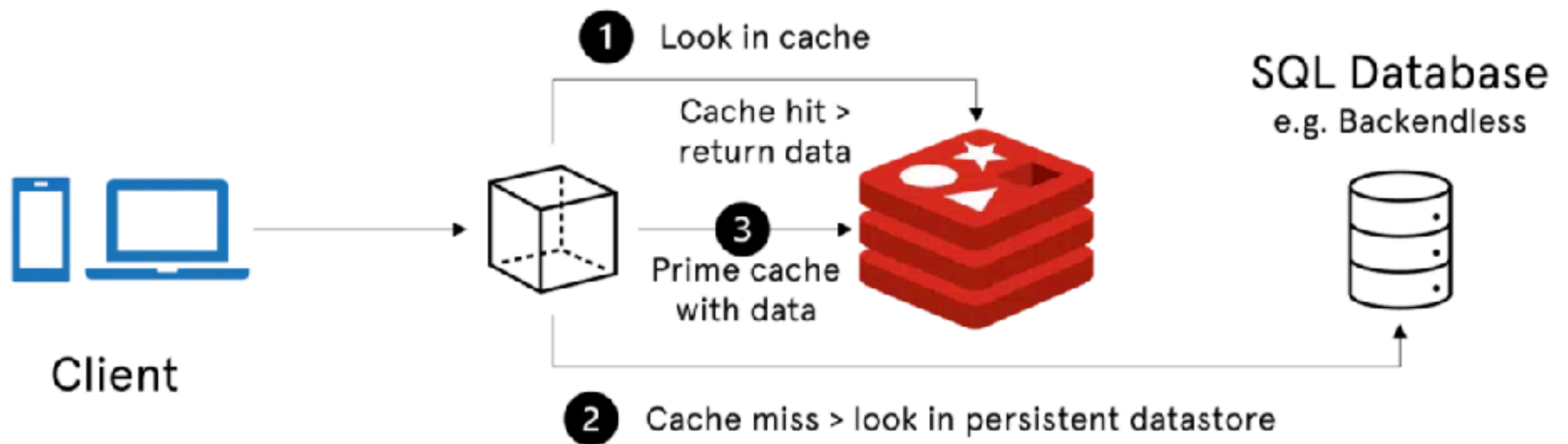
Use connection pool in client

Open and close connections cause overhead !!



Simple Use Case

Caching Data for applications
Reduce workload of database



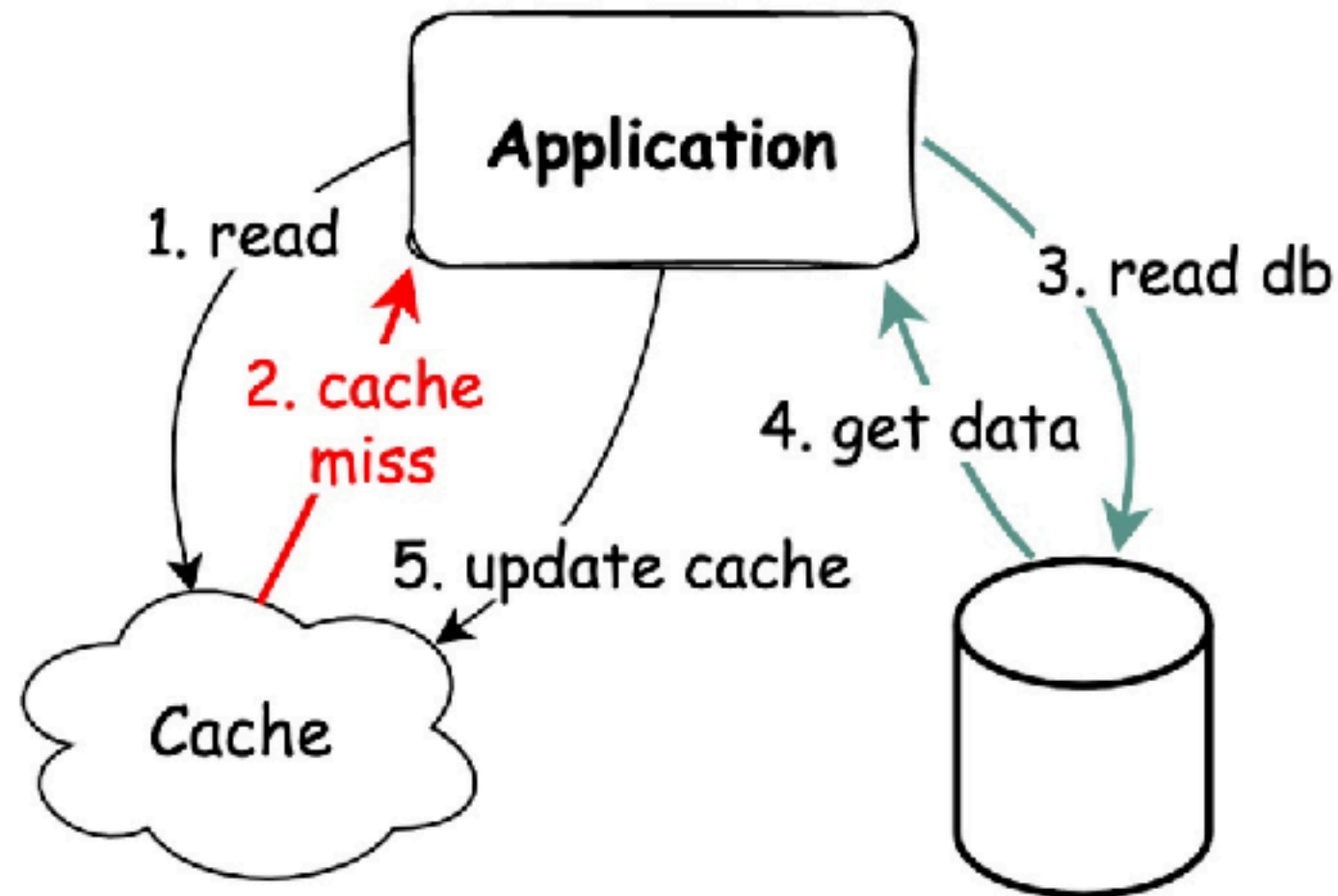
Caching Strategies ?

Read

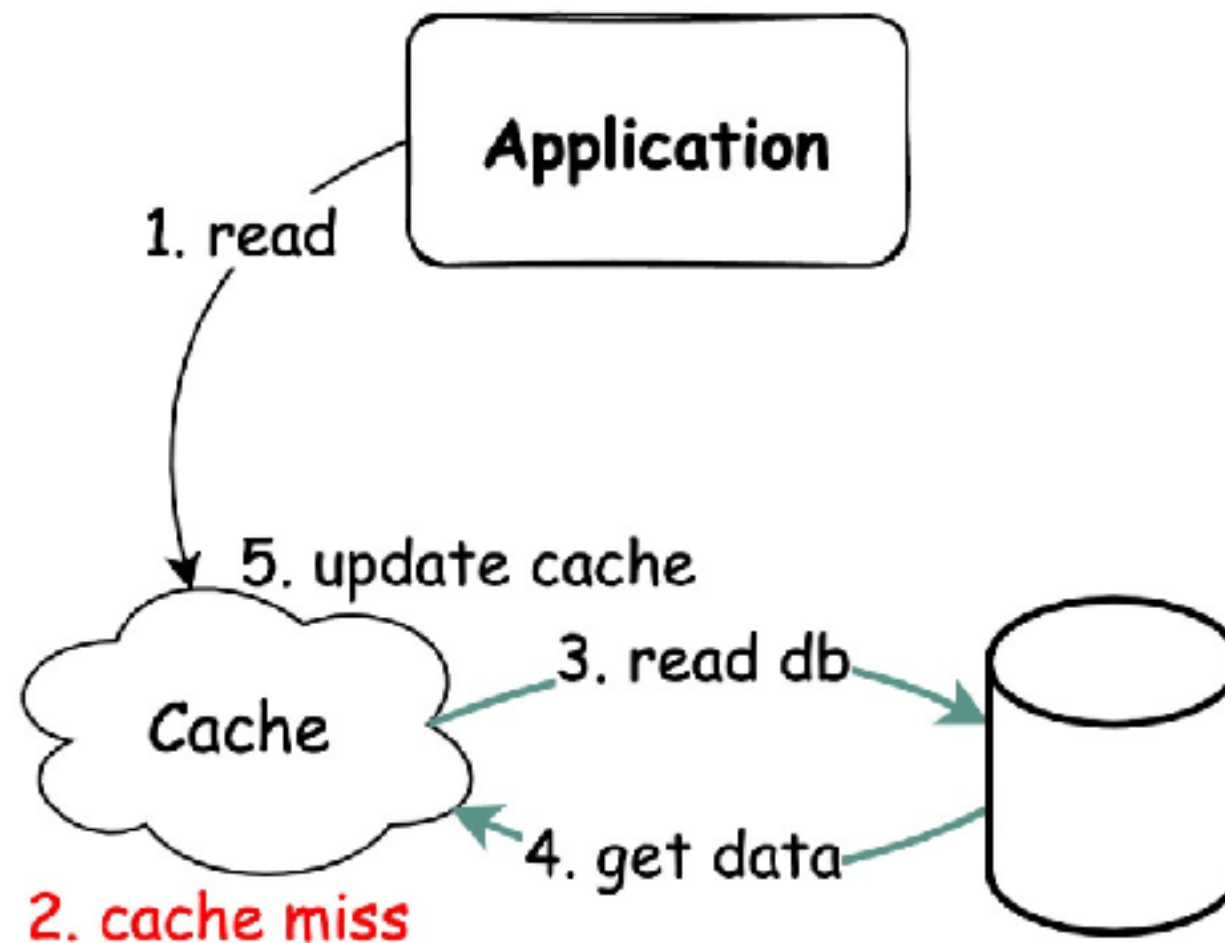
Write



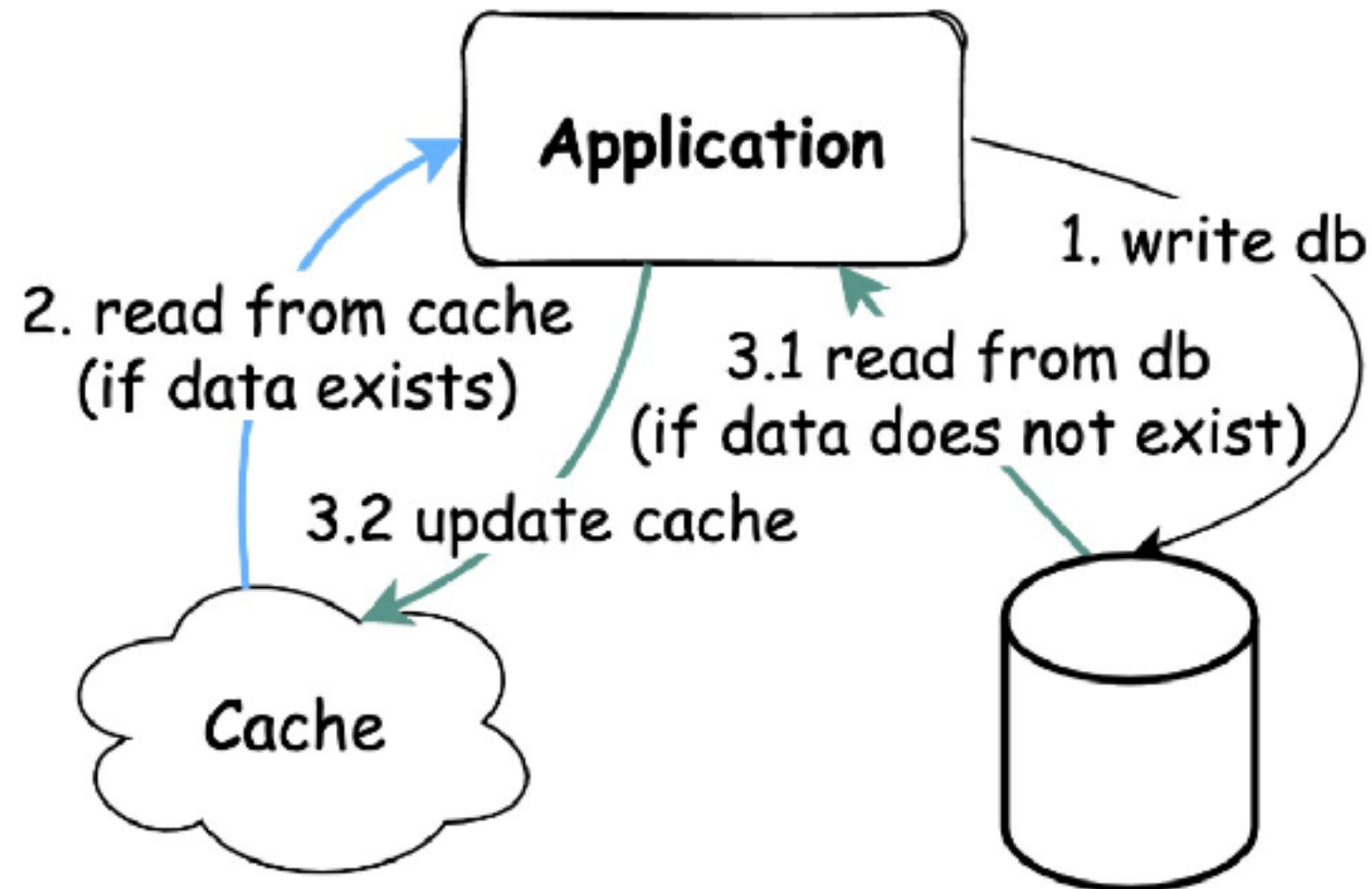
Read strategy - Cache Aside



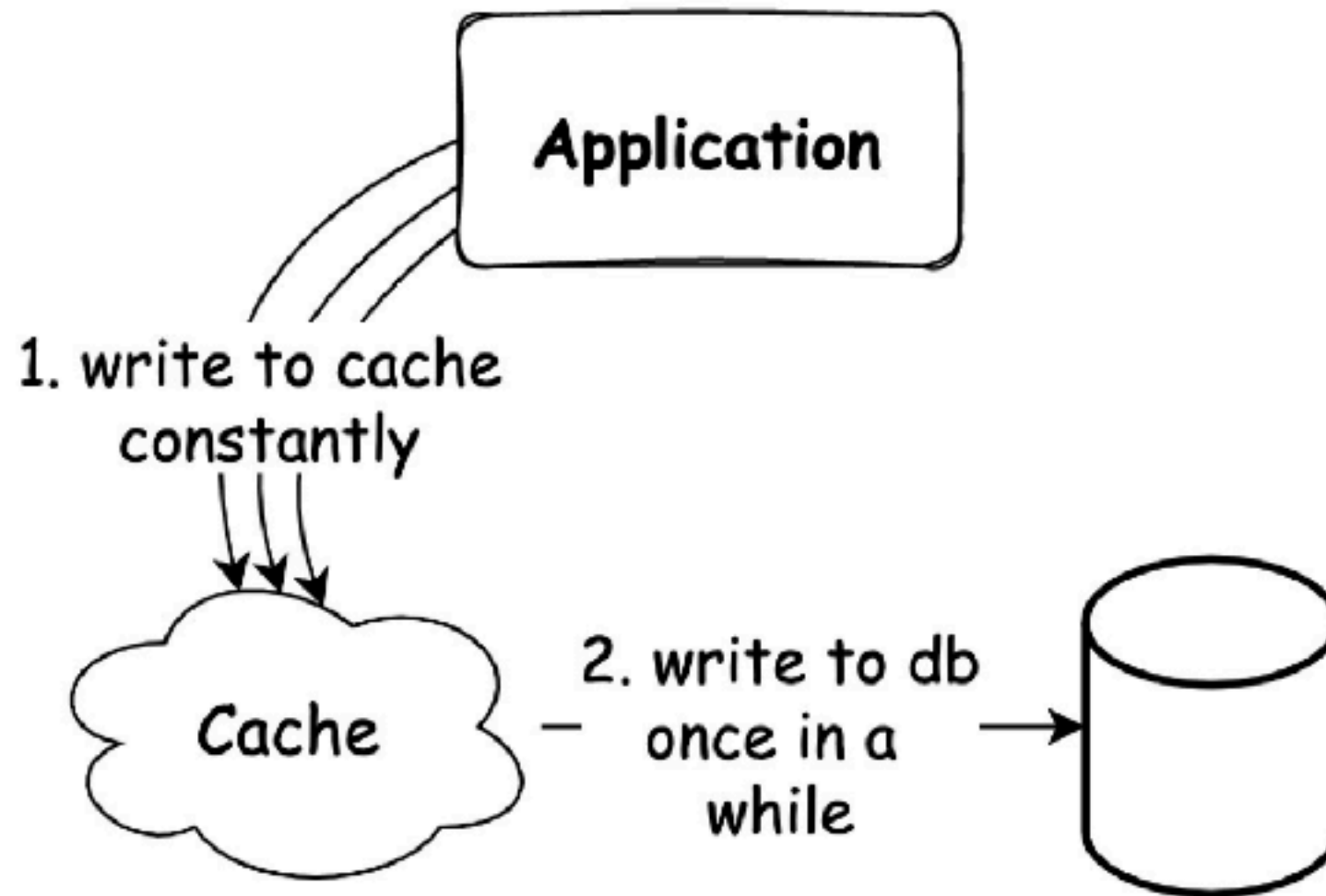
Read strategy - Read through



Write strategy - Write around

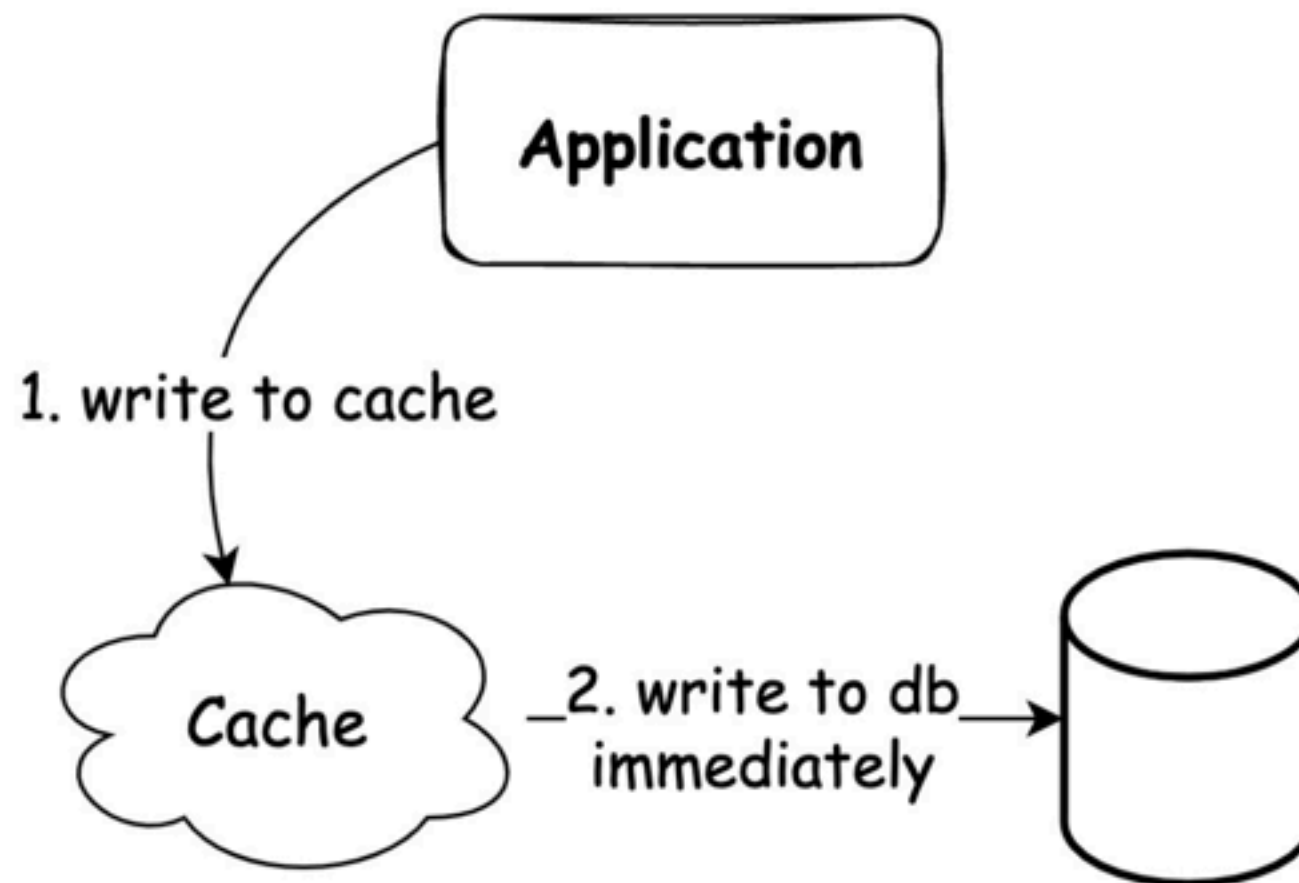


Write strategy - Write back



Write strategy - Write through

Write Strategy - Write Through



<https://redis.io/learn/howtos/solutions/caching-architecture/write-through>



More Use Cases

Session or Token data
Message broker (pub/sub)
Job queue
Realtime analytic
Fraud detection
Gaming



Sales leaderboard

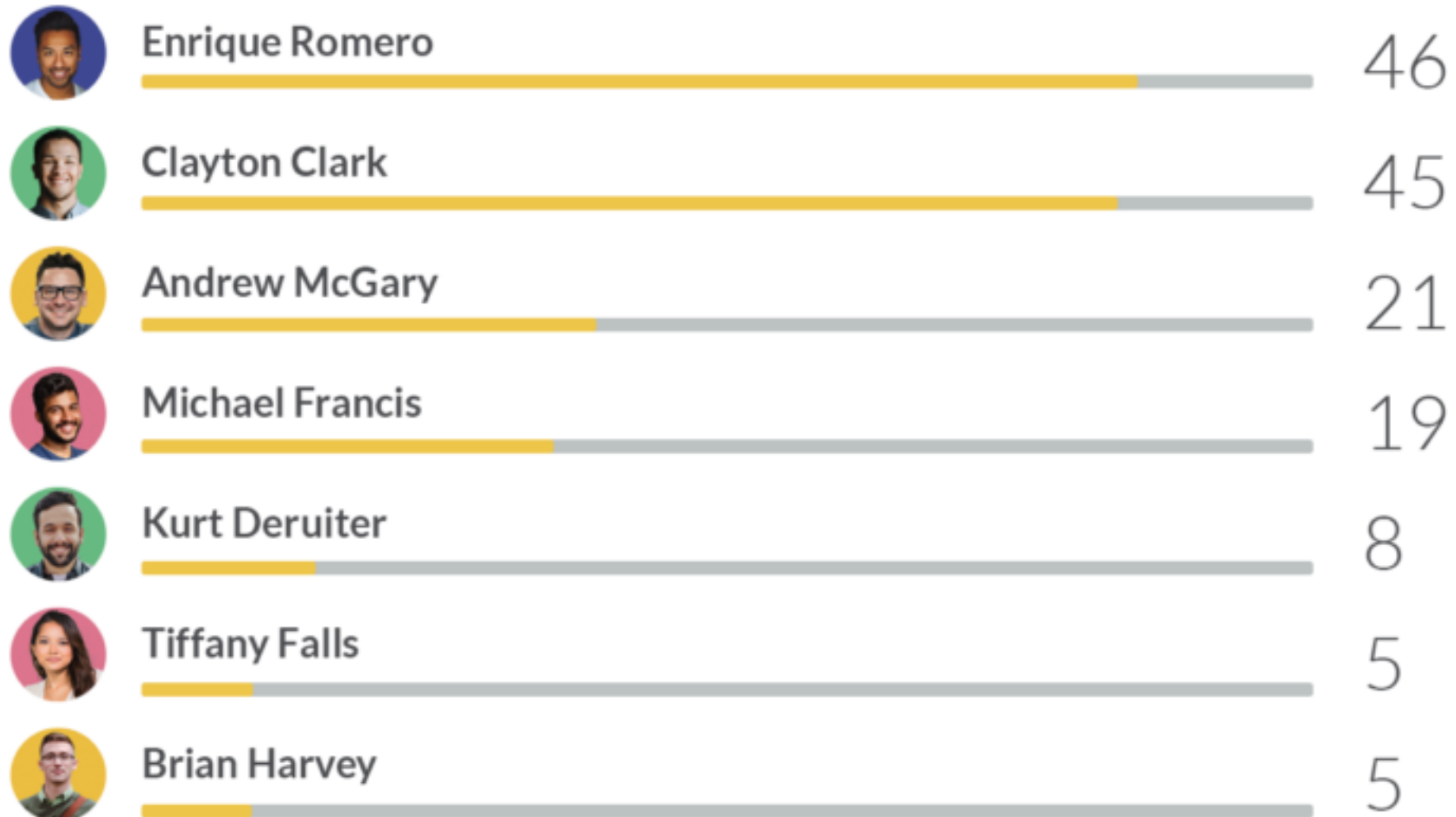
BY WIN RATE

BY LEADS WON

BY VALUE WON

BY ACTIVITIES

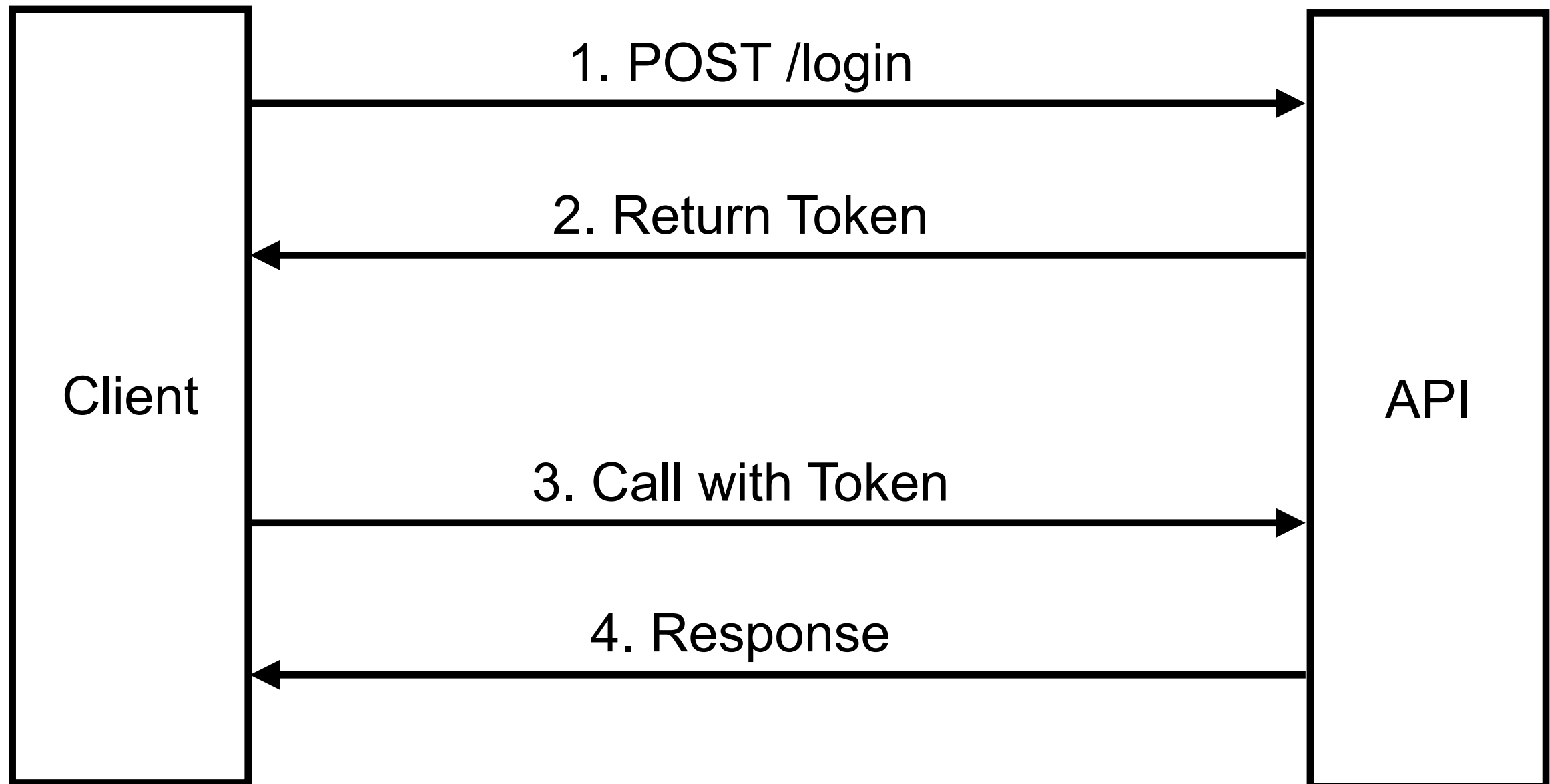
Sales Reps



Authentication Authorization

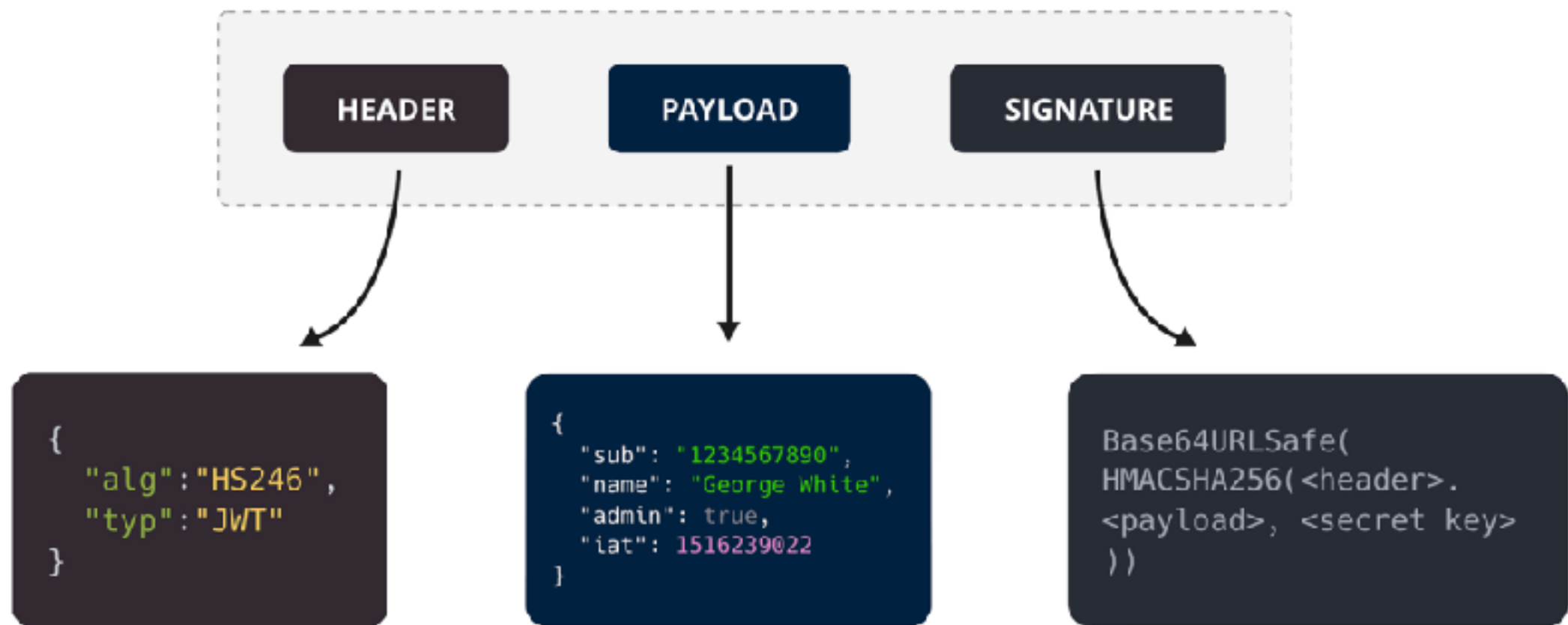


Simple Flow



JWT Token

Structure of a JSON Web Token (JWT)

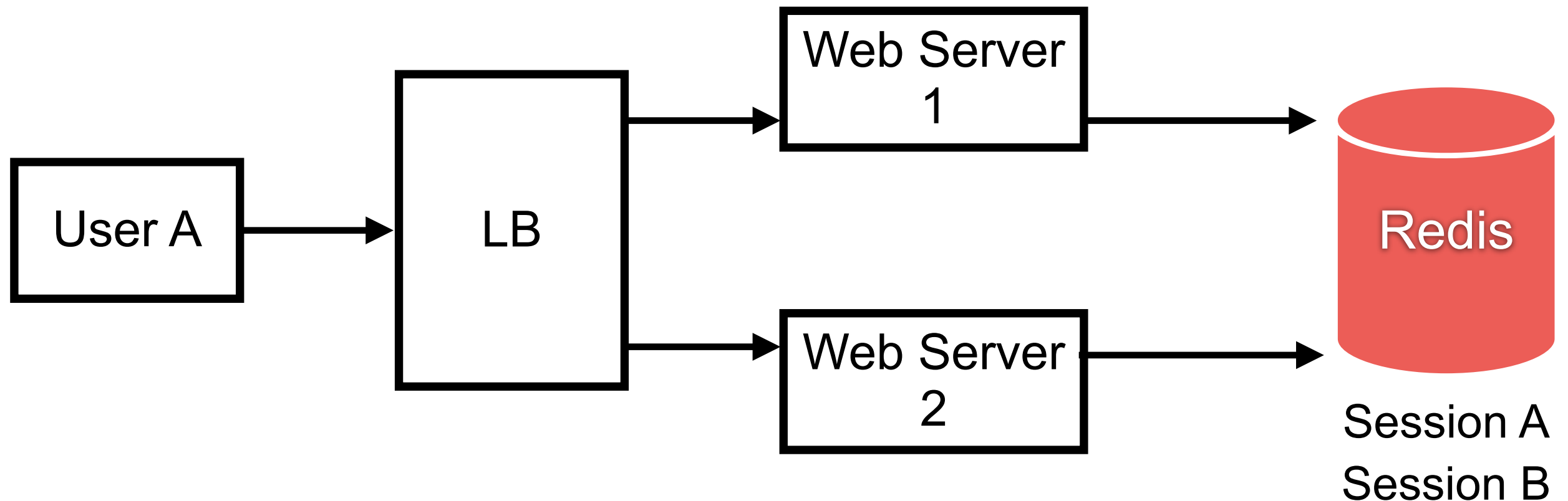


<https://jwt.io/>



Session or Token data

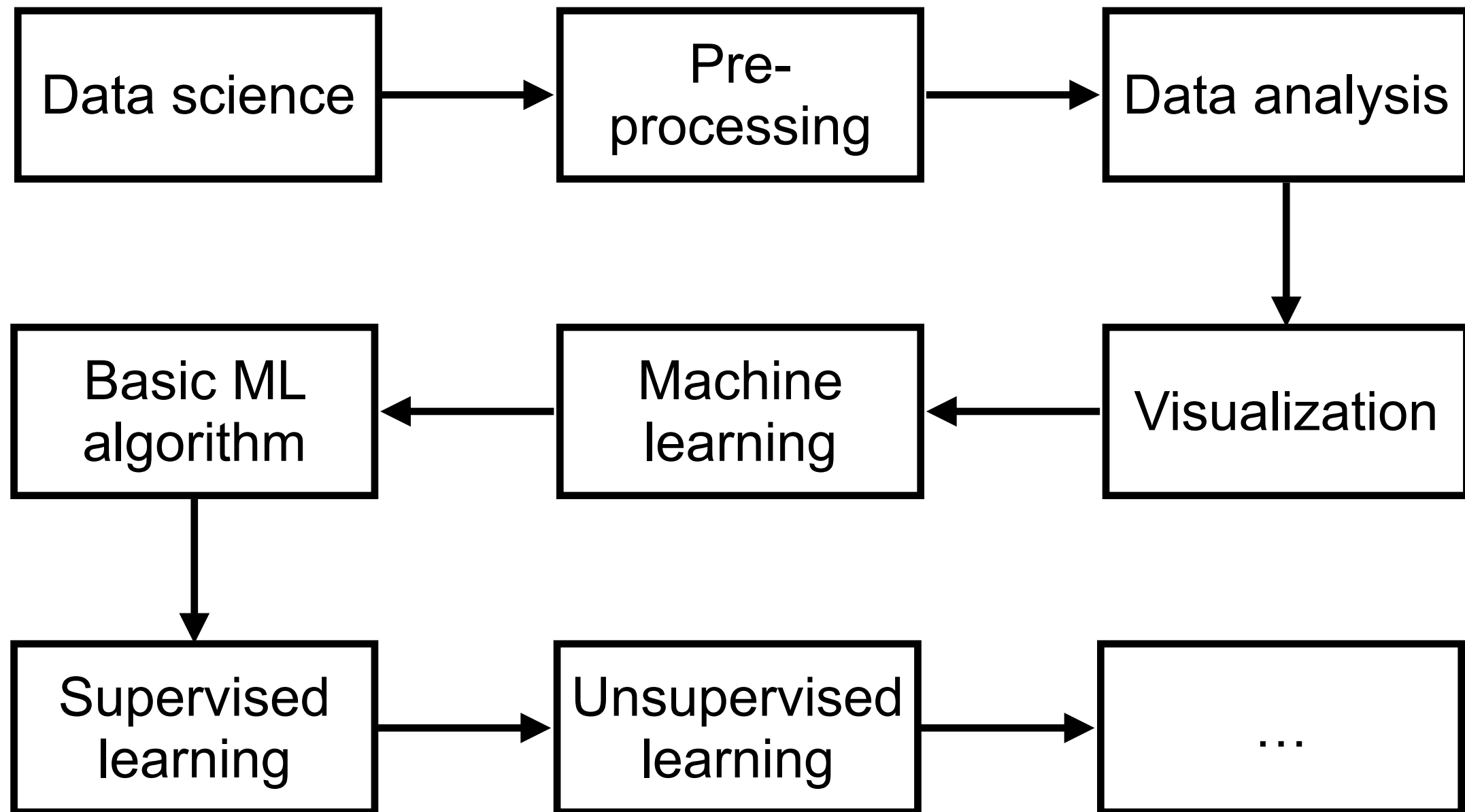
Data with expire time (automatically delete)
key=value(String)



Part 3



Learning Path 3

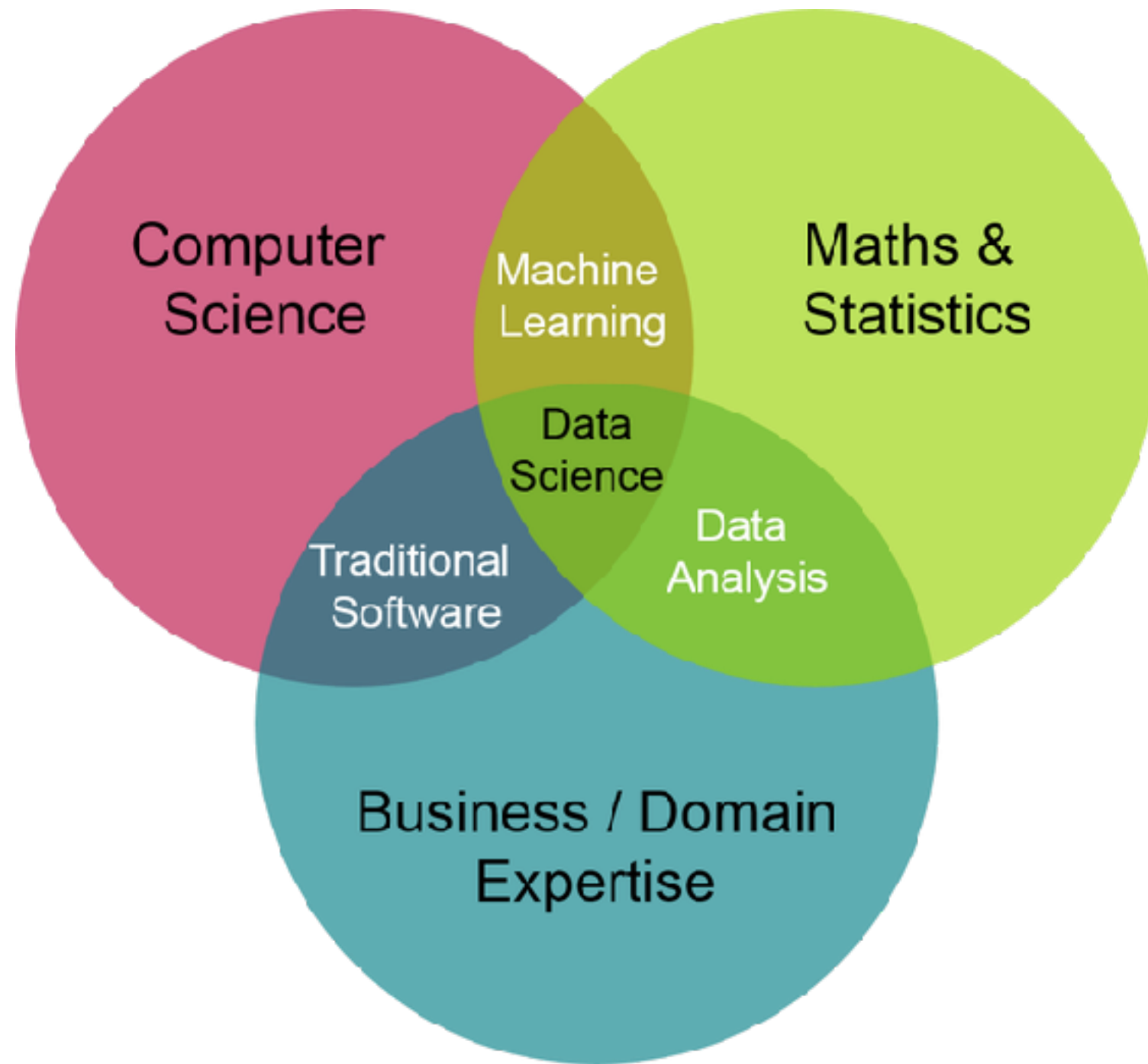


Data Science

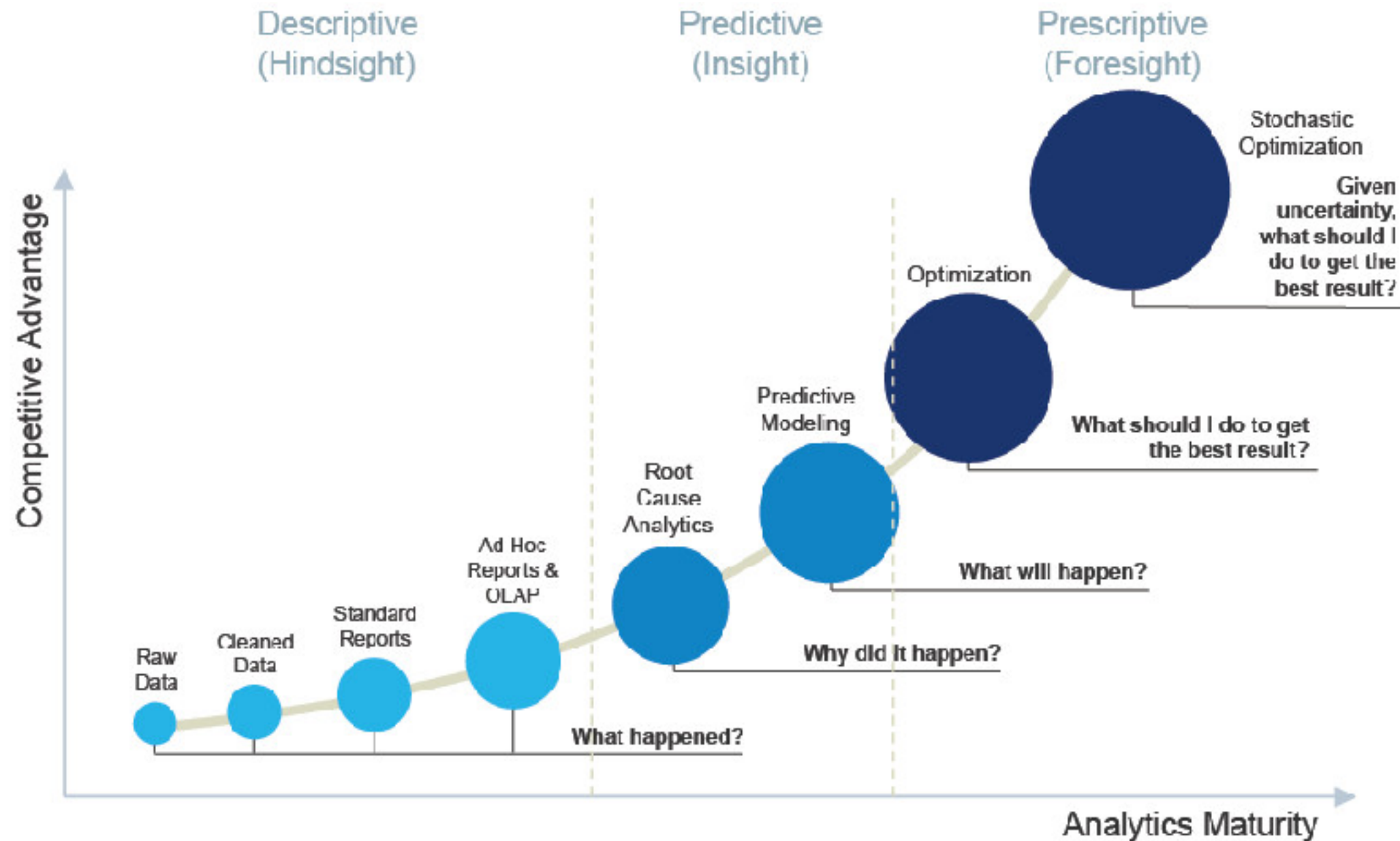
The study of formulating and rigorously answering questions using **data-centric process** that emphasizes clarity, reproducibility, effective communication and ethical practices



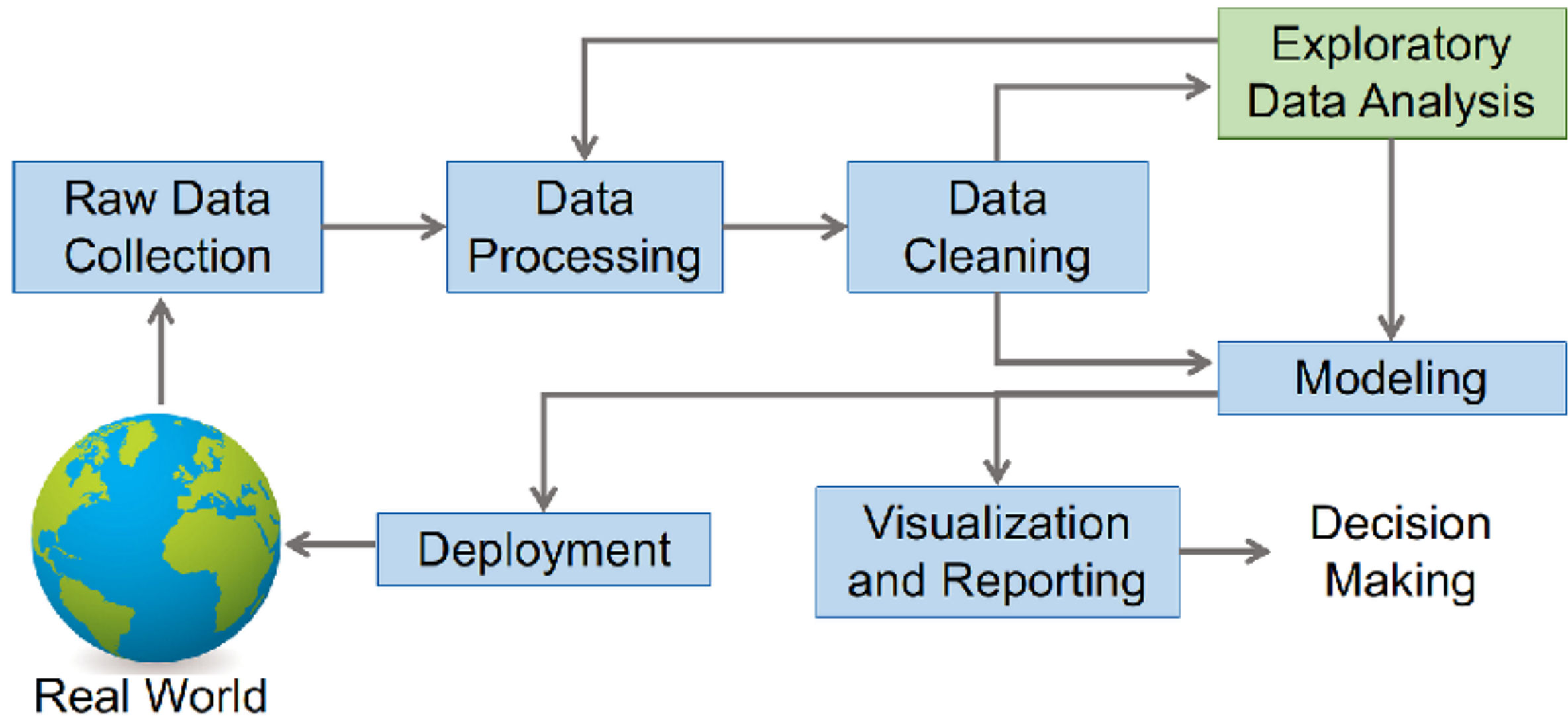
Data Science



Data Science



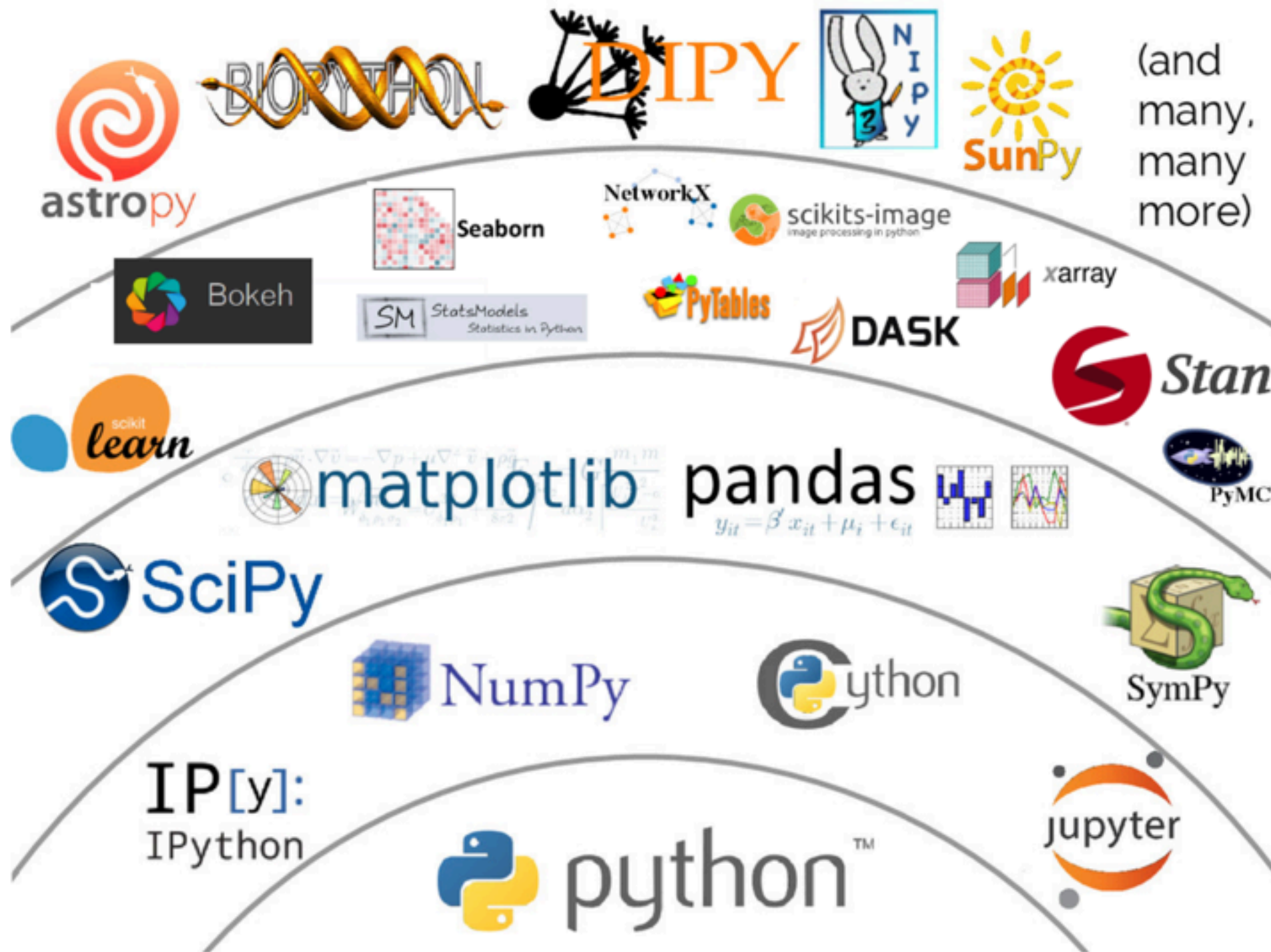
Data Science Process



<https://www.markovml.com/blog/exploratory-data-analysis>



Data Science in Python

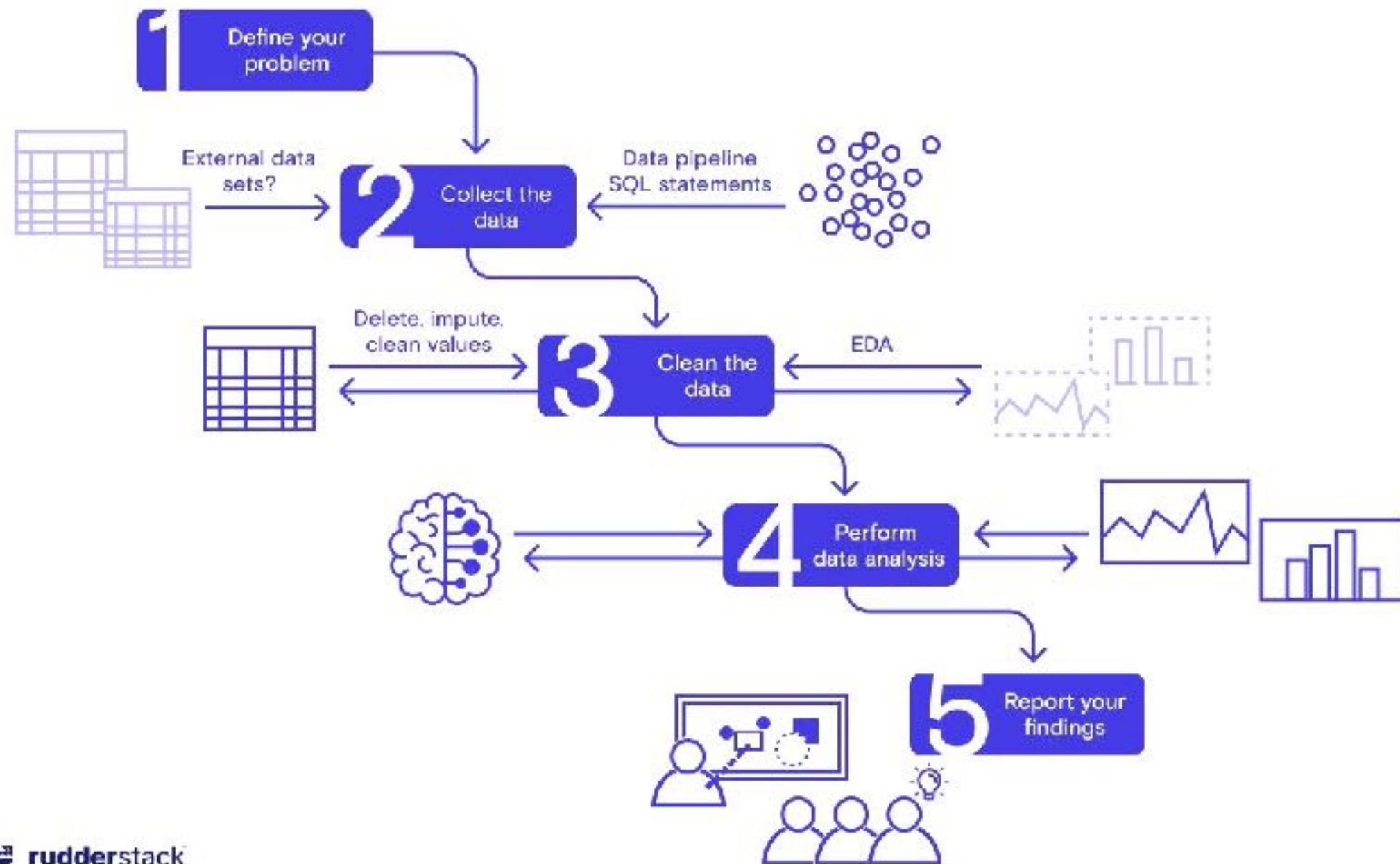


Data Analysis

The process of inspecting, cleaning, transforming, and modeling data to discover useful information, draw conclusions, and **support decision-making**



Data analysis process



 **rudderstack**

<https://www.rudderstack.com/learn/data-analytics/data-analytics-processes/>



Exploratory Data Analysis

Statistical approach that uses visual and numerical methods to **understand** the main characteristics of a dataset, often before applying more advanced analysis techniques



Python for Data analysis



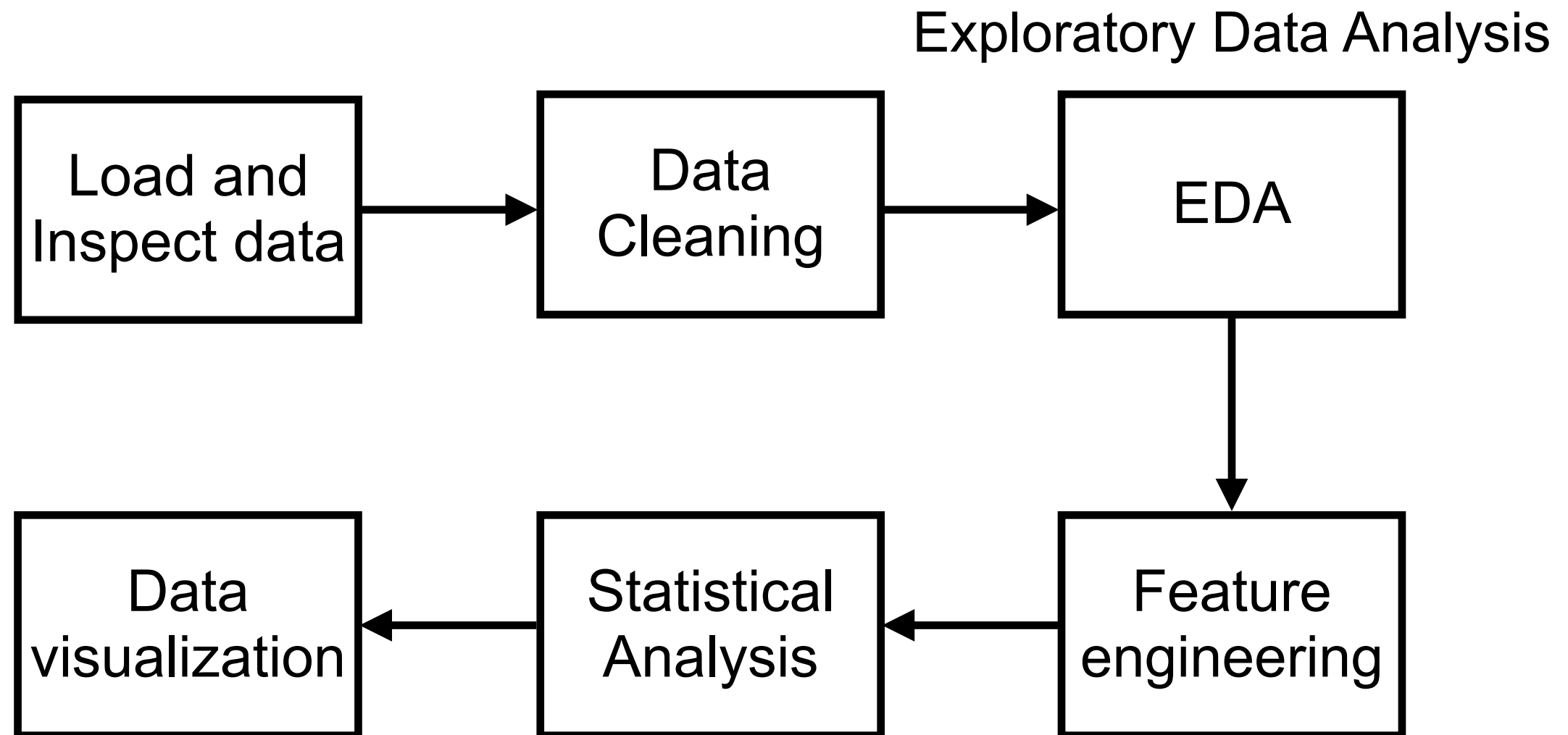
Visualization with Python



<https://matplotlib.org/>



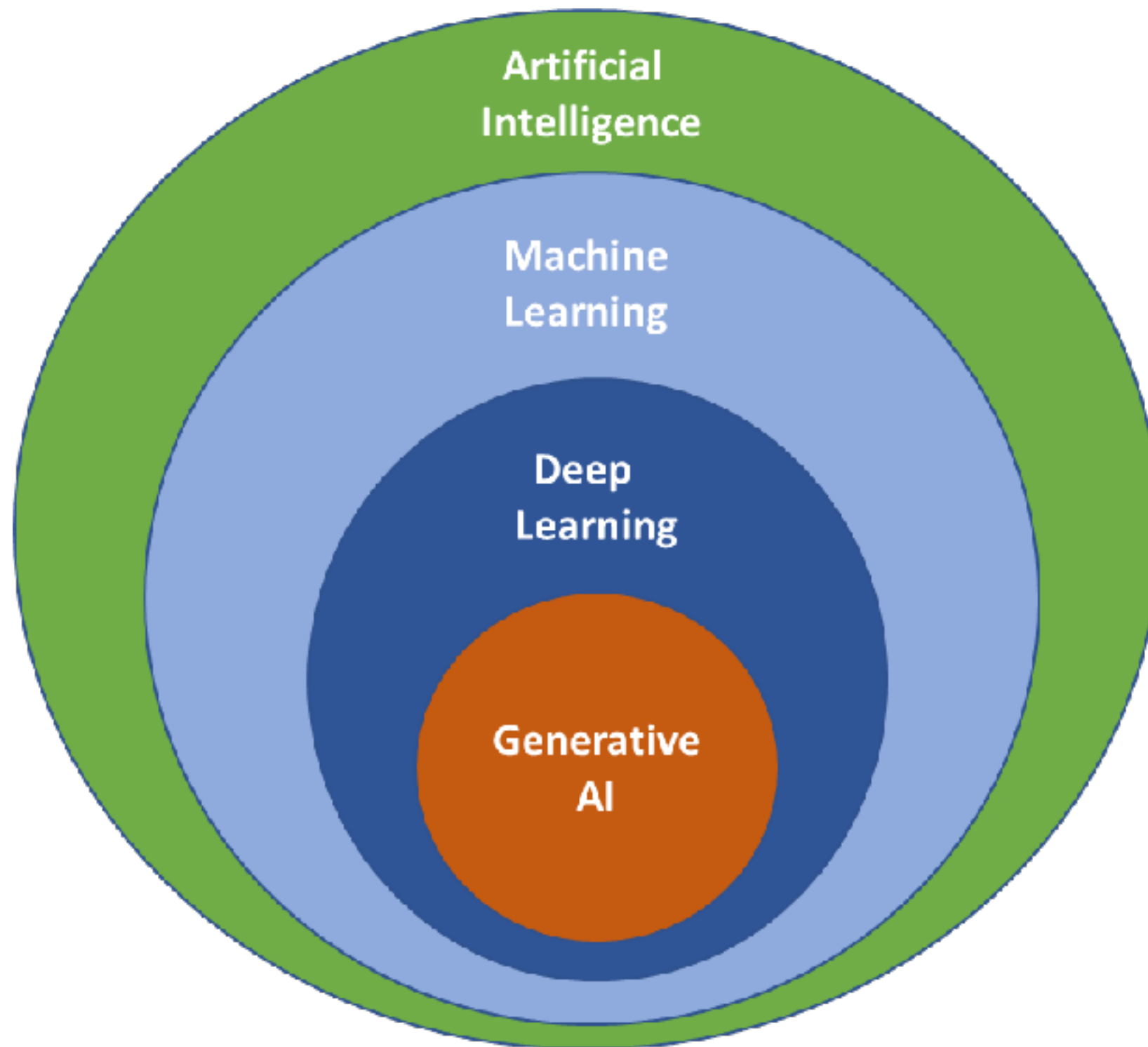
Workshop with Data Analysis



Machine Learning



Machine Learning ?



Artificial Intelligence (AI)

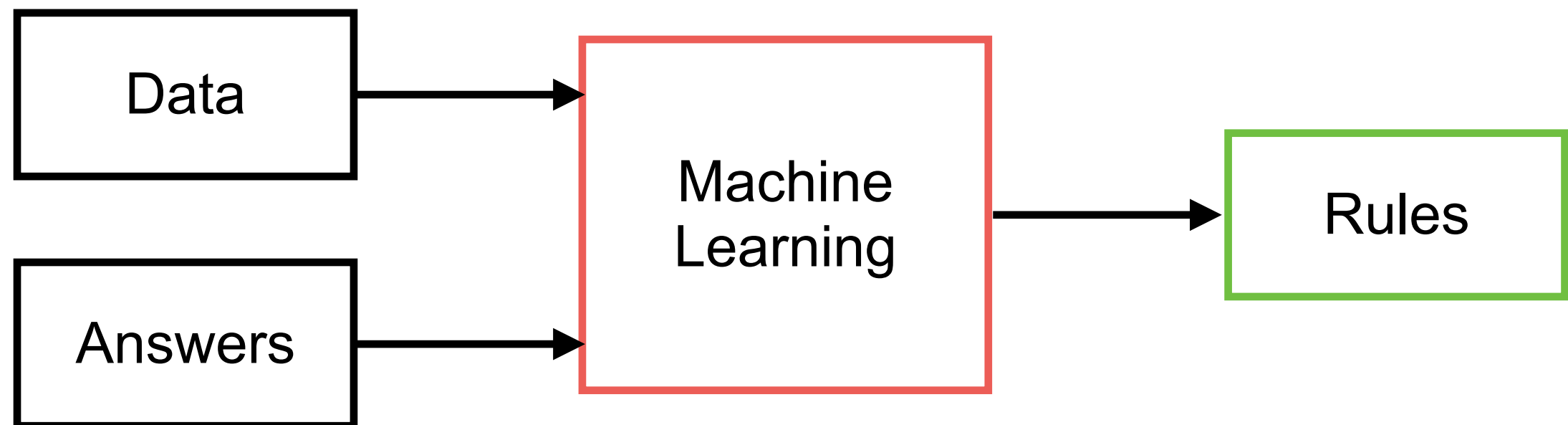
The effort to automate intellectual tasks normally performed by humans

Simple and complex tasks



Machine Learning (ML)

A subset of AI that can learn by themselves
ML models take in data and fit data to algorithm



ML Use Cases

Image recognition

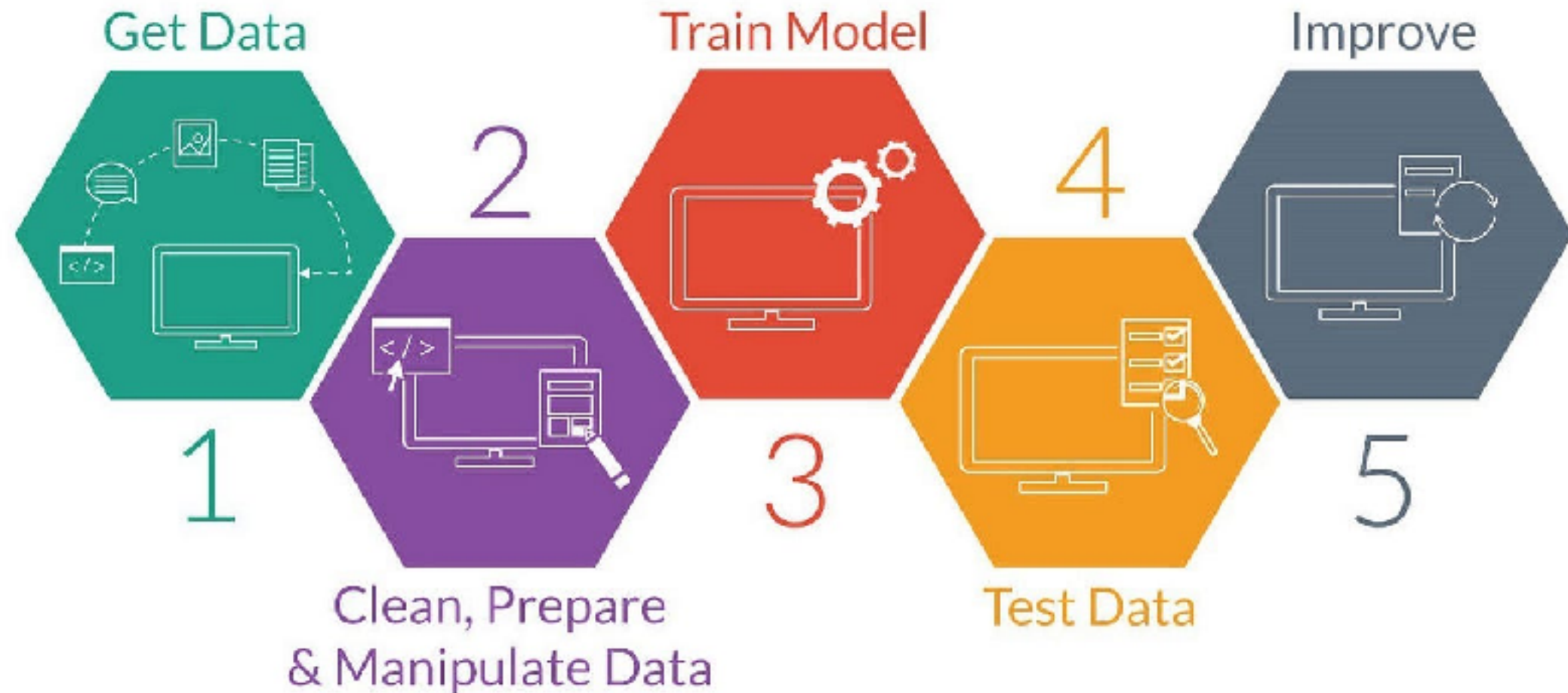
Natural Language
Processing

Recommendation
System

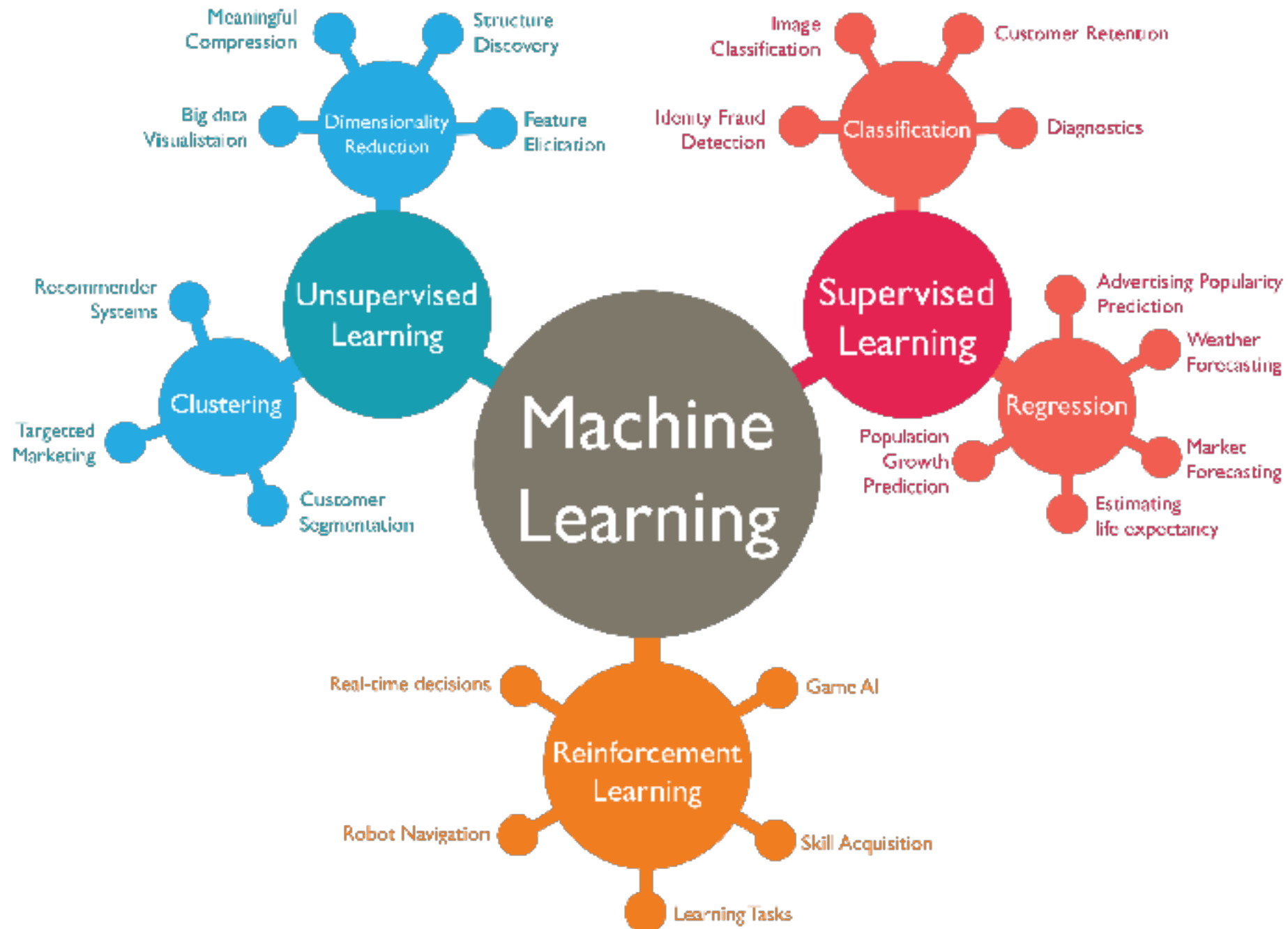
Predictive
Maintenance



Machine learning workflow



Machine Learning



<https://www.datacamp.com/blog/what-is-machine-learning>



Supervised Learning

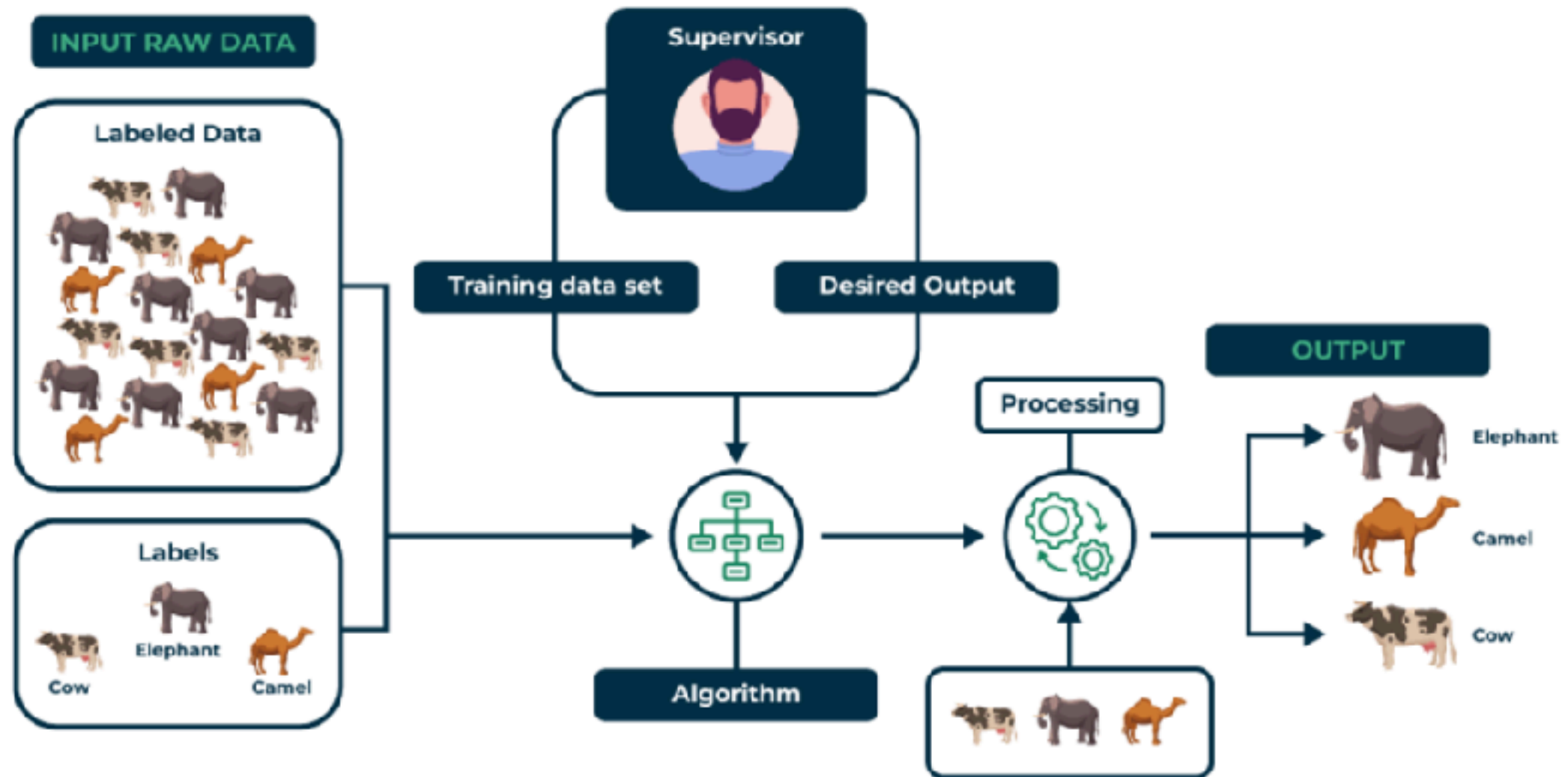
Trains models on labeled data to predict or classify new, unseen data.

Classification

Regression



Supervised Learning



<https://www.geeksforgeeks.org/machine-learning/>



Unsupervised Learning

Finds patterns or groups in unlabeled data, like clustering or dimensionality reduction.

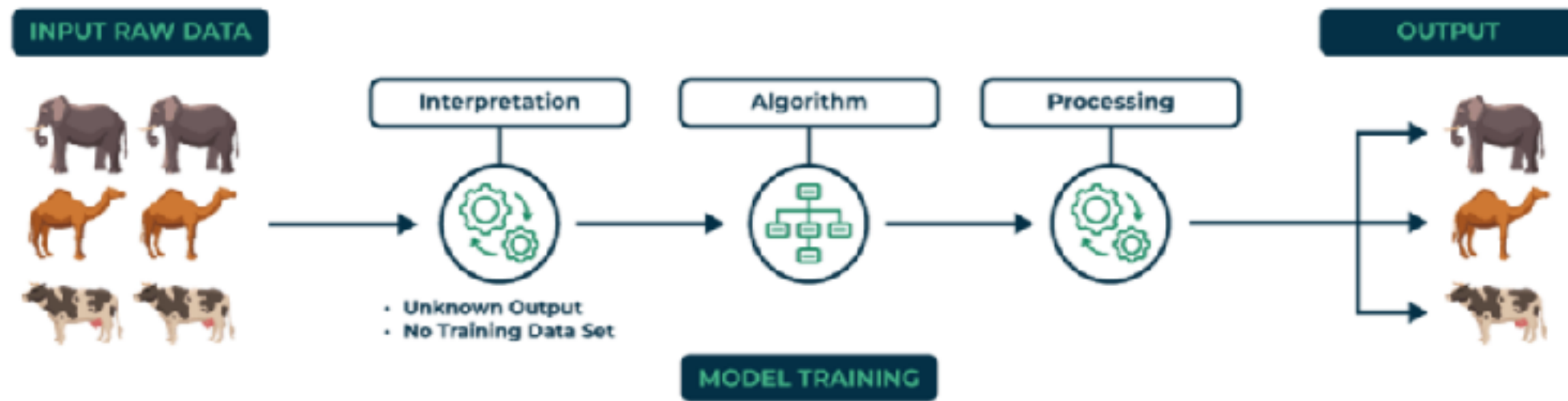
Clustering

Association Rule

Dimensionality
Reduction



Unsupervised Learning

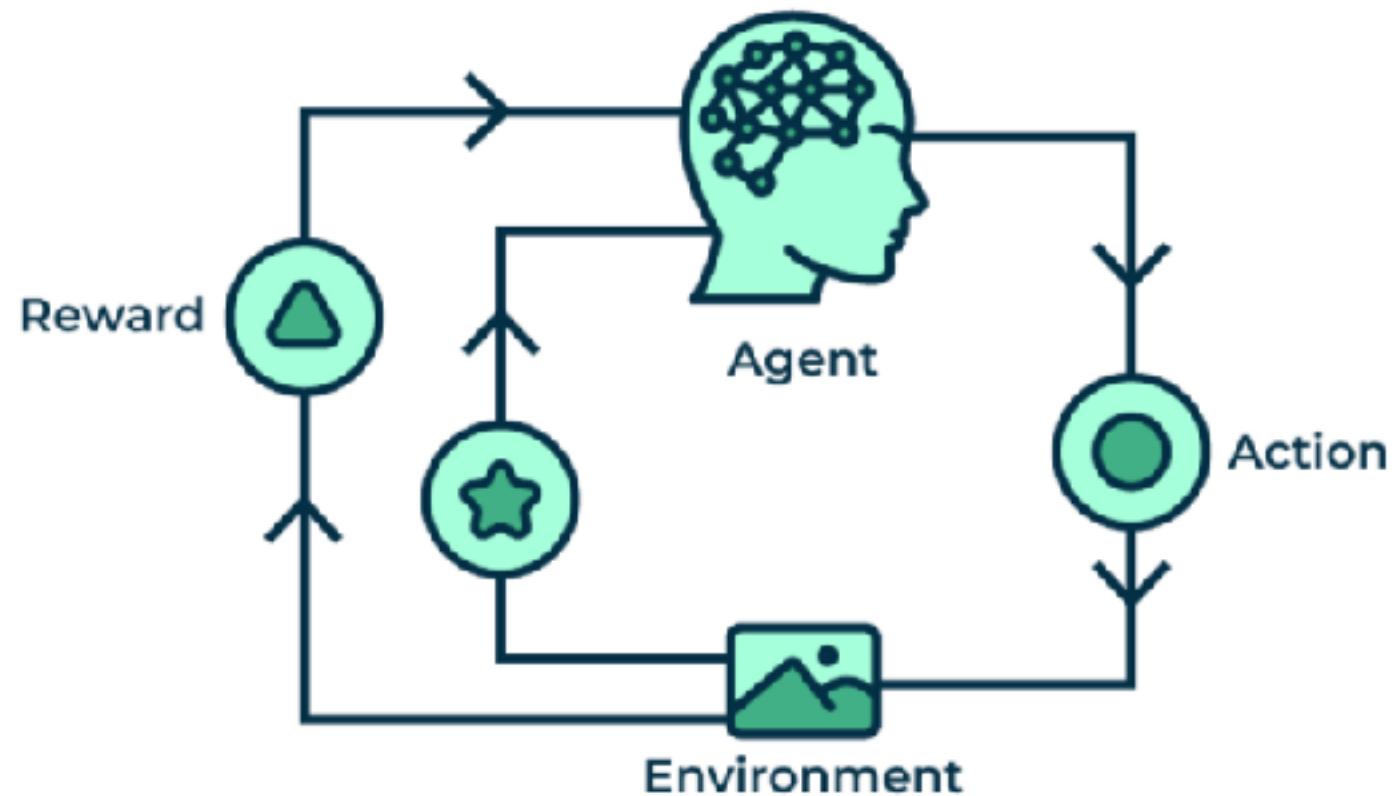


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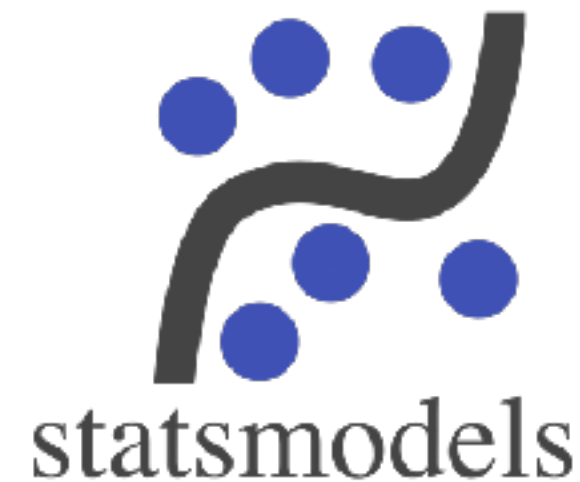


Reinforcement Learning

Learns through trial and error to maximize rewards, ideal for decision-making tasks.



Python for Machine learning



Let's go !!

